

Multimodal Retrieval-Augmented Generation for Chart and Document Understanding: A Comprehensive Survey

1 Introduction, Scope, and Problem Formulation

1.1 Defining the Scope of Multimodal RAG in Visually Rich Domains

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The rapid evolution of Retrieval-Augmented Generation (RAG) has fundamentally transformed how artificial intelligence systems access and synthesize external knowledge. However, the prevailing paradigm of standard text-only RAG, while effective for unstructured textual corpora, encounters significant limitations when applied to visually rich domains such as financial reports, scientific literature, technical manuals, and complex data visualizations. In these domains, information is not merely encoded in linear text but is intrinsically embedded within a heterogeneous interplay of layout geometry, visual semantics, and textual content. Consequently, defining the scope of Multimodal RAG (MM-RAG) for charts and documents requires a rigorous delineation that distinguishes it from both traditional text-centric retrieval and unimodal Visual Question Answering (VQA). The core distinction lies in the necessity for MM-RAG systems to perform joint reasoning over multi-granular structural cues and visual evidence, rather than treating images as mere illustrative supplements or relying solely on Optical Character Recognition (OCR) transcriptions. This definition sets the stage for understanding why the field must evolve beyond the linear assumptions of text-centric paradigms discussed in the following section.

Standard text-only RAG operates under the assumption that knowledge can be effectively chunked, embedded, and retrieved based on semantic similarity in a vector space. This approach fails catastrophically in visually rich document understanding (VRDU) because it inherently discards the spatial and structural relationships that define meaning in documents like tables, forms, and charts. For instance, in a financial report, the semantic relationship between a row header and a numerical value is defined by their relative spatial positioning and alignment, not just their textual proximity. Research has demonstrated that layout geometry is highly predictive of entity relations and can even serve as a feasible unimodal approach in certain contexts, suggesting that ignoring layout leads to a fundamental loss of information [1]. Furthermore, traditional pipelines often rely on flattening documents into text streams, which destroys the hierarchical semantic relations among multiple pages and multimodal components, a challenge explicitly addressed by datasets like PDF-MVQA that aim to retrieve entire paragraphs and visually rich entities simultaneously [2]. Therefore, the scope of MM-RAG must extend beyond simple text retrieval to include the indexing and retrieval of structured visual regions, requiring models that can encode and reason over layout-aware representations [3]. The inability of flat text embeddings to capture these non-linear dependencies directly motivates the shift toward multi-granular and hierarchical retrieval strategies explored subsequently.

Conversely, unimodal Visual Question Answering (VQA) tasks, while capable of processing images, typically operate in a closed-world setting where the model relies entirely on its parametric knowledge or the immediate visual context of a single image without access to an external, scalable knowledge base. Unimodal VQA often struggles with the “long-tail” knowledge required for complex document analysis, such as specific regulatory details in legal contracts or historical data trends in multi-page scientific papers. Moreover, many existing VQA benchmarks focus on natural

images or isolated charts, neglecting the document-level context where charts are embedded. The introduction of tasks like Document-Level Chart Question Answering (DCQA) highlights the gap in current systems, emphasizing the need to first extract charts via document layout analysis before performing complex reasoning that integrates common-sense understanding with document-wide context [4]. Unlike standard VQA, which might answer a question based on a single plot, MM-RAG for documents must retrieve relevant pages or sections from a massive corpus and then ground its generation in the specific visual evidence found within those retrieved segments, effectively bridging the gap between large-scale retrieval and fine-grained visual reasoning.

The unique scope of MM-RAG in visually rich domains is further characterized by the challenge of handling multimodal inputs where visual content is not auxiliary but essential. In domains such as product manual understanding, visual illustrations often contain critical procedural information that cannot be fully captured by text alone. The Multimodal Product Manual Question Answering (MPMQA) task exemplifies this by requiring models to process both visual contents and provide multimodal answers consisting of textual sentences and related visual regions, thereby enforcing a strict dependency on joint modality processing [5]. Similarly, in scientific domains, the ability to interpret complex figures, diagrams, and chemical structures alongside text is paramount. Benchmarks like MMMU and CMMMU have been developed to evaluate models on college-level multi-discipline tasks that demand deliberate reasoning over heterogeneous image types, including charts, maps, and tables, underscoring that high-level expertise requires a seamless integration of visual perception and textual knowledge retrieval [6; 7].

Furthermore, the scope of MM-RAG encompasses the mitigation of unimodal biases that plague current Multimodal Large Language Models (MLLMs). These models often exhibit an over-reliance on language priors or superficial visual features, leading to hallucinations when faced with complex, data-intensive queries. A causal perspective on these biases reveals that effective MM-RAG systems must employ strategies to decompose, verify, and answer queries by actively retrieving and grounding evidence, rather than relying on probabilistic generation alone [8]. This necessitates a shift from passive recognition to active visual search and reasoning mechanisms, such as those proposed in V*, which utilize LLM-guided visual search to focus on important visual details in high-resolution, crowded images [9].

In summary, the scope of Multimodal RAG for charts and document understanding is defined by its capacity to navigate non-linear, structured, and visually dense information spaces. It transcends the limitations of text-only retrieval by incorporating layout-aware indexing and exceeds the boundaries of unimodal VQA by enabling open-world knowledge access through retrieval. This domain demands architectures capable of unified structure learning, where models can parse, retrieve, and reason over the intricate interdependencies between text, layout, and visual semantics to deliver faithful, grounded, and logically consistent responses [10]. As the field advances, the definition of MM-RAG will continue to expand, encompassing increasingly complex reasoning topologies and interactive agents capable of dynamic exploration within massive, visually rich corpora. This foundational understanding underscores the critical need to evolve from rigid, text-centric paradigms to the flexible, multi-granular retrieval frameworks detailed in the subsequent section.

1.2 The Evolution from Text-Centric to Multimodal Retrieval Paradigms

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Building upon the defined scope of Multimodal RAG in visually rich domains, the field has undergone a critical transformation, shifting from rigid text-centric paradigms to flexible, multimodal

retrieval frameworks. Traditional information retrieval has long been dominated by approaches that match textual queries against corpora of textual documents, relying heavily on lexical signals or dense vector representations that compress entire passages into single, flat embeddings. While effective for homogeneous natural language tasks, these classical methods encounter severe limitations when applied to the non-linear, spatially complex nature of charts and document layouts. The fundamental flaw lies in the assumption of linearity; traditional dense retrievers typically fragment documents into short passages, inevitably breaking the holistic semantic structure required to understand visual relationships [11]. In visually rich corpora, where meaning is derived not merely from word sequences but from spatial arrangement, color coding, and geometric dependencies, such flat text embeddings fail to capture the intricate interplay between layout and content. This inadequacy creates a significant retrieval gap, necessitating the paradigm shift toward multi-granular and multi-view strategies that can navigate the structural complexities outlined in the previous section.

To overcome the constraints of single-vector representations, the evolution of retrieval has moved toward multi-granular frameworks capable of operating at varying levels of detail, ranging from the document level down to specific regions or atomic propositions. Research underscores that the choice of retrieval unit is pivotal for downstream performance, with finer granularity often yielding superior results for complex reasoning tasks. The concept of “proposition-based retrieval,” for instance, advocates indexing atomic expressions rather than whole passages, thereby enabling more condensed and question-relevant retrieval while reducing noise [12]. This is particularly vital for chart and document understanding, where an answer may hinge on a single data point within a bar chart or a specific cell in a table. Consequently, modern systems increasingly adopt hierarchical indexing strategies that integrate macroscopic document-level semantics with microscopic region-level features. Frameworks like Dense Hierarchical Retrieval (DHR) exemplify this by employing a two-stage process: a document-level retriever first identifies relevant candidates, followed by a passage or region-level retriever that calibrates rankings based on fine-grained relevance, effectively mitigating the context fragmentation inherent in flat models [11].

Parallel to the shift in granularity is the transition from single-representation models to multiple-representation architectures designed to preserve local details while maintaining global coherence. Traditional dense retrieval often encodes an entire passage into a single vector (e.g., via the [13] token), which can obscure critical local interactions essential for visual analysis. In contrast, multi-vector models assign distinct embeddings to individual tokens or visual patches, enabling late-interaction mechanisms that capture fine-grained alignments between query terms and specific document regions. Studies confirm that these multiple representations are statistically more effective than single representations, particularly for queries demanding definitional precision or involving complex information needs [14]. In the context of document images, this evolution enables models to simultaneously encode page-level, region-level, and word-level information. Frameworks such as MGDoc illustrate this capability by utilizing unified text-visual encoders that project multi-granular features into a shared hyperspace, leveraging cross-granular attention to reinforce hierarchical relationships between regions and words, thus overcoming the limitations of models restricted to word-level or region-level features [15].

The complexity of non-linear document structures further necessitates the adoption of multi-view retrieval strategies. A single document often contains multiple potential answers interpretable from diverse semantic perspectives, a reality that single-vector representations fail to accommodate due to semantic mismatch issues. To address this, emerging paradigms propose multi-view document representation learning, generating multiple embeddings for a single document to align with varied

query intents. This approach prevents the collapse of distinct semantic views into a single averaged vector, ensuring that queries targeting specific aspects of a complex chart or layout retrieve the appropriate visual evidence [16]. Moreover, the integration of structural information has become paramount. Unlike plain text, documents and charts possess inherent spatial hierarchies and graph-like structures. Contemporary retrieval mechanisms are increasingly incorporating graph-based structural representations to model inter-cell dependencies and nested headers, allowing for robust retrieval across non-linear visual structures that flat embeddings cannot decipher.

Despite these architectural advances, efficiency and scalability remain central challenges driving the development of hybrid and adaptive retrieval systems. While dense retrieval offers superior semantic matching, it often incurs high computational costs and memory overhead, especially when utilizing multi-vector models. To balance effectiveness with efficiency, researchers are exploring hybrid approaches that combine sparse lexical matching with dense semantic search, or employ hierarchical clustering to reduce the search space. For instance, hybrid inverted indexes leverage both embedding clusters and salient terms to accelerate dense retrieval without sacrificing quality, ensuring relevant documents are not missed due to clustering loss [17]. Furthermore, adaptive strategies that dynamically select between sparse, dense, or hybrid retrieval based on query characteristics are becoming essential for handling the diverse nature of multimodal queries in real-world applications [18]. As the field progresses, the convergence of hierarchical indexing, multi-granular dense retrieval, and multi-view representation learning marks a definitive departure from text-centric limitations. However, while these theoretical frameworks provide a robust foundation for accurate reasoning, their practical deployment faces severe implementation barriers related to layout heterogeneity and OCR dependency, challenges that are critically examined in the following section.

1.3 Core Challenges in Layout Complexity and Fine-Grained OCR Dependency

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While the evolution toward multi-granular and multi-view retrieval paradigms outlined in the previous section provides a robust theoretical framework for handling non-linear data, its practical realization in chart and document understanding faces severe implementation barriers. Specifically, the deployment of Multimodal Retrieval-Augmented Generation (MM-RAG) systems is fundamentally constrained by two interlocking pillars of difficulty: the extreme heterogeneity of document layouts and the fragile dependency on high-fidelity Optical Character Recognition (OCR). Unlike standard text-only retrieval scenarios where data is inherently structured and linear, chart and document understanding requires models to navigate non-Euclidean spatial arrangements where semantic meaning is encoded not just in textual content, but in the precise geometric positioning of elements. This layout complexity is exacerbated by the vast variability found in real-world documents, ranging from historical manuscripts to modern financial reports. For instance, scientific articles, newspapers, and legal contracts possess distinct spatial configurations that models trained on narrow distributions often fail to generalize to. Research has demonstrated that layout-infused language models, while effective on in-distribution data, suffer significant performance degradation—up to a 20 F1 drop—when encountering layout distribution shifts, such as new combinations of text sizes or unfamiliar spatial configurations [19]. This lack of robustness highlights a critical gap: current MM-RAG systems struggle to disentangle semantic content from structural noise when the layout deviates from training norms, thereby undermining the hierarchical indexing strategies discussed previously.

The challenge is further compounded in domains characterized by dense, uneven, or highly de-

formable layouts, such as handwritten manuscripts or multi-column newspapers. In these scenarios, standard segmentation approaches often fail to correctly identify reading orders or separate overlapping content blocks, breaking the logical continuity required for effective retrieval. The difficulty of parsing such complex structures is evident in the digitization of newspapers, where articles span multiple columns and are frequently interrupted by images, necessitating sophisticated layout segmentation as a precursor to any meaningful text recognition [20]. Similarly, historical handwritten documents present unique challenges due to dense and uneven layouts prone to complex deformations, requiring specialized deep networks capable of deformation-aware instance segmentation to avoid catastrophic parsing errors [21]. When an MM-RAG system fails to accurately parse these layouts, the retrieval mechanism may attend to irrelevant regions or miss crucial contextual cues, leading to hallucinated or incomplete generations that violate the fine-grained alignment goals essential for precise reasoning.

Parallel to layout complexity is the critical reliance on accurate OCR, which serves as the primary bridge between visual perception and linguistic reasoning in most current pipelines. The accuracy of OCR is paramount, as even minor recognition errors can propagate exponentially through downstream tasks, corrupting the retrieval index and misleading the generative model. Despite advancements, OCR systems remain imperfect, particularly when dealing with degraded inputs, low-resolution scans, or domain-specific typography. For example, historical printings often feature variant typography and complex layouts that challenge even state-of-the-art engines, necessitating comprehensive workflows that integrate preprocessing, layout recognition, and continuous model training to achieve acceptable character error rates [22]. Furthermore, the quality of OCR is heavily influenced by input image fidelity; documents captured via mobile devices often suffer from non-uniform lighting, blur, and compression artifacts, creating a “long tail” of difficult cases where word error rates can skyrocket to over 70% [23]. These imperfections directly threaten the integrity of the “atomic propositions” and region-level features that modern retrieval frameworks seek to exploit.

The propagation of OCR errors into MM-RAG systems creates a “garbage-in, garbage-out” scenario that is particularly detrimental to tasks requiring fine-grained reasoning, such as key information extraction or numerical analysis in charts. When OCR misrecognizes critical entities or numerical values, the retrieval component may fail to locate relevant documents, or the generator may fabricate facts based on corrupted text. Studies have shown that OCR errors significantly impact downstream Natural Language Processing tasks like Named Entity Recognition, causing substantial accuracy drops that can only be mitigated through specialized text restoration models trained on synthetic degraded data [24]. Moreover, the presence of non-textual elements or visual noise in documents can lead to erroneous text detections, further degrading the quality of the extracted content [25]. In business contexts, where documents may contain sensitive information and cannot be processed by cloud-based services, the need for lightweight yet robust local OCR models becomes acute, as standard solutions often yield poor results on visually damaged inputs [26].

The interplay between layout and OCR errors creates a compounding effect that severely hampers the transition to truly multimodal reasoning. A failure in layout analysis can lead to incorrect text line grouping, which in turn confuses the OCR engine, resulting in garbled output that loses its semantic coherence. This is particularly problematic in documents with complex reading orders or mixed content types, where the spatial relationship between text blocks defines the logical flow. Approaches that attempt to harmonize heterogeneous aspects of documents, such as syntactic, semantic, density, and visual information through graph convolutional networks, underscore the necessity of jointly modeling these factors to achieve robust layout analysis [27]. Without such

holistic understanding, MM-RAG systems remain vulnerable to the cascading failures initiated by minor perceptual errors, preventing them from fully leveraging the multi-view and multi-granular advantages described earlier. Consequently, addressing the dual challenges of layout complexity and OCR dependency is not merely an optimization problem but a fundamental prerequisite for building trustworthy and reliable multimodal reasoning systems. These perceptual bottlenecks ultimately manifest as the profound semantic disconnects and alignment barriers explored in the following section, where the failure to accurately ground visual evidence leads to a breakdown in cross-modal coherence. Future advancements must therefore focus on developing end-to-end architectures that can jointly reason over visual layout and textual content while exhibiting robustness to the noisy, imperfect inputs characteristic of real-world document corpora.

1.4 The Modality Gap and Semantic Alignment Barriers

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While the previous section identified layout complexity and OCR dependency as critical perceptual bottlenecks, these low-level errors ultimately manifest as a more profound structural challenge: the modality gap. This gap represents the fundamental semantic disconnect between visual representations—characterized by spatial continuity, color gradients, and geometric layouts—and textual queries, which rely on discrete, syntactic, and logical structures. In the context of complex charts and dense documents, this misalignment is not merely a difference in data format but a cognitive divergence that prevents models from effectively grounding generated answers in specific visual evidence. The challenge is exacerbated by the inherent diversity of visual features, which lack the grammatical rules and structured logic of natural language, making direct mapping between the two modalities exceptionally difficult without sophisticated intervention [28]. Consequently, even when perceptual inputs are accurate, the failure to bridge this semantic divide leads to the reasoning failures that plague current systems.

The core of this alignment barrier stems from the heterogeneity of feature spaces utilized by standard encoders. Traditional approaches often employ pre-trained backbones for vision and language separately, resulting in latent representations that occupy distinct manifolds. When these disparate features are fused, the resulting representations often retain domain-specific biases rather than achieving true semantic unification. For instance, models may struggle to distinguish between similar visual objects or chart elements when the textual query relies on nuanced relational descriptions, as the direct matching of language and visual modalities has limited capacity to comprehend complex referential relationships [29]. This limitation is particularly acute in document understanding, where the spatial arrangement of text blocks, tables, and charts carries semantic weight that pure pixel-based or token-based embeddings fail to capture jointly. As a result, systems may successfully retrieve relevant document sections based on keyword matching yet fail to interpret the visual context correctly, setting the stage for the hallucinations where generated text contradicts visual data—a phenomenon detailed in the subsequent section.

Furthermore, the insufficiency of cross-modal alignment is often rooted in the reliance on coarse-grained matching strategies that ignore fine-grained inter-modality relationships. Many existing models assume that modality alignment emerges automatically through standard cross-entropy losses during training; however, empirical evidence suggests that such implicit learning is inadequate for tasks requiring precise reasoning [30]. Without explicit supervision or specialized architectural components, models tend to exploit statistical shortcuts and spurious correlations rather than performing genuine visual-textual reasoning. This results in systems that can predict correct

answers based on dataset biases but cannot explain the rationale or point to the specific visual region that supports the conclusion. The failure to establish robust word-object or phrase-region alignments means that when a user asks a question about a specific trend in a line chart or a value in a table cell, the model cannot reliably ground its response to that specific visual coordinate, thereby undermining the trustworthiness of the retrieval-augmented generation process.

To mitigate these barriers, recent research has highlighted the necessity of moving beyond simple feature concatenation toward structured and hierarchical alignment mechanisms. Approaches that leverage graph representations to model the internal relations within both visual and textual modalities have shown promise in capturing deep connections that linear embeddings miss [31]. By converting entities into sequential nodes and utilizing adjacency graphs, these methods attempt to bridge the structural gap, enabling the model to reason over relations rather than isolated features. Similarly, multi-granularity alignment strategies propose splitting the alignment process into different levels—ranging from global scene understanding to local object attributes—to better learn correlations without requiring excessive external annotations [28]. These methods acknowledge that a single-level alignment is insufficient for the complexity of real-world documents, where a query might require synthesizing information from a chart title, axis labels, and specific data points simultaneously.

Despite these advancements, the modality gap remains a persistent challenge, particularly when dealing with out-of-distribution data or noisy inputs common in real-world document corpora. The domain gap between single-modal pre-training objectives and the multi-modal reasoning required for downstream tasks often leads to inconsistent feature extraction, where the visual backbone produces features that are agnostic to the specific query intent [32]. This inconsistency necessitates dynamic refinement mechanisms, such as query-modulated attention, to adjust visual features based on linguistic context dynamically. Moreover, the lack of causal reasoning in many alignment frameworks means that models often fail to disentangle genuine causal structures from spurious correlations, further degrading their ability to perform robust reasoning across modalities [33].

Ultimately, overcoming the modality gap and semantic alignment barriers requires a paradigm shift from treating vision and language as separate streams to be fused, toward viewing them as interconnected components of a unified reasoning process. This involves not only improving the fidelity of cross-modal embeddings but also ensuring that the alignment is interpretable and grounded in explicit visual evidence. Techniques that introduce weak supervision for word-object alignment or utilize contrastive learning to enforce intra-modal and inter-modal consistency are critical steps forward [34][30]. However, as the complexity of charts and documents increases, the demand for more sophisticated, structure-aware, and causally robust alignment mechanisms becomes paramount. Without resolving these foundational alignment issues, the errors propagated from the retrieval stage inevitably lead to the severe hallucinations and faithfulness breakdowns explored in the following section, where the inability to robustly align fine-grained visual features with textual reasoning processes creates a persistent gap between the generated narrative and the visual evidence.

1.5 The Pervasiveness of Hallucinations in Chart and Document Reasoning

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While the modality gap and semantic alignment barriers discussed in the previous section establish the structural difficulties in unifying visual and textual representations, these foundational misalignments directly precipitate a critical downstream consequence: the pervasiveness of hallucinations in

Multimodal Retrieval-Augmented Generation (MM-RAG). In visually rich domains such as financial analysis, scientific research, and legal documentation, these hallucinations manifest not merely as generic factual errors but as specific, high-stakes misinterpretations of visual data and structural layouts. These errors range from the fabrication of non-existent data points in time-series charts to the citation of document sections that do not exist or contain contradictory information. The root of this instability often lies in the complex interplay between a model’s internal parametric knowledge and the retrieved visual context, creating a “tug-of-war” where the model may prioritize its pre-trained priors over the evidence presented in the retrieved document [35]. This tension is particularly acute in domains requiring high precision, where off-the-shelf large language models (LLMs) have been empirically shown to exhibit serious hallucination behaviors, such as inventing stock prices or misinterpreting regulatory concepts [36].

In the specific context of chart reasoning, these failures frequently take the form of visual hallucinations, where the model imagines incorrect details about the image, such as attributing a rising trend to a declining line or fabricating numeric values that are not present in the legend or axes [37]. This issue is exacerbated by the model’s excessive reliance on language priors; as the generation process proceeds, the influence of the visual prompt diminishes, leading the model to generate plausible-sounding but visually ungrounded answers [38]. For instance, in neural chart summarization, hallucinations often arise because training datasets contain target summaries with additional information not strictly grounded in the input chart, effectively teaching the model to “fill in the gaps” with fabricated data [39]. Furthermore, recent studies indicate that spurious images or highly relevant but inconsistent visual inputs can trigger an “instinctive bias” in Multi-modal Large Language Models (MLLMs), causing them to hallucinate answers that align with the textual query but contradict the visual evidence [40]. The granularity of these errors is significant; models may correctly identify the existence of a chart element but fail to accurately describe its attributes or relations, a failure mode categorized as attribute or relation hallucination [41].

Similarly, in document understanding, the manifestations of hallucination are equally critical, often involving the generation of non-factual references or the misattribution of claims to specific sections of a report. A notable form of this is the hallucination of references, where models cite non-existent books or articles, yet paradoxically, they often possess sufficient internal consistency checks to “know” when they are fabricating these citations if queried directly [42]. In Retrieval-Augmented Generation (RAG) settings, this problem is compounded by knowledge conflicts. When the retrieved document contradicts the model’s parametric memory, the model may fail to update its knowledge, instead recapitulating the error from its internal prior or ignoring the correct retrieved information [43]. This failure to ground responses in retrieved evidence is a primary driver of hallucination in domain-specific tasks, such as scientific document reasoning, where models have been observed to justify predictions with fabricated evidence even when provided with relevant scientific corpora [44]. The situation is further complicated in legal and medical domains, where the stakes of hallucinating a precedent or a diagnosis are severe, prompting the creation of specialized benchmarks like Med-HALT to test reasoning and memory-based hallucinations [45].

To understand why these errors persist despite retrieval mechanisms, it is necessary to examine the internal model dynamics driving them. Research suggests that non-factual hallucinations stem from insufficient subject attribute knowledge in lower layers and a failure to select correct object attributes in upper layers [46]. Additionally, the internal states of LLMs retain signals that can predict hallucinations; for example, correct generations tend to exhibit sharper context activations in hidden states compared to incorrect ones, suggesting that the “sharpness” of internal representations correlates with faithfulness [47]. In the embedding space, hallucinated text often displays

a low-rank spectrum that vanishes quickly across sentences, unlike ground-truth text, indicating a fundamental divergence in how factual and fabricated information is represented [48].

Addressing these challenges requires moving beyond simple retrieval augmentation, as the very mechanism designed to mitigate hallucinations can introduce new vulnerabilities. While RAG is intended to ground responses in external knowledge, it introduces the risk of “external hallucinations” if the retrieved information is irrelevant or if the model cannot effectively integrate it with its parametric knowledge [49]. The reliability of RAG systems is thus contingent on the model’s ability to discern when to rely on retrieved context versus internal memory, a balance that is often disrupted in complex multimodal scenarios [50]. Consequently, the pervasiveness of hallucinations in chart and document reasoning underscores a fundamental limitation in current MM-RAG architectures: the inability to robustly align fine-grained visual features with textual reasoning processes—a direct extension of the alignment barriers previously discussed—leading to a persistent gap between the generated narrative and the visual evidence. As the field advances, understanding these specific manifestations, from numeric discrepancies in charts to entity-level errors in documents, is crucial for developing targeted mitigation strategies that ensure the trustworthiness of automated reasoning systems in high-stakes environments.

2 Foundations of Multimodal Representation and Alignment

2.1 Unified Latent Spaces and Modality-Agnostic Encoders

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Before addressing the explicit spatial hierarchies and layout cues detailed in the subsequent section, the foundational challenge for Multimodal Retrieval-Augmented Generation (MM-RAG) in chart and document understanding is to project heterogeneous data modalities—such as textual queries, visual chart elements, and complex document layouts—into a coherent, shared latent space. Without such unification, retrieval mechanisms struggle to bridge the semantic disconnect between a user’s natural language question and the visual evidence embedded within a report or graph. The evolution of representation learning has bifurcated into two primary architectural paradigms designed to minimize this “modality gap,” where representations of different data types remain geometrically separated: two-stream contrastive frameworks that align separate encoders, and single-tower unified transformers that process multimodal inputs jointly.

Two-stream architectures, exemplified by the widely adopted CLIP model, typically employ dual encoders—one for vision and one for text—that are trained via contrastive objectives to maximize the similarity of matching pairs while pushing non-matching pairs apart. While effective for coarse-grained retrieval, these models often suffer from a persistent modality gap, where image and text embeddings occupy distinct manifolds within the shared space, limiting the fine-grained alignment necessary for detailed document reasoning. To address this, recent advancements have introduced refined contrastive strategies. For instance, the [51] framework hypothesizes that standard joint embeddings fail to preserve inter-class dynamics while retaining intra-class relationships. By drawing inspiration from Contrastive Predictive Coding and Noise Contrastive Estimation, [51] explicitly optimizes for both inter- and intra-class relationships, demonstrating a significant reduction in the modality gap and generating a more robust, task-agnostic embedding space suitable for diverse downstream applications. Furthermore, theoretical analyses suggest that this gap is not merely an optimization artifact but is influenced by model initialization and the temperature parameters of the contrastive loss, as detailed in [52]. Addressing these geometric constraints is vital for MM-

RAG systems, as a large modality gap can lead to retrieval failures when querying specific data points within a dense chart.

In contrast to the dual-encoder paradigm, single-tower or unified transformer architectures aim to eliminate the modality gap at the architectural level by processing all inputs through a shared network, thereby laying the groundwork for the sophisticated spatial reasoning discussed later. The [53] model epitomizes this approach by utilizing a single transformer network capable of handling unimodal and multimodal inputs interchangeably. By enforcing multi-task learning—including image-text contrastive loss, matching loss, and masked language modeling—within a unified architecture, [53] reduces task conflict and achieves state-of-the-art performance in visual question answering and captioning. This architecture is particularly relevant for document understanding, where the boundary between text and layout is often blurred, requiring a model that does not treat them as entirely separate streams until the final fusion stage. Similarly, [54] leverages large-scale pre-training with conditional masking and Optimal Transport-based Word-Region Alignment to enforce fine-grained alignment between words and image regions. The success of [54] across nine datasets underscores the importance of explicit fine-grained alignment objectives in unified models, which are essential for grounding answers in specific visual regions of a chart.

Pushing the boundary of modality agnosticism further, the [55] framework proposes a bold hypothesis: that an image and its caption can be treated as two views of the same underlying mutual information, encoded by a completely modality-agnostic single-tower transformer. [55] demonstrates that such a unified space enables unique capabilities like zero-shot object localization and text-guided visual reasoning without the need for modality-specific branches. This suggests that for MM-RAG applications, a truly unified encoder could simplify the retrieval pipeline by allowing direct semantic matching between query tokens and visual patches without intermediate projection layers. Complementing this, [56] investigates the spectrum of parameter sharing in transformers, finding that a mostly unified encoder outperforms separated variants. Their findings indicate that sharing parameters forces semantic concepts from different modalities to be encoded more closely, facilitating the transfer of attention patterns from language to vision, a mechanism crucial for interpreting the structured logic of financial tables or scientific diagrams.

The drive toward unlimited modality integration is also evident in models like [57], which employs modality adapters and shared self-attention layers to align vision, audio, and language. While primarily focused on broader modalities, the architectural principles of [57]—specifically the use of shared layers for fusion and modality-specific adapters for input processing—offer a scalable blueprint for incorporating diverse document elements like thermal maps or 3D structural diagrams into standard RAG pipelines. Additionally, the [58] approach challenges the image-centric alignment of previous models by using Large Language Models to create a modality-agnostic embedding center. By aligning all modalities to this LLM-augmented center, [58] achieves a balanced representation space that avoids the bias toward dominant modalities, ensuring that textual semantics in documents are not overshadowed by visual features during retrieval.

Ultimately, the choice between two-stream and single-tower architectures involves a trade-off between retrieval efficiency and alignment depth. While two-stream models like those enhanced by [51] offer computational advantages for indexing massive document corpora, single-tower models such as [53] and [55] provide superior fine-grained alignment necessary for complex reasoning tasks. Establishing this unified latent space is a prerequisite for the next layer of abstraction: encoding the specific geometric and structural relationships inherent in documents. Consequently, having secured a robust semantic foundation, MM-RAG systems must now integrate layout-aware representations to fully capture the spatial dependencies that define document logic, a topic explored in

the following subsection.

2.2 Layout-Aware Representations and Spatial Reasoning

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Having established a unified latent space that bridges the modality gap between text and vision in Section 2.1, the next critical step for Multimodal Retrieval-Augmented Generation (MM-RAG) is to infuse this semantic foundation with explicit spatial intelligence. In the domain of Visually-Rich Document Understanding (VRDU) and chart analysis, the structural geometry of a page is not merely a container for content but a primary carrier of semantic meaning. Unlike natural images where spatial relationships are often implicit, documents and charts rely on explicit layout cues—such as proximity, alignment, containment, and reading order—to define the logical connections between text blocks, tables, and graphical elements. Consequently, effective MM-RAG systems must evolve from modality-agnostic embeddings to sophisticated layout-aware representations that encode these spatial dependencies. This subsection details the progression from coordinate-infused transformers to fine-grained patch-level alignment strategies, providing the geometric context necessary for the higher-level semantic parsing discussed in Section 2.3.

A foundational challenge in layout-aware modeling is the integration of spatial coordinates into the transformer architecture without disrupting the pre-trained language semantics secured in the previous stage. Early approaches demonstrated that augmenting input sequences with token bounding box coordinates could significantly enhance information extraction capabilities. For instance, the introduction of layout-aware language models that modify the Transformer encoder to accept coordinate features allows the system to understand non-trivial layout influences on local semantics without re-learning language from scratch [59]. Building upon this, more advanced architectures have sought to optimize how positional information is encoded. The [60] framework addresses a critical limitation in existing approaches: the reliance on noisy reading orders generated by standard Optical Character Recognition (OCR) tools. By proposing an Augmented XY Cut algorithm to derive proper reading orders and introducing a Dilated Conditional Position Encoding module, [60] effectively captures rich layout information from both textual and visual modalities. This approach ensures that the model respects the two-dimensional nature of documents, handling variable input lengths while extracting local layout features that are essential for accurate reasoning.

Beyond static coordinate injection, recent research has emphasized the importance of learning robust reading orders and spatial relationships through specialized pre-training tasks, thereby deepening the spatial awareness initiated by unified encoders. The [3] model exemplifies this trend by integrating layout knowledge enhancement throughout the entire workflow. It employs a spatial-aware disentangled attention mechanism and introduces specific pre-training tasks, such as reading order prediction and replaced regions prediction, to systematically mine layout-centered knowledge. Similarly, the [61] architecture highlights the necessity of layout enrichment for scene text visual question answering. By utilizing a single-objective pre-training scheme that relies solely on text and spatial cues, [61] demonstrates that scanned documents can serve as effective training grounds for learning spatial relations (e.g., “left-of,” “below”), thereby improving robustness against OCR errors and eliminating the need for external object detectors.

While global layout understanding is crucial, many document understanding tasks require fine-grained alignment at the patch or textline level to resolve local ambiguities that coarse coordinates cannot address. The [62] model addresses this by focusing on the document textline as an intrinsic granularity. It introduces textline-region contrastive learning to achieve precise alignment between

visual regions and the texts within document textlines. Furthermore, [62] employs masked region modeling and textline-grid matching to enhance visual and layout representations, empowering the model with superior localization abilities. Complementing this, the [63] approach proposes AETNet, which adapts pre-trained document image models to downstream tasks through alignment-enriched tuning. AETNet incorporates extra visual and text transformers to enforce alignment at three distinct levels: document-level, global-local, and local-level. This multi-granular strategy allows the model to capture localized structural information and patch-level details, consistently outperforming state-of-the-art baselines on various downstream tasks.

The complexity of document structures also necessitates mechanisms that can handle long-range dependencies and hierarchical arrangements efficiently, moving beyond pairwise spatial relations to holistic structural comprehension. The [64] mechanism draws inspiration from human reading strategies, utilizing a novel attention mechanism that attends only to the two-dimensional positions of words. This approach not only reduces computational costs but also encourages the emergence of explicit document structure representations within the model. Moreover, the [65] presents a two-stream architecture that leverages grid-based models for 2D token-level and segment-level semantic understanding, effectively bridging the gap between document pre-trained models and grid-based approaches for layout analysis.

Spatial reasoning also extends to handling geometric deformations and irregular text structures, which are common in real-world document images and pose challenges for rigid coordinate systems. The [66] architecture simultaneously localizes and recognizes irregular text by employing a scale-aware attention mechanism and a perspective RoI transform layer, ensuring that spatial features are preserved for recognition even in the presence of curvature or distortion. Additionally, for tasks requiring the reconstruction of document layouts, generative approaches like [67] utilize recursive neural networks to map structural representations to compact codes, enabling the generation of plausible 2D layouts that respect both local semantic elements and global spacing constraints.

Finally, the robustness of these layout-aware representations to distribution shifts remains a critical consideration for deploying MM-RAG systems in diverse environments. Studies indicate that while layout-infused models excel in familiar domains, their performance can degrade significantly when encountering documents with unfamiliar spatial configurations or layout distributions [19]. This underscores the need for training strategies that increase layout diversity and the development of models capable of generalizing across varying structural patterns. By integrating these diverse techniques—from coordinate-aware encoders and reading order optimization to fine-grained patch alignment and generative layout modeling—modern MM-RAG systems are increasingly capable of navigating the complex spatial hierarchies inherent in charts and documents. Having secured this geometric fidelity, the focus now shifts to abstracting these continuous spatial signals into discrete, logical structures. This transition enables the explicit semantic reasoning and relational modeling explored in the following subsection through scene graphs.

2.3 Scene Graphs and Structured Semantic Parsing

2.3 Scene Graphs and Structured Semantic Parsing

While Section 2.2 established the importance of encoding raw spatial geometry and layout coordinates, effective Multimodal Retrieval-Augmented Generation (MM-RAG) for charts and documents requires a higher level of abstraction: explicit semantic reasoning. Scene graphs serve as the critical bridge between the continuous, coordinate-based representations discussed previously and the discrete, logical structures needed for complex querying. By formalizing visual inputs into graph

structures where nodes represent objects, attributes, and textual entities, and edges denote semantic relationships, scene graphs enable compositional reasoning that flat vector embeddings often obscure. This structured parsing allows systems to move beyond mere proximity detection to understand interactions, logical dependencies, and the specific topology of data visualizations, laying the necessary groundwork for the fine-grained alignment strategies detailed in the subsequent section.

The fundamental utility of scene graphs lies in their ability to decompose complex visual inputs into interpretable semantic units, addressing the “compositional gap” where models fail to bind attributes to objects or infer relationships between distant chart elements. To overcome this, researchers have developed frameworks that leverage scene graphs to enhance vision-language compositionality. For instance, the integration of coarse-to-fine contrastive learning in image-text-graph spaces has proven effective in aligning sentences of varying complexities with corresponding visual regions, significantly improving attribute binding and relation understanding [68]. By parsing visual scenes into graph structures, these models can perform systematic generalization, a capability essential for interpreting novel chart types or document layouts not seen during training. Furthermore, the generation of scene graphs can be optimized through iterative message passing, where contextual cues are leveraged to refine predictions of objects and their relationships jointly, thereby outperforming methods that treat object detection and relation classification as disjoint tasks [69].

In the specific context of document and chart understanding, where Optical Character Recognition (OCR) tokens must be tightly integrated with visual elements, specialized architectures like SceneGATE have emerged. SceneGATE utilizes scene graph-based co-attention networks to reveal semantic relations among objects, OCR tokens, and query words, effectively capturing the intra-modal interplay that simple transformer encoders often miss [70]. This approach is particularly vital for Financial Intelligence and Scientific Literature Synthesis, where the relationship between a data point in a chart and its corresponding label or annotation determines the correctness of the retrieved answer. By constructing a TextVQA-based scene graph, such frameworks discover underlying semantics that guide the retrieval and generation processes, ensuring that answers are grounded in specific visual evidence rather than parametric hallucinations.

However, the effectiveness of scene graphs in MM-RAG pipelines is contingent upon the quality of the graph generation itself, which faces challenges such as data bias and the long-tail distribution of relationship categories. Models trained on biased datasets tend to predict common, generic predicates (e.g., “on,” “near”) rather than informative, specific relations (e.g., “blocking,” “looking at”), leading to a loss of precise information crucial for document analysis. To mitigate this, debiasing strategies such as the Cognition Tree (CogTree) loss have been proposed, which organize relationships into a hierarchical cognitive structure to focus attention on tail relationships in a coarse-to-fine mode [71]. Similarly, Compositional Feature Augmentation strategies enrich the diversity of relation triplet features by decomposing and recombining intrinsic and extrinsic features, thereby enhancing the robustness of scene graph generation against bias [72]. These advancements ensure that the structured representations used for retrieval are both accurate and semantically rich, providing high-quality nodes and edges for the contrastive learning objectives discussed next.

Moreover, the alignment between generated scene graphs and textual queries requires sophisticated embedding techniques that complement the contrastive strategies explored in Section 2.4. Recent innovations employ graph Transformers combined with image Transformers to align scene graphs and images in a shared latent space via contrastive learning, enabling the measurement of similarity beyond simple triplet matching [73]. This is particularly relevant for retrieval mechanisms where the query may describe a complex relational scenario rather than a single object. Additionally, the concept of differentiable scene graphs allows for end-to-end optimization, where the

graph representation is learned directly from downstream task supervision, bypassing the need for discrete, non-differentiable intermediate steps [74]. This facilitates seamless integration into generative pipelines, allowing the model to reason over the graph structure while optimizing for the final generation objective.

The application of scene graphs also extends to handling low-shot and open-vocabulary scenarios, which are common in dynamic document corpora. Strategies involving semantic compositional learning enable the construction of realistic relation triples by fusing components from different images, significantly improving performance in few-shot settings [75]. Furthermore, frameworks like OvSGTR expand scene graph boundaries to fully open-vocabulary settings by learning visual-concept alignment for both nodes and edges, allowing systems to recognize unseen categories in evolving document types [76]. In the context of MM-RAG, this adaptability ensures that the retrieval system can handle diverse and emerging document structures without requiring extensive re-annotation.

Finally, the interpretability offered by scene graphs provides a mechanism for verifying the reasoning path of MM-RAG systems. By visualizing the learned visual features for relationships and explicitly investigating the contribution of spatial and semantic features, models can offer transparent explanations for their generated answers [77]. This transparency is crucial for high-stakes domains such as legal precedent retrieval and clinical decision support, where the provenance of an answer must be traceable to specific relational evidence within the document. As the field progresses, the integration of scene graphs with large multimodal models via prompting strategies, such as Compositional Chain-of-Thought, further demonstrates the potential of leveraging these structured representations to extract compositional knowledge without extensive fine-tuning [78]. Thus, scene graphs remain a foundational component in advancing the fidelity, reasoning capability, and trustworthiness of multimodal retrieval-augmented generation for visually rich domains, serving as the structured semantic backbone that enables the advanced contrastive alignment techniques described in the following section.

2.4 Advanced Contrastive Learning Strategies for Fine-Grained Alignment

2.4 Advanced Contrastive Learning Strategies for Fine-Grained Alignment

Building upon the structured semantic representations established by scene graphs in Section 2.3, the efficacy of Multimodal Retrieval-Augmented Generation (MM-RAG) for chart and document understanding ultimately hinges on the precision with which these discrete logical structures are aligned with continuous visual and textual embeddings. While scene graphs provide the necessary topological backbone for reasoning, standard contrastive learning frameworks often struggle to map these complex, hierarchical relationships into a shared latent space due to the intricate spatial layouts and dense textual information inherent in visually rich documents. Consequently, recent research has shifted from coarse-grained image-text matching toward sophisticated, fine-grained alignment strategies that enforce consistency across intra-modal and inter-modal levels. These advanced objectives fundamentally redefine how visual regions, graph nodes, and textual tokens are associated, moving beyond simple pair-wise matching to hierarchical, dense, and multi-view optimization paradigms that directly address the limitations of flat vector embeddings.

A pivotal advancement in this domain is the integration of intra-modal self-supervision alongside traditional cross-modal alignment to preserve the structural coherence introduced by scene graphs.

Standard approaches that maximize mutual information strictly between matched image-text pairs often neglect the internal consistency required to maintain relational integrity within a single modality. The concept of Triple Contrastive Learning (TCL) addresses this deficiency by introducing an intra-modal contrastive objective that complements cross-modal alignment. This approach ensures that similar inputs within the same modality remain proximal in the embedding space, which is critical for handling noisy pre-training data common in document corpora. Furthermore, TCL maximizes the average mutual information between local regions of an image or text and their global summaries, explicitly leveraging localized and structural information to enhance representation quality [79]. By simultaneously optimizing cross-modal, intra-modal, and local-global consistencies, such frameworks provide a more robust foundation for retrieving specific chart elements or document sections based on nuanced queries, effectively grounding the abstract relations defined in scene graphs.

Beyond the tripartite structure, the granularity of alignment has been pushed to the region and patch level through Dense Contrastive Visual-Linguistic Pretraining (DCVLP), a necessary evolution for handling the fine-grained details often lost in coarse graph representations. Traditional methods relying on region regression or label classification often suffer from noisy labels and sparse semantic annotations, particularly when dealing with crowd-sourced datasets that lack the precision required for technical chart analysis. DCVLP overcomes these hurdles by replacing regression tasks with cross-modality region contrastive learning that operates independently of object annotations. This self-supervised paradigm utilizes data augmentation strategies, such as mask perturbation and adversarial perturbation, to generate high-quality negative samples, enabling the model to learn dense, fine-grained correspondences between visual patches and textual descriptions [80]. For MM-RAG systems, this capability is essential for grounding generated answers in specific data points within a chart or distinct clauses in a legal document, bridging the gap between the semantic nodes of a scene graph and their precise pixel-level manifestations.

Hierarchical alignment strategies further refine this process by addressing the multi-level nature of semantic information in documents, mirroring the layered abstraction of scene graphs. Real-world image-text data often exhibits varying degrees of semantic correlation, where some pairs align at a superficial level while others require high-level abstraction. The SemVLP framework proposes a joint alignment of both low-level and high-level semantics by iteratively employing single-stream pre-training for fine-grained feature alignment and two-stream pre-training for high-level semantic alignment [81]. Similarly, hierarchical semantic alignment strategies expand the views generated from a single instance to include cross-sample and multi-level representations, explicitly pulling semantically similar images or patches together at different network layers [82]. This hierarchical approach is particularly beneficial for documents with nested structures, such as tables within reports or multi-axis charts, allowing the retrieval mechanism to navigate different levels of abstraction effectively and ensuring that the logical dependencies captured in Section 2.3 are preserved during embedding.

To further enhance the precision of these alignments, recent works have explored the geometry of the embedding space and the handling of noisy correspondences, which is crucial when scene graph generation itself introduces uncertainty. The CyCLIP framework demonstrates that standard contrastive objectives can lead to inconsistent representations between image and text spaces. By explicitly symmetrizing the similarity between mismatched image-text pairs (cross-modal consistency) and image-image versus text-text pairs (in-modal consistency), CyCLIP ensures geometric consistency that translates to significant gains in robustness [83]. Moreover, the challenge of annotation noise, ubiquitous in large-scale document datasets, is addressed by frameworks like CRCL,

which utilize active complementary learning and self-refining correspondence correction to mitigate the risks of overfitting to erroneous supervision [84]. These mechanisms are vital for MM-RAG systems that must operate reliably even when retrieved documents contain OCR errors or imperfect layouts, providing a resilient buffer before the text recognition challenges discussed in the subsequent section.

Additionally, the regulation of negative samples and the optimization of mutual information have been identified as key factors in improving alignment quality. Research indicates that improperly minimizing mutual information involving negative samples can be detrimental when noise is present. Strategies that regulate cross-modal similarity to balance the effects of partial false negatives have shown to systematically improve performance across diverse tasks [85]. Complementary to this, the use of hard sample mining, as seen in HELIP, refines pre-trained models by focusing on challenging text-image pairs, thereby sharpening the decision boundaries in the joint vision-language space [86]. Finally, the introduction of multi-grained cross-modal alignment frameworks, such as MGCA, explicitly bridges the granularity gap by learning pixel-level, object-level, and region-level alignments simultaneously, which is crucial for tasks requiring open-vocabulary semantic segmentation in complex document images [87]. Collectively, these advanced contrastive learning strategies form the bedrock of modern multimodal representation learning, enabling MM-RAG systems to achieve the fine-grained precision necessary for accurate chart and document understanding. However, while these alignment strategies optimize the interaction between visual and textual modalities, they have historically relied on external text recognition pipelines that introduce their own bottlenecks. This reliance sets the stage for the paradigm shift explored in Section 2.5, where OCR-free techniques eliminate intermediate transcription errors to create a truly end-to-end reasoning framework.

2.5 OCR-Free and Robust Text Recognition Techniques

2.5 OCR-Free and Robust Text Recognition Techniques

While advanced contrastive learning strategies have significantly improved fine-grained visual-textual alignment, as discussed in the previous section, the efficacy of Multimodal Retrieval-Augmented Generation (MM-RAG) for chart and document understanding remains historically constrained by a reliance on traditional, multi-stage pipelines that decouple text recognition from semantic reasoning. Conventional approaches typically depend on off-the-shelf Optical Character Recognition (OCR) engines to transcribe visual text into digital strings before feeding them into language models. Although effective in controlled environments, this paradigm suffers from critical limitations when applied to complex, visually rich domains. Errors in the initial OCR stage, such as misrecognized characters or lost layout information, propagate irreversibly to downstream reasoning modules, leading to hallucinations and factual inconsistencies that even sophisticated alignment mechanisms cannot fully rectify. Furthermore, standard OCR systems often struggle with non-standard fonts, handwritten annotations, mathematical formulas, and dense table structures common in scientific and financial documents. Consequently, the field is witnessing a decisive paradigm shift toward end-to-end OCR-free approaches. These modern techniques leverage the inherent capabilities of Multimodal Large Language Models (MLLMs) to directly perceive, recognize, and reason over visually situated text without explicit intermediate transcription steps, thereby creating a more robust and unified foundation that complements the fine-grained alignment strategies outlined previously.

The cornerstone of this OCR-free revolution is the integration of shallow text recognition capabilities directly within the architecture of MLLMs. Rather than treating text recognition as a standalone

preprocessing task, OCR-free models treat it as an intrinsic part of the visual comprehension process. A pioneering example of this approach is [88], which proposes a universal framework for visually-situated language understanding without relying on external OCR tools. By leveraging the shallow text recognition abilities inherent in pre-trained MLLMs, [88] demonstrates that fine-tuning only a small fraction of parameters (approximately 1.2%) is sufficient to achieve state-of-the-art performance across diverse domains, including documents, tables, charts, and webpage screenshots. The model employs a unified instruction format that jointly trains the system on text reading and key point generation tasks, allowing it to adapt to high-resolution images through a shape-adaptive cropping module. This design effectively bypasses the error propagation issues of traditional pipelines, enabling the model to reason over visual text with a fidelity that closely mirrors human perception.

Beyond general document understanding, achieving robustness in text recognition requires addressing the specific challenges of feature alignment between visual cues and semantic representations, a concern that parallels the fine-grained alignment goals of contrastive learning. In difficult text scenarios involving rare character shapes, severe deformation, or poor linguistic support, standard attention mechanisms often fail to correctly associate visual features with their corresponding character identities. To mitigate this, advanced techniques have been developed to enhance the perceptual granularity of recognition models. For instance, [89] introduces a novel module known as Multi-Domain Character Distance Perception (MDCDP). This approach explicitly models the distance and spatial relationships between characters across both visual and semantic domains. By generating content-aware position embeddings that perceive character spacing, orientation variants, and semantic affinities, [89] establishes a tightly coupled alignment between visual features and character tokens. This method significantly improves recognition accuracy in challenging conditions where traditional models falter, demonstrating that robust text recognition is not merely about seeing pixels but about understanding the structural and semantic distances that define textual entities.

The push toward OCR-free systems is further supported by the development of specialized architectures designed to handle the unique structural complexities of text-rich images. Traditional sequence-to-sequence models often struggle with the repetitive patterns and complex layouts found in academic papers and technical reports. Addressing these issues, [90] integrates location guiding directly into the transformer architecture during autoregression. By training on massive datasets comprising tens of millions of text-location pairs, [90] effectively handles various formatting elements and drastically reduces repetition errors, a common failure mode in end-to-end OCR. Similarly, [91] proposes an end-to-end information extraction model that differs from prior works by its ability to better extract discriminative character features directly from the image, achieving superior results while maintaining higher computational efficiency than two-step pipelines. These models illustrate that OCR-free approaches are not just theoretical alternatives but practical solutions that offer tangible improvements in speed and accuracy for real-world applications.

Furthermore, the transition to OCR-free paradigms is being accelerated by the realization that MLLMs can benefit from cross-modal transfer, where knowledge gained from visual tasks enhances pure language understanding and vice versa. Models like [92] have shown that training on web-scale multimodal corpora enables zero-shot OCR-free NLP capabilities, allowing the model to process document images directly without gradient updates. This suggests that the boundary between “vision” and “language” is increasingly blurring, with MLLMs developing a unified representational space where text is treated as a visual pattern that carries semantic meaning. Additionally, efforts to create comprehensive instruction-tuning datasets, such as those used in [93], are critical for

strengthening the OCR-free capabilities of these models. By jointly training on language-only, general vision-and-language, and specific document instruction datasets, models like [93] learn to ignore fine-grained OCR features less and instead develop a deep, structural understanding of document layouts.

Despite these advancements, challenges remain in scaling OCR-free techniques to extremely high-resolution images and diverse linguistic contexts. While models like [94] employ content filtering mechanisms to efficiently process high-resolution inputs, and others like [95] address multilingual gaps, the community continues to seek more efficient and universal solutions. The emergence of frameworks that treat OCR as a retrieval problem rather than a sequence generation task, as seen in [96], offers alternative pathways that prioritize sample efficiency and adaptability. Ultimately, the shift toward OCR-free and robust text recognition represents a fundamental rethinking of how machines interact with textual information in the wild. By eliminating the brittle dependency on external OCR engines and fostering direct, end-to-end reasoning over visual text, these techniques lay the groundwork for the next generation of MM-RAG systems. This evolution is crucial not only for overcoming current limitations but also for enabling the scalable, modality-agnostic representation spaces discussed in the following section, which aim to integrate an ever-expanding universe of multi-modal information sources with high fidelity and minimal supervision.

2.6 Scaling to Multi-Modal and Open-World Representation Spaces

2.6 Scaling to Multi-Modal and Open-World Representation Spaces

Building upon the robust, OCR-free foundations established in the previous section—which eliminate brittle dependencies on external text recognition engines—the evolution of Multimodal Retrieval-Augmented Generation (MM-RAG) now faces the challenge of scaling beyond binary image-text alignment. While end-to-end visual-textual reasoning resolves immediate perception errors, the complexity of real-world documents and scientific charts often encompasses a heterogeneous array of auxiliary information streams, including audio narrations, 3D structural models, thermal imaging, and time-series sensor data. To address the limitations of image-centric representation spaces that may lead to unbalanced semantic alignment among such diverse modalities, recent research has pioneered modality-agnostic approaches. These methods leverage Large Language Models (LLMs) as central hubs for unified knowledge integration, creating scalable, open-world representation spaces capable of encompassing this expanding universe of data. A premier example of this paradigm is [58], which constructs a flexible and efficient unified representation space for seven distinct modalities, including images, text, audio, point clouds, thermal data, video, and event data. Unlike prior arts that treat images as the central anchor, [58] introduces a modality-agnostic alignment center empowered by LLMs. This approach constructs a knowledge base of text embeddings and adaptively builds class-wise embedding centers, allowing all modalities to align to a shared semantic space via contrastive learning. This results in a balanced representation that significantly enhances zero-shot recognition performance, a capability crucial for interpreting novel chart types or document layouts without extensive fine-tuning.

Beyond the creation of unified spaces from scratch, there is a critical need for methodologies that can extend existing Multi-modal Contrastive Representations (MCR) to new modalities without the prohibitive cost of collecting massive, high-quality paired datasets. The challenge of data scarcity is particularly acute in specialized domains like medical imaging or industrial diagnostics, where paired multi-modal data is rare. To overcome this, [97] proposes Ex-MCR, a training-efficient and paired-data-free method designed to flexibly learn unified contrastive representation spaces for

more than three modalities. By integrating the knowledge of existing MCR spaces, Ex-MCR aligns multiple pre-trained representations into a common base, effectively preserving original semantic alignments while enabling emergent connections between previously unpaired modalities, such as audio and 3D shapes. This technique demonstrates that scalable open-world understanding can be achieved by leveraging overlapping modalities as bridges, thereby facilitating the inclusion of complex data types like 3D molecular structures or thermal heatmaps into document understanding pipelines without requiring direct paired training data.

Complementing these extension strategies, the concept of connecting disparate contrastive spaces through overlapping modalities offers a robust pathway for scaling. [98] introduces C-MCR, a method that projects existing MCRs into a new space and utilizes data from an overlapping modality to align them. This approach allows the transfer of alignment learned from well-resourced pairs (e.g., image-text and audio-text) to non-overlapping pairs (e.g., image-audio), effectively synthesizing a unified semantic space from disjointed components. Such mechanisms are vital for MM-RAG systems that must reason across documents containing mixed media, ensuring that a query about a “rising temperature trend” can retrieve relevant information from both a textual report and a corresponding thermal chart, even if direct training pairs for such specific combinations were unavailable. Furthermore, the push toward unlimited modality scalability is exemplified by [57], which employs a highly extensible architecture comprising modality adapters and shared self-attention layers. By utilizing modality-agnostic pretraining tasks, [57] enables the seamless integration of new modalities simply by adding adapters, fostering a scalable environment where diverse data sources can be incorporated dynamically.

The pursuit of open-world understanding also extends to handling high-modality scenarios where heterogeneity quantification becomes essential. [99] addresses the challenge of scaling to large sets of diverse modalities by proposing information-theoretic metrics to measure modality and interaction heterogeneity. This allows the model to automatically prioritize the fusion of modalities that provide unique information, scaling effectively up to ten modalities including tables and time-series data, which are intrinsic to complex financial and scientific documents. Additionally, frameworks like [100] demonstrate the feasibility of performing unified learning across twelve modalities with unpaired data by mapping raw inputs into a shared token space processed by a frozen encoder. This parameter-efficient strategy underscores the potential for building general-purpose representation models that can adapt to the evolving landscape of document and chart data without catastrophic forgetting or excessive retraining.

In the context of 3D understanding, which is increasingly relevant for interactive technical manuals and spatial data visualization, methods like [101] and [102] illustrate how scaling data and leveraging large multimodal models can generate holistic language descriptions for 3D shapes automatically. These approaches eliminate the need for manual annotations, thereby scaling 3D representation learning to open-world settings. Similarly, [103] facilitates efficient omni-modal learning by projecting novel modalities into a pre-defined space using a pretrained Vision Transformer, enabling emergent downstream capabilities for rare modalities. Collectively, these advancements signify a transformative shift in MM-RAG foundations. By moving beyond rigid, binary alignment schemes to flexible, modality-agnostic, and knowledge-transfer-driven architectures, these frameworks enable the construction of robust, scalable representation spaces. This scalability is indispensable for next-generation document and chart understanding systems that must operate in open-world environments, dynamically ingesting and reasoning over an ever-expanding universe of multi-modal information sources with high fidelity and minimal supervision.

3 Advanced Retrieval Mechanisms for Structured Visual Corpora

3.1 Hierarchical and Multi-Granular Indexing Strategies

3.1 Hierarchical and Multi-Granular Indexing Strategies

Before advancing to graph-based relational modeling, it is fundamental to establish how visually rich documents are indexed and accessed. The transition from flat, text-centric retrieval paradigms to sophisticated multimodal systems necessitates a rethinking of indexing strategies for charts, tables, and complex layouts. In these domains, the assumption that a document can be treated as a linear sequence of tokens is often invalid; instead, artifacts possess intricate spatial hierarchies where macroscopic document-level semantics govern the interpretation of microscopic region-level features. Consequently, effective Multimodal Retrieval-Augmented Generation (MM-RAG) systems must employ dense hierarchical retrieval frameworks that explicitly model these multi-granular relationships. By integrating macroscopic context with microscopic details, such frameworks mitigate the risk of retrieving semantically disjointed fragments that lack the necessary structural grounding for accurate reasoning, laying the essential groundwork for the more complex relational analyses discussed in the following section.

A primary challenge in indexing structured visual corpora is the “scope hypothesis,” which posits that long documents or complex charts encompass multiple distinct topics and visual contexts. Traditional dense retrievers, which often split documents into short, fixed-length passages, struggle with this heterogeneity. As noted in foundational research on text retrieval, splitting documents into short passages can yield inaccurate and misleading hidden representations because local contexts may be biased or partial without the guiding signal of the broader document structure [11]. This limitation is even more pronounced in visual domains, where a chart’s title, axis labels, and data points form a cohesive semantic unit that cannot be understood in isolation. To address this, hierarchical frameworks propose a two-stage retrieval process: a document-level retriever first identifies relevant documents based on global semantics, followed by a passage or region-level retriever that extracts specific fine-grained evidence [11]. This coarse-to-fine strategy ensures that the retrieved regions are not only locally relevant to the query but also globally consistent with the document’s overarching theme.

To effectively implement such strategies, models must be capable of encoding information across varying levels of granularity simultaneously. The MGDoc framework exemplifies this approach by introducing a multi-modal, multi-granular pre-training architecture designed specifically for document image understanding [15]. Unlike previous methods that focused exclusively on word-level or region-level features, MGDoc utilizes a unified text-visual encoder to project page-level, region-level, and word-level information into a shared hyperspace. By employing a cross-granular attention mechanism, the model reinforces the learning of hierarchical dependencies between regions and words, allowing it to capture the spatial relationships crucial for interpreting tables and charts [15]. This capability is essential for MM-RAG, as it enables the system to retrieve not just a specific data point from a chart, but also the associated column headers and the chart’s caption, thereby preserving the semantic integrity required for generation.

Furthermore, the choice of retrieval unit significantly impacts the efficacy of dense retrieval in complex domains. Recent investigations suggest that moving beyond standard sentence or passage units to more atomic “propositions” can enhance retrieval precision [12]. In the context of charts and tables, a proposition might correspond to a specific trend, a comparison between two bars, or a cell value linked to its row and column headers. By indexing these atomic expressions, systems

can reduce the inclusion of extraneous information and minimize the token overhead during the generation phase [12]. Complementing this, research into phrase retrieval has demonstrated that fine-grained supervision can naturally entail better coarse-level retrieval, suggesting that indexing at the most granular level possible—such as individual visual elements or text spans within a layout—can serve as a robust foundation for retrieving larger structural blocks [104].

The integration of hierarchical structures into the indexing process also extends to the optimization of the retrieval mechanism itself. End-to-end Hierarchical Indexing (EHI) represents a significant advancement by jointly learning document embeddings and the approximate nearest neighbor search (ANNS) structure [105]. Traditional approaches treat embedding learning and index construction as disjoint stages, often resulting in embeddings that are ill-suited for the specific tree structures used in efficient search. EHI overcomes this by learning a dense path embedding that captures the position of a document or region within a hierarchical tree, ensuring that the semantic space is optimized for the retrieval topology [105]. For multimodal corpora, this implies that the index can reflect the logical hierarchy of a document (e.g., Section > Subsection > Figure > Caption), allowing the retrieval system to navigate the corpus with structural awareness rather than relying solely on vector similarity.

Moreover, the application of hierarchical modeling is not limited to spatial structures but extends to semantic taxonomies and topic hierarchies inherent in scientific and technical documents. Approaches like Hierarchical Concept Discovery Models leverage concept pyramids to uncover granular concept information residing in patch-specific regions, enabling interpretable decision-making across multiple levels of abstraction [106]. Similarly, in the realm of video and sequential data, hierarchical sequence embedding has proven effective in mapping multi-granular events to semantic spaces, a technique that is directly transferable to the temporal or logical sequences found in multi-page reports or animated data visualizations [107]. By adopting these multi-granular pre-training and indexing strategies, MM-RAG systems can better handle the non-linear, heterogeneous nature of charts and documents, ensuring that the retrieved context provides a faithful and comprehensive basis for downstream generation tasks. While these strategies successfully organize information across scales, they primarily address hierarchy and granularity; the subsequent section explores how explicitly modeling the non-linear dependencies and relations between these hierarchical nodes via graph structures further enhances understanding.

3.2 Graph-Based Structural Representation and Relation Mining

3.2 Graph-Based Structural Representation and Relation Mining

While the hierarchical indexing strategies discussed in Section 3.1 effectively organize information across scales, they primarily address granularity and containment. To fully harness the potential of Multimodal Retrieval-Augmented Generation (MM-RAG) for visually rich corpora, it is essential to move beyond hierarchical containment and explicitly model the non-linear, complex dependencies between these granular nodes. This necessitates a paradigm shift from flat, sequence-based representations to graph-structured modeling. Traditional retrieval mechanisms, which often rely on linearized text or global image embeddings, inherently fail to capture the intricate inter-cell dependencies, nested headers, and cross-modal relations present in tables, documents with complex layouts, and hierarchical charts. Graph Neural Networks (GNNs) have thus emerged as a critical architectural component, enabling retrieval systems to reason over data topology with unprecedented fidelity. By explicitly modeling documents and tables as graphs—where nodes represent semantic entities (e.g., cells, paragraphs, visual elements) and edges encode spatial, syntactic, or logical

relationships—these approaches provide the structural grounding required for accurate retrieval, setting the stage for the semantic identifier generation explored in the subsequent section.

A primary application of this graph-based structural representation lies in the domain of tabular data, where semantic meaning is derived not just from cell content but from the complex interplay between rows, columns, and headers. Standard deep learning models often treat tables as simple grids or flattened sequences, ignoring the hierarchical and paratactic relationships that define their structure. To overcome this, architectures like [108] propose unified neural backbones that simultaneously extract spatial information via pooling mechanisms and relational information through Graph Convolutional Networks (GCNs). By constructing graphs based on semantic hierarchies, such models effectively capture the dependencies between cells that are crucial for answering complex queries involving multi-hop reasoning across table structures. Furthermore, the challenge of retrieving complex tables with nested headers and distributed relevant content is addressed by frameworks such as [109], which convert tables into tabular graphs containing cell, row, and column nodes. This multi-granular approach allows the model to capture layout structures and content semantics simultaneously, significantly improving generalization to novel table formats and diverse query intents that plague traditional dataframe-assumption-based systems.

Beyond individual tables, graph-based methods are indispensable for analyzing document layouts where the spatial arrangement of components dictates semantic flow. In visually rich documents, the relationship between text blocks, images, and tables is often non-sequential, requiring a structural understanding that transcends simple coordinate proximity. The [27] framework exemplifies this by constructing graphs that integrate four distinct aspects of document information: syntactic, semantic, density, and visual appearance. By applying GCNs to these heterogeneous graphs, the system can harmonize disparate modalities to classify layout components accurately, a prerequisite for retrieving specific document sections based on structural queries. Similarly, [110] introduces a relation module based on GNNs to model the interactions between layout components after fusing visual and semantic features. This approach addresses the limitations of purely vision-based or NLP-based methods by explicitly encoding the relational topology of the document, thereby enhancing the retrieval of relevant regions in complex, multi-column, or multi-modal documents.

The utility of graph matching and structural similarity further extends to retrieving documents or interface elements based on their layout geometry rather than just textual content. For instance, [111] leverages Graph Matching Networks to predict structural similarity between 2D layouts, such as floorplans or UI designs. By learning a layout metric through neural graph matching within a triplet network setting, this approach compensates for the lack of structure awareness in standard pixel-wise metrics, enabling retrieval systems to find documents with similar organizational structures even when textual content differs. This capability is particularly valuable in enterprise knowledge management where users may search for forms or reports based on their visual schema. In the context of Graphical User Interfaces (GUIs), [112] exploits GNNs to capture semantic-visuo-spatial constraints among interface elements, facilitating the retrieval and completion of design patterns based on structural logic.

Moreover, graph-based approaches are pivotal in modeling long-range dependencies and inter-sentence relations that are often lost in window-limited transformer architectures. [113] demonstrates the efficacy of building labeled edge graphs on a document level to capture both local and non-local dependencies, utilizing multi-instance learning to predict relations between entity pairs separated by significant textual distance. This is complemented by [114], which addresses word ambiguity and dynamic contextual dependency by learning sparse graph structures that jointly exploit local and global contextual information. Such sparse structure learning is essential for inductive

retrieval tasks where the system must generalize to unseen document types without retraining on dense, static co-occurrence graphs.

To maximize the efficacy of these representations, advanced retrieval systems are increasingly focusing on learning optimal graph structures dynamically rather than relying on predefined topologies. [115] proposes a bi-level optimization approach to learn the optimal graph structure by directly learning the propagation matrix, mitigating issues arising from noise or discrepancies in the initial graph topology. Additionally, [116] introduces a novel mechanism for detecting dependencies in datasets, operating on covariance matrices to respect the geometry of symmetric positive definite manifolds, which enhances the robustness of graph structure inference in the presence of complex, non-linear dependencies. For scenarios involving heterogeneous information, [117] presents a scalable architecture that models both node-node and relation-relation interactions, allowing for personalized attention mechanisms that are critical when retrieving information from large-scale heterogeneous graphs containing multiple entity and relation types.

Finally, the integration of these graph structures with generative and retrieval tasks marks a convergence toward more sophisticated modeling frameworks. Frameworks like [118] combine pre-trained language models with graph semantics through a joint encoder and a link prediction approach. This method reconstructs document graphs to prioritize neighborhood restoration, accelerating convergence and improving performance on information extraction and question-answering tasks. Similarly, [119] addresses the heterogeneity of table structures by employing stacked collaborative blocks that alternate between extracting intra-modality context and modeling inter-modality interactions, ensuring that the collaborative patterns of different modalities are modulated according to the specific table case. These advancements collectively underscore that graph-based structural representation is not merely an auxiliary tool but a foundational requirement for robust, relation-aware retrieval. By explicitly capturing the relational depth of visually rich corpora, these graph-based approaches provide the structured semantic foundation necessary for the next evolutionary step: the internalization of retrieval into generative models via semantic identifiers, as discussed in Section 3.3.

3.3 Generative Retrieval with Semantic Identifiers

3.3 Generative Retrieval with Semantic Identifiers

Building upon the explicit structural modeling capabilities of graph-based approaches discussed in Section 3.2, the paradigm of Generative Retrieval (GR) represents a further architectural evolution by internalizing the retrieval process entirely within the model parameters. Moving away from the traditional “index-retrieve” framework where documents are stored in external structures and accessed via similarity matching, GR conceptualizes retrieval as a sequence-to-sequence generation task. In this setting, a model directly autoregressively generates unique identifiers (docIDs) for relevant documents given a query, enabling end-to-end optimization of both indexing and retrieval. However, the efficacy of generative retrieval is intrinsically linked to the design of these document identifiers. Naive integer-based or random identifiers often fail to capture the semantic richness derived from the complex graph structures of visually rich corpora, leading to poor generalization. Consequently, recent research has pivoted toward developing Semantic Identifiers—structured, meaningful, and hierarchical representations that encode document semantics directly into the ID space, thereby bridging the gap between the relational depth captured by graphs and the generative fluency of language models.

A primary strategy in constructing effective semantic identifiers involves leveraging hierarchical

clustering to impose a tree-like structure on the document space, mirroring the multi-granular relationships inherent in visual domains. For instance, [120] introduces SEATER, a framework that learns semantic tree-structured item identifiers through contrastive learning. By employing a balanced k-ary tree structure, SEATER allocates semantic space to individual tokens, ensuring that identifiers at the same level maintain semantic consistency while different levels correspond to varying granularities. This hierarchical allocation is crucial for visually rich domains where a chart might be identified first by its domain (e.g., finance), then by its type (e.g., line chart), and finally by its specific data trends, effectively translating the structural hierarchy modeled in Section 3.2 into a generative token sequence. Similarly, [121] proposes a category-guided hierarchical clustering scheme that integrates both semantic and efficiency information to facilitate docID generation. The hierarchical nature of these identifiers not only aids in organizing massive corpora but also enables coarse-to-fine retrieval strategies, which are essential when navigating the non-linear structures inherent in complex documents.

Beyond hierarchical structures, the semantic content of the identifiers themselves plays a pivotal role in enhancing interpretability and alignment. Abstractive identifiers, which utilize natural language or keyphrases rather than opaque integers, have shown significant promise in bridging the gap between query intent and document representation. The [122] study demonstrates that using abstractive keyphrases generated by large language models as document IDs significantly outperforms traditional integer sequences. By composing IDs from human-readable, content-based phrases, the model can better leverage its pre-trained linguistic knowledge to align queries with relevant documents. This is particularly relevant for chart understanding, where an identifier composed of semantic tags like “quarterly-revenue-growth” is more intuitively retrievable than a numeric hash. Furthermore, [123] presents GenRet, a method that learns to tokenize documents into short discrete representations via a discrete auto-encoding approach. This end-to-end learning process ensures that the resulting docIDs capture the complete semantic information of the document, overcoming the limitations of fixed rule-based identifiers that often miss nuanced visual or textual cues.

To optimize the learning of these semantic identifiers, contrastive learning has emerged as a critical technique, helping to align the latent representations of queries and documents while enforcing discriminative boundaries between different identifier tokens. [124] highlights the limitations of binary relevance in contrastive frameworks and proposes learning from fine-grained rankings, a concept vital for distinguishing between visually similar charts with subtle data differences. Moreover, [125] and [126] illustrate how contrastive objectives can be adapted to create robust cluster structures in unsupervised settings. These methods ensure that the semantic space occupied by the identifiers is smooth and uniform, reducing the risk of “memory confusion” as the corpus scales. The integration of contrastive losses with generation tasks, as seen in [120], allows the model to simultaneously learn the hierarchical relationships between tokens and the semantic dependencies within the identifier sequence.

Despite these advancements, scaling generative retrieval to massive corpora remains a significant challenge, creating a natural transition toward the hybrid solutions discussed in the subsequent section. [127] reveals that while GR is competitive on small corpora, naive scaling often fails due to memory constraints and the difficulty of memorizing fine-grained features for millions of items. To address this, hybrid approaches like [128] propose a coarse-to-fine process where generative models handle inter-cluster matching (generating cluster IDs) and dense retrieval handles intra-cluster matching. This synthesis leverages the deep interaction capabilities of GR for high-level semantic indexing while maintaining the scalability of dense retrieval for fine-grained document access. Additionally, [129] addresses the issue of evolving corpora, proposing mechanisms like Incremental

Product Quantization to update semantic identifiers for new documents without catastrophic forgetting. This is essential for dynamic document environments where new charts and reports are constantly added.

In the context of structured visual corpora, the ability to generate semantic identifiers that encapsulate layout, textual content, and visual semantics is transformative. Methods like [130] further bridge the gap by using residual quantization to generate semantic virtual cluster IDs before performing fine-grained searches, effectively combining the benefits of generative and embedding-based retrieval. By replacing arbitrary indexing with semantically grounded, hierarchically organized identifiers optimized through contrastive learning, generative retrieval offers a promising pathway for achieving high-precision, scalable, and interpretable retrieval. However, as the limitations of pure parametric memory become apparent at scale, the field is increasingly moving toward the cluster-assisted and hybrid architectures detailed in Section 3.4, which seek to balance the semantic depth of generative models with the computational efficiency required for enterprise-scale deployment.

3.4 Hybrid and Cluster-Assisted Retrieval Architectures

3.4 Hybrid and Cluster-Assisted Retrieval Architectures

As highlighted in Section 3.3, while Generative Retrieval (GR) with semantic identifiers offers profound semantic alignment by internalizing the index within model parameters, it faces inherent scalability barriers when applied to massive, visually rich corpora. Naive scaling of purely parametric models often leads to memory confusion, degraded accuracy for fine-grained visual features, and prohibitive costs for updating indices with new documents [128]. To bridge the gap between the deep semantic interaction of generative models and the computational efficiency required for enterprise-scale deployment, the field has pivoted toward Hybrid and Cluster-Assisted Retrieval Architectures. These paradigms synthesize the strengths of coarse-to-fine clustering strategies with hybrid inverted indexing, effectively balancing the semantic depth necessary for complex chart reasoning with the scalability demands of large-scale document retrieval.

A foundational strategy in this domain involves decoupling the retrieval process into hierarchical stages, where cluster-based intermediate matching serves as a critical bridge between global semantics and local details. The Generative Dense Retrieval (GDR) paradigm exemplifies this approach by utilizing limited model memory to perform inter-cluster matching, effectively narrowing the search space from a massive corpus to relevant document clusters [128]. Once the relevant clusters are identified, a memorizing-free dense retrieval mechanism is employed to conduct fine-grained intra-cluster matching. This coarse-to-fine methodology is particularly advantageous for chart and document understanding, where a query may first require identifying a specific report or figure type (cluster level) before pinpointing exact data points or text regions (document/region level). By maximizing the deep interaction capabilities of generative models at the cluster level and leveraging the scalability of dense retrieval for intra-cluster operations, such frameworks mitigate the scalability issues inherent in purely parametric retrieval systems [128]. Furthermore, the Model-enhanced Vector Index (MEVI) reinforces this concept by employing Residual Quantization to generate semantic virtual cluster identifiers in few steps, followed by a fine-grained search within candidate clusters using adapted embedding vectors [130]. This two-stage process significantly reduces inference time compared to decoding unique document identifiers sequentially, making it viable for real-time applications in enterprise knowledge management involving large visual archives.

Complementing hierarchical clustering, hybrid inverted indexes provide a robust mechanism for accelerating dense retrieval while preserving retrieval quality, a necessity when dealing with the

heterogeneous nature of visual documents. Traditional inverted file structures accelerate search by clustering documents based on embeddings and probing nearby clusters; however, this clustering is often lossy, leading to the omission of relevant documents [17]. To overcome this, the Hybrid Inverted Index (HI²) framework introduces a collaborative mechanism where embedding clusters and salient lexical terms work in tandem [17]. In the context of MM-RAG, this is vital because visual documents often contain critical textual anchors (e.g., axis labels in charts, headers in tables) that pure dense embeddings might obscure. HI² employs a cluster selector and a term selector to construct compact inverted lists, ensuring that both semantic similarity and exact keyword matches contribute to the retrieval decision [17]. This dual-selection process ensures that the system does not miss relevant chart sections due to embedding quantization errors, thereby maintaining high recall rates even in massive collections.

The synergy between generative and dense paradigms is further optimized through unified frameworks that treat these seemingly distinct approaches as complementary components of a single system. Research into sponsored search and query rewriting has shown that generative and dense retrieval methods often retrieve unique sets of high-quality results, with roughly 40% of retrieved items being exclusive to one method [131]. The CLOVER-Unity model demonstrates that unifying these methods within a single architecture can outperform ensembles of separate models while halving computational costs [131]. For chart and document understanding, this implies that a hybrid architecture can generate abstractive descriptions of visual layouts while simultaneously performing dense vector matching on encoded visual features, ensuring comprehensive coverage of the query intent. Additionally, the concept of blending semantic search with hybrid query strategies, as seen in Blended RAG, has set new benchmarks by effectively fusing dense vector indexes with sparse encoder indexes to handle scaling challenges [132].

Efficiency in these hybrid architectures is also enhanced through advanced indexing techniques that learn discrete representations via constrained clustering. The RepCONC model, for instance, jointly trains dual-encoders with Product Quantization to learn discrete document representations, supporting end-to-end optimization of both the quantization method and the encoders [133]. This approach allows for highly efficient vector search on CPUs using a vector-based inverted file system, which is crucial for deploying MM-RAG systems in resource-constrained environments where GPU acceleration for every query is impractical. Similarly, the construction of tree-based indexes through joint optimization of the index structure and query encoding, as proposed in JTR, ensures that the index possesses a maximum heap property that supports effective beam search [134]. These tree-structured identifiers, when combined with contrastive learning, enable the system to maintain semantic consistency across different levels of granularity, from entire documents down to specific visual regions [120], thereby extending the hierarchical principles introduced in Section 3.3 into a scalable retrieval infrastructure.

In conclusion, hybrid and cluster-assisted retrieval architectures represent a pivotal evolution in handling structured visual corpora, effectively resolving the tension between the deep semantic understanding required for complex chart reasoning and the efficiency demands of large-scale document retrieval. By integrating cluster-based intermediate matching, as seen in GDR and MEVI, with hybrid inverted indexing strategies like HI², these systems provide a robust foundation that pure generative models alone cannot sustain. However, while these architectures address static scalability, they often presuppose a relatively fixed corpus or rely on batch updates. As real-world applications in financial reporting and scientific literature demand continuous adaptation to evolving document landscapes, the focus must shift toward dynamic indexing and continual learning. The following section, Section 3.5, explores how these hybrid foundations can be extended to sup-

port incremental updates and prevent catastrophic forgetting, ensuring retrieval systems remain resilient in the face of constantly changing multimodal data streams.

3.5 Dynamic Indexing and Continual Learning for Evolving Corpora

3.5 Dynamic Indexing and Continual Learning for Evolving Corpora

While the hybrid and cluster-assisted architectures detailed in the previous subsection effectively balance semantic depth with computational scalability, they often presuppose a relatively static corpus or rely on batch updates. However, the deployment of Multimodal Retrieval-Augmented Generation (MM-RAG) systems in real-world environments—such as financial reporting, scientific literature, and legal compliance—necessitates a fundamental shift from static assumptions to dynamic, evolving document landscapes. In these domains, charts and documents are continuously updated, rendering traditional retrieval indices obsolete shortly after construction. The core challenge, therefore, extends beyond efficient search to enabling retrieval systems to incrementally incorporate new visual and textual content without incurring the prohibitive computational costs of full re-indexing or suffering from catastrophic forgetting, where the model loses proficiency in retrieving previously indexed documents. This subsection reviews the emerging paradigms of dynamic indexing and continual learning specifically tailored for generative retrieval and structured visual corpora, bridging the gap between the architectural efficiencies of Section 3.4 and the layout-aware mechanisms required for future adaptability discussed in Section 3.6.

Generative Retrieval (GR), which encodes document identifiers directly into model parameters, represents a transformative approach by merging indexing and retrieval stages. Yet, early GR models were predominantly designed for static collections, struggling significantly when faced with dynamic corpora where new documents are constantly added. The inability to efficiently update the parametric index without retraining the entire model has been a primary bottleneck. Recent advances have addressed this through continual learning frameworks specifically designed for generative retrieval. For instance, the CLEVER model introduces a novel approach to continual learning for generative retrieval by utilizing Incremental Product Quantization to encode new documents into identifiers with low computational overhead, alongside a memory-augmented learning mechanism that preserves connections between old and new knowledge to prevent forgetting [129]. Similarly, DSI++ tackles the continual indexing challenge for Differentiable Search Indices by optimizing for flatter loss basins to stabilize learning and employing a generative memory to sample pseudo-queries, thereby significantly mitigating the forgetting of previously indexed documents during incremental updates [135].

The scalability of these generative approaches to massive, evolving corpora remains a critical area of investigation, echoing the efficiency concerns raised in hybrid architectures. While generative retrieval shows promise on smaller datasets, scaling to millions of passages reveals limitations in naive parameter scaling and highlights the necessity of synthetic queries for effective document representation during indexing [127]. To bridge the gap between the deep interaction capabilities of generative models and the scalability of dense retrieval in dynamic settings, hybrid architectures like Generative Dense Retrieval (GDR) have been proposed. GDR employs a coarse-to-fine strategy, using limited model memory for inter-cluster matching and a memorizing-free dense retrieval mechanism for fine-grained intra-cluster matching, thus enhancing scalability while maintaining retrieval accuracy [128]. Furthermore, the Model-enhanced Vector Index (MEVI) leverages Residual Quantization to connect sequence-to-sequence deep retrieval with embedding-based models, generating semantic virtual cluster IDs to reduce inference time while supporting efficient document

updates [130]. These approaches demonstrate that the cluster-assisted strategies discussed previously can be adapted to support incremental growth, ensuring that the “coarse” cluster layer can evolve without destabilizing the “fine” retrieval layer.

Beyond generative paradigms, traditional vector indexing methods are also evolving to support dynamic environments, providing a robust backbone for the layout-sensitive features that will be explored in the following section. The concept of “learned indices” has been extended to support precise updates; for example, the LIPP framework introduces a dynamic adjustment strategy for learned tree indices, allowing for efficient insertions and deletions without sacrificing lookup performance, a capability crucial for maintaining up-to-date indices in fast-changing document repositories [136]. In the realm of adaptive retrieval, Graph-based Adaptive Re-ranking (GAR) demonstrates how clustering hypotheses can be utilized to dynamically expand candidate pools during re-ranking, improving recall in scenarios where initial retrieval might miss relevant, newly added documents due to index lag [137]. Such graph-based adaptability is particularly pertinent for visually rich documents, where the addition of new pages or charts may alter the structural context of existing content.

The integration of continual learning techniques from other domains, such as computer vision and natural language processing, offers valuable insights for MM-RAG, especially as systems begin to handle more complex modality-specific data. Strategies like experience replay, knowledge distillation, and architectural expansion are being adapted to handle the unique demands of multimodal data. For instance, methods that utilize generative replay to synthesize pseudo-data for past tasks have shown efficacy in preventing catastrophic forgetting in classification tasks, suggesting potential applications for regenerating visual features of archived charts or documents to maintain retrieval fidelity [138][139]. Additionally, the principle of selecting maximally interfered samples for replay, originally developed for online continual learning, could be adapted to prioritize the re-indexing of document regions most susceptible to representation drift in dynamic corpora [140]. This focus on preserving specific visual regions aligns closely with the region-level matching and spatial reasoning mechanisms that define the next generation of layout-aware retrieval.

Despite these advancements, significant challenges persist. The trade-off between plasticity (the ability to learn new information) and stability (the ability to retain old knowledge) is particularly acute in multimodal settings where visual layouts and OCR dependencies add layers of complexity. Existing surveys on continual learning for Large Language Models highlight that while pre-trained models possess inherent anti-forgetting capabilities, specialized mechanisms are still required for task-specific adaptation in dynamic streams [141][142]. Moreover, the evaluation of these dynamic systems requires new benchmarks that reflect temporal persistence and the ability to handle non-stationary data distributions, as static evaluations fail to capture the longitudinal robustness required in production environments [143][144].

Future directions in this area point towards fully autonomous, self-updating indexing systems capable of handling heterogeneous data streams. The development of frameworks like CorpusBrain++, which extends generative pre-training to continual document learning scenarios, underscores the move towards systems that can adapt to knowledge-intensive tasks in real-time without explicit retraining cycles [145]. As MM-RAG systems increasingly operate in open-world settings, the synthesis of adaptive quantization, generative memory, and graph-based structural updates will be paramount in building resilient retrieval backbones. These dynamic foundations are essential prerequisites for the advanced layout-aware and modality-specific mechanisms discussed in the subsequent section, as a system cannot effectively reason about evolving spatial structures unless its underlying index can fluidly accommodate the continuous influx of new visual and textual elements.

3.6 Layout-Aware and Modality-Specific Retrieval Mechanisms

3.6 Layout-Aware and Modality-Specific Retrieval Mechanisms

Building upon the dynamic indexing and continual learning foundations established in Section 3.5, which ensure retrieval systems can fluidly accommodate evolving corpora, this subsection addresses the critical need for spatial intelligence within those indices. While dynamic mechanisms solve the problem of *when* and *how* to update indices, they do not inherently resolve the challenge of *what* semantic signals to capture in visually rich domains. Traditional retrieval mechanisms have historically relied heavily on textual similarity, treating documents as flat sequences of tokens. However, in domains such as scientific charts, financial reports, and engineering diagrams, the spatial arrangement of content carries semantic weight equal to or greater than the text itself. Consequently, advanced Multimodal Retrieval-Augmented Generation (MM-RAG) systems must evolve beyond keyword matching to incorporate layout-aware and modality-specific mechanisms that directly interpret structural cues, spatial relationships, and visual semantics. This shift enables retrieval based on structural sketches, sub-layout patterns, and complex spatial queries, transforming the static or dynamically updated index into a spatially cognizant knowledge base.

A pivotal advancement in this domain is the capability to perform retrieval using structural sketches or sub-layout queries, allowing users to search for documents based on their visual organization rather than solely their textual content. This approach is particularly valuable when textual information is unknown, ambiguous, or irrelevant to the user’s intent. For instance, researchers have proposed efficient graph-based matching algorithms integrated with hash-based indexing to prune large search spaces based on layout entities [146]. These systems enable users to specify combinations of sub-layouts of interest through sketch-based queries, supporting partial matching for unspecified entities. By handling segmentation pre-processing errors through symmetry maximization strategies, such frameworks demonstrate robustness in retrieving relevant sections from unstructured corpora like newspaper images, where the arrangement of columns, headers, and images defines the document’s logic [146]. Furthermore, the concept of “query by semantic sketch” extends this capability by allowing users to draw spatial distributions of concept labels, effectively merging sketch-based retrieval with semantic label searching [147]. This method accounts for both the semantic concepts and their spatial context, enabling the retrieval of images or scenes that exhibit similar distributions of elements, a technique that is increasingly applicable to complex document layouts where the relative positioning of charts and text blocks is critical [147].

Beyond sketch-based inputs, modality-specific retrieval mechanisms increasingly rely on the fusion of vision, semantics, and relations to achieve high-precision matching. The limitations of pure NLP-based or CV-based methods have led to the development of unified frameworks that adaptively fuse visual features from document images with semantic text embedding maps [110]. In these architectures, a relation module based on graph neural networks is incorporated to model the intricate dependencies between layout components, ensuring that retrieval decisions are informed by the holistic structure of the document rather than isolated features [110]. This vision-semantics-relation fusion is essential for addressing the information incompleteness often found in visually rich documents, where contextualizing retrieved information requires understanding the interplay between ad-hoc visual cues and textual spans [148]. Frameworks like Juno exemplify this by aligning text spans with relational tuples in a multimodal encoding space without prior knowledge of the document schema, thereby augmenting heterogeneous documents with supplementary information for more actionable insights [148].

The granularity of retrieval has also shifted from document-level to region-level and even point-level

matching, driven by the need for fine-grained spatial reasoning—a necessity that complements the region-level replay strategies discussed in the context of continual learning. Hierarchical frameworks now propose searching from “area to point,” establishing initial search spaces based on matched image areas containing prominent semantics before narrowing down to precise point matches [149]. This hierarchical strategy utilizes semantic priors and geometry consistency to establish accurate area matches, significantly alleviating accuracy limitations in transformer-based matching methods [149]. Similarly, region-based approaches reformulate key information extraction as a region prediction problem in 2D space, utilizing layout-aware BERT models with soft layout attention masking to incorporate spatial coordinates directly into the retrieval representation [150]. By generating candidate regions and selecting the most probable ones, these models can seamlessly handle diverse document types, including those with mixed table-text structures that defy linear sequence tagging [150].

Moreover, the integration of geometric pre-training has emerged as a critical factor in enhancing retrieval performance for relation extraction tasks. Models like GeoLayoutLM explicitly model geometric relations during the pre-training phase through specialized geometry-related tasks, bridging the objective gap between pre-training and fine-tuning for visual information extraction [151]. This explicit modeling of geometric relations allows the system to capture the spatial dependencies between entity blocks that are often overlooked by implicit learning methods, leading to significant improvements in retrieving correct relational tuples from complex forms [151]. Complementing this, methods that leverage local indexing and deep feature confidence scores have demonstrated the ability to combine local and global descriptors for fast and robust image-to-video and document search, adapting to geometric deformations such as scale and orientation changes [152].

Finally, the evolution of retrieval mechanisms includes the ability to handle multimodal queries that combine spoken natural language with spatial guidance, such as mouse traces over a canvas, to express both “what” and “where” simultaneously [153]. This form of interaction allows for significantly more accurate retrieval results compared to text-only systems by naturally incorporating localization preferences into the query process [153]. Additionally, the use of synthetic document generators to train layout recognition models without extensive manual annotation further supports the scalability of these layout-aware retrieval systems, ensuring they can adapt to diverse and evolving document structures [154]. Together, these layout-aware and modality-specific mechanisms represent a paradigm shift towards retrieval systems that truly understand the structural and visual grammar of complex documents, providing the necessary spatial grounding for the next generation of autonomous, self-updating MM-RAG agents.

4 Generative Architectures and Reasoning Frameworks

4.1 Fusion Paradigms and Multimodal Alignment Strategies

4.1 Fusion Paradigms and Multimodal Alignment Strategies

Before large language models can engage in the complex, non-linear reasoning topologies described in the following section, they must first establish a robust foundation for integrating heterogeneous data streams. The efficacy of Multimodal Retrieval-Augmented Generation (MM-RAG) systems in chart and document understanding hinges critically on this strategic integration, where the core architectural decision lies in selecting an appropriate fusion paradigm. This choice must balance the preservation of modality-specific nuances with the necessity of creating a unified semantic space capable of supporting intricate relational queries. Historically, the field has oscillated between early

fusion, where modalities are combined at the input or shallow feature levels, and late fusion, which maintains separate processing pipelines until the final decision stage. In the context of visually rich documents, early fusion has demonstrated significant advantages by allowing the model to capture low-level interactions between visual layouts and textual semantics immediately. Research indicates that immediate fusion of inputs, such as audio and visual data in initial network layers, results in networks that are more robust to noise and better capable of handling incomplete information from individual modalities [155]. This principle extends to document analysis, where the “Multimodal Core” hypothesis suggests that early integration provides the fundamental operations necessary for solving complex relational queries that holistic, late-fusion models often miss [156].

However, naive early fusion can suffer from feature heterogeneity and high sample complexity, necessitating the evolution toward progressive or dynamic fusion strategies. Progressive fusion introduces backward connections that make late-stage fused representations available to earlier layers, thereby refining expressiveness while retaining the structural benefits of late fusion designs [157]. Similarly, dynamic fusion mechanisms utilize intra- and inter-modality attention flows to alternately pass information between visual and language domains, robustly capturing high-level interactions essential for accurate visual question answering [158]. These approaches are particularly vital for MM-RAG, where the retrieved context may consist of fragmented chart regions or disjointed document pages that require iterative refinement to establish coherent semantic relationships—a prerequisite for the multi-hop reasoning paths explored later.

Beyond the timing of fusion, the mechanism of alignment is paramount for grounding generated responses in specific visual evidence, a capability that directly informs the quality of intermediate thoughts in reasoning chains. Scene graph representations have emerged as a powerful tool for structuring this alignment. By decomposing visual scenes into objects, attributes, and relationships, scene graphs enable models to perform fine-grained reasoning that simple joint embeddings cannot achieve. For instance, hierarchical graph reasoning allows for the disentanglement of complex scenes into multi-level semantic structures, guiding the learning of diverse video or document representations to capture both global context and local details [159]. In the domain of text-rich images, scene graph-based co-attention networks explicitly model the semantic relations between objects, OCR tokens, and query words, significantly outperforming standard transformer encoders that fail to capture these underlying semantics [70]. Furthermore, self-supervised adaptations of scene graphs have shown promise in circumventing the need for costly annotated data, enhancing the generalization of graph representations in visual reasoning tasks [160].

Cross-modal attention mechanisms serve as the computational engine driving these alignment strategies. Traditional co-attention models often focus solely on inter-modal interactions, neglecting the rich intra-modal context. Unified attention networks address this by simultaneously capturing both intra- and inter-modal interactions, outputting attended representations that seamlessly integrate visual and textual features for tasks like visual grounding [161]. To further refine this alignment, recent works propose structured alignment techniques that convert modality entities into sequential nodes and adjacency graphs, thereby capturing deep connections that entity-level alignments miss [31]. Additionally, relation-level alignment is critical for compositional generalization; enforcing congruence between directed language attention and spatial visual attention ensures that models correctly interpret relational phrases such as “mug in grass” by matching semantic relations with spatial configurations [162].

The challenge of modality mismatch, where fixed architectures struggle with diverse input types, has led to the development of adaptive models like Switch-BERT, which learns to optimize attention from a set of modes representing different intra- and inter-modal interactions [163]. This

adaptability is crucial for MM-RAG systems that must handle a wide variety of chart types and document layouts. Moreover, multi-granularity alignment architectures split the alignment process into different levels to learn better correlations without requiring additional annotations, effectively bridging the gap between the unstructured nature of image features and the grammatical rules of language [28].

Ultimately, establishing a unified semantic space for reasoning in MM-RAG requires a synergistic combination of these paradigms. Whether through the hierarchical decomposition of scene graphs, the dynamic flow of cross-modal attention, or the strategic timing of feature fusion, the goal remains to create a representation where visual chart elements and textual queries are inextricably linked. This alignment not only facilitates accurate retrieval but also grounds the generative component, ensuring that the synthesized answers are faithful to the retrieved visual evidence and logically consistent with the document’s structure. As models evolve, the integration of these sophisticated fusion and alignment strategies provides the essential representational backbone upon which advanced structural thought topologies—such as Trees and Graphs of Thought—can be constructed to navigate the complex dependencies inherent in chart and document understanding.

4.2 Evolution from Linear Chains to Structural Thought Topologies

4.2 Evolution from Linear Chains to Structural Thought Topologies

While Section 4.1 established the foundational fusion paradigms and alignment strategies necessary to create a unified semantic space, these robust representations alone are insufficient for navigating the complex, non-linear reasoning topologies inherent in chart and document understanding. The journey of enhancing reasoning capabilities in foundation models has been predominantly defined by the Chain-of-Thought (CoT) paradigm, which guides models to decompose problems into linear, sequential steps. Although CoT proves effective for general tasks, its inherent linearity imposes significant constraints when addressing the intricate dependencies found in multimodal contexts. Human experts rarely reason in a strictly sequential manner; instead, they engage in high-order multi-hop reasoning and comparative judgments that require revisiting previous conclusions, exploring alternative paths, and synthesizing disparate pieces of information simultaneously. To bridge the gap between the linear outputs of standard models and expert-level cognitive processes, recent research has pivotally shifted towards structural thought topologies, including Tree-of-Thoughts, Graph-of-Thoughts, and Hypergraph-of-Thoughts. These frameworks transcend the rigid “left-to-right” generation of standard CoT, offering flexible architectures that better model the complex spatial and logical hierarchies present in visually rich documents.

The limitations of linear CoT are particularly acute in multi-step reasoning scenarios where later stages depend on results from several steps prior, not just the immediately preceding one. Standard CoT prompts struggle to capture these complex reasoning graphs, often leading to error propagation where an early mistake derails the entire sequence. To address this, the Tree-of-Thoughts (ToT) framework was introduced to enable deliberate decision-making by allowing language models to explore multiple reasoning paths, self-evaluate intermediate choices, and backtrack when necessary. Unlike the singular path of CoT, ToT generalizes prompting to consider a tree of coherent text units, significantly enhancing performance on tasks requiring strategic lookahead and exploration. For instance, in complex planning tasks, ToT has demonstrated a dramatic improvement in success rates compared to CoT, proving that allowing models to look ahead and revise decisions is crucial for solving elaborate problems [164]. Furthermore, recognizing that intermediate thoughts in such trees possess inherent uncertainties, the Tree of Uncertain Thoughts (TouT) framework incorporates

Monte Carlo Dropout to quantify local uncertainty, marrying this with global search algorithms to enhance precision in response generation [165]. This evolution from a single chain to a branching tree structure is essential for document analysis, where a single chart may support multiple conflicting interpretations that must be evaluated before reaching a conclusion.

Pushing beyond the hierarchical constraints of trees, Graph-of-Thoughts (GoT) represents a further abstraction by modeling the information generated by large language models as an arbitrary graph. In this topology, units of information serve as vertices, and edges represent dependencies, enabling the combination of arbitrary thoughts into synergistic outcomes and the distillation of entire networks of reasoning. This approach captures the non-sequential nature of human thinking, allowing for feedback loops and the aggregation of diverse reasoning strands. Research indicates that GoT can significantly outperform both CoT and ToT baselines on reasoning benchmarks by effectively managing the complexity of interdependent sub-tasks [166]. The utility of graph structures is further emphasized in multimodal settings, where the Aggregation-Graph-of-Thought (AGoT) mechanism models each reasoning step as an aggregation graph to cope with multiple aspects of thinking often overlooked in single-step linear reasoning. This has shown promising results in visual question answering and text-image retrieval, domains critical for chart understanding [167]. Additionally, graph-guided reasoning methods have been developed to construct “question/rationale graphs” that diagnose and filter irrelevant rationales, ensuring that multi-hop queries in document retrieval remain focused on relevant information [168].

The most recent and sophisticated evolution in this trajectory is the Hypergraph-of-Thought (HoT), designed specifically to emulate the high-order reasoning patterns of domain experts. While graphs connect pairs of nodes, hypergraphs utilize hyperedges to connect various vertices simultaneously, making them naturally suitable for modeling high-order relationships and multimodal comparative judgments. The HoT paradigm constructs textual hypergraphs using triples as primary thoughts and generates hyperedges through multi-hop walking paths. Crucially, it introduces a visual hypergraph-of-thought that interacts with the textual component via cross-modal co-attention, enabling robust comparative verification between visual chart elements and textual queries. Empirical evaluations have shown that HoT-based models can outperform CoT-based variants and even match the performance of larger models like GPT-4 on complex scientific reasoning benchmarks, all while utilizing smaller model sizes [169]. This capability is indispensable for chart understanding, where reasoning often requires comparing multiple data series, axes, and legends simultaneously—a task that linear chains and even standard graphs may fail to capture fully.

Complementary to these structural expansions, other approaches like Residual Connection Prompting (RESPROMPT) attempt to reconstruct reasoning graphs within prompts by integrating missing links from the reasoning graph into the linear flow, effectively morphing CoT into a graph representation to handle complex dependencies [170]. Similarly, recursive approaches like Socratic Questioning and Recursion of Thought (RoT) introduce divide-and-conquer strategies that break problems into sub-contexts, mimicking the recursive nature of human cognition to handle problems that exceed standard context limits [171; 172]. Collectively, these advancements signify a fundamental paradigm shift from viewing reasoning as a linear trajectory to treating it as a dynamic, structural topology. For Multimodal RAG systems dealing with charts and documents, adopting these non-linear topologies is not merely an optimization but a necessity to accurately navigate multi-hop dependencies and validate evidence across modalities. However, while these structural frameworks excel at organizing logical flows and mitigating hallucination through backtracking, they do not inherently resolve the fundamental limitation of Large Language Models in performing precise numerical computation. Even when guided by sophisticated Graph-of-Thought

or Hypergraph-of-Thought frameworks, LLMs remain probabilistic generators that frequently suffer from arithmetic errors when tasked with data-intensive queries. To bridge this gap between high-level structural reasoning and rigorous quantitative accuracy, the research community has increasingly shifted towards Program-Aided and Symbolic Reasoning paradigms, as detailed in the following section.

4.3 Program-Aided and Symbolic Reasoning for Precision

4.3 Program-Aided and Symbolic Reasoning for Precision

While the structural thought topologies discussed in the previous section effectively address the non-linear dependencies and multi-hop reasoning required for complex chart and document analysis, they do not inherently solve the fundamental limitation of Large Language Models (LLMs) in performing precise numerical computation. Even when guided by sophisticated Graph-of-Thought or Hypergraph-of-Thought frameworks, LLMs remain probabilistic generators that frequently suffer from arithmetic hallucinations and logical inconsistencies when tasked with data-intensive queries derived from financial reports, scientific plots, or statistical tables. To bridge this gap between high-level semantic reasoning and rigorous quantitative accuracy, the research community has increasingly shifted towards Program-Aided and Symbolic Reasoning paradigms. These strategies fundamentally decouple the reasoning process from the calculation process by delegating the latter to external, deterministic interpreters, thereby ensuring factual accuracy and preventing the propagation of numerical errors that can undermine even the most advanced structural reasoning architectures.

The core philosophy behind Program-Aided Language models (PAL) is to leverage the LLM’s capacity for natural language understanding to decompose a complex query into a sequence of executable code steps, typically in Python, rather than generating a natural language chain of thought. In this framework, the LLM acts as a semantic parser that translates the user’s intent and the retrieved visual data into a formal program, which is then executed by an external runtime environment to derive the final answer. This approach has demonstrated superior performance compared to standard Chain-of-Thought (CoT) prompting, particularly in tasks requiring rigorous arithmetic. For instance, [173] explicitly proposes the “Program of Thoughts” (PoT) method to disentangle computation from reasoning, showing an average performance gain of around 12% across various mathematical and financial datasets compared to CoT. By offloading the computational burden to a computer, PoT effectively eliminates the arithmetic errors that plague purely neural approaches, a capability that is critical when interpreting precise data points from charts or tables in retrieved documents.

This paradigm extends beyond simple arithmetic to encompass complex logical deductions and symbolic manipulation, offering a complementary mechanism to the structural explorations described previously. The integration of code as an intermediate reasoning step allows models to handle tasks that involve conditional logic, iterative processing, and state tracking, which are often necessary when analyzing multi-page documents or dynamic data visualizations. [174] explores this by triggering code as intermediate steps, finding that code prompting generally outperforms natural language CoT across symbolic and arithmetic reasoning benchmarks. Furthermore, the concept of “Chain of Code” introduces a flexible pseudocode format where the model can emulate an interpreter for semantic sub-tasks that cannot be directly executed, broadening the scope of reasoning questions that LLMs can answer correctly by “thinking in code” [175]. This flexibility is particularly valuable in MM-RAG systems where the retrieved context may contain unstructured

or semi-structured data that requires custom parsing logic before quantitative analysis can occur.

In the specific context of financial and scientific document analysis, where precision is paramount, program-aided strategies have shown remarkable efficacy. Financial reports often require multi-hop numerical reasoning across different sections of a document, a task where standard LLMs frequently falter due to calculation errors or misinterpretation of numerical magnitudes. [176] demonstrates that encoding required reasoning into a Python program or domain-specific language significantly improves accuracy over baselines by mitigating the limitations of LLMs in performing accurate arithmetic. Similarly, in the realm of chart understanding, where models must extract data variables and perform trend analysis, methods like [177] utilize external tools like Python to derive final answers from step-by-step sub-questions, achieving state-of-the-art accuracy on datasets like ChartQA and PlotQA. This step-wise generation, powered by external execution, ensures that the visual insights derived from charts are mathematically sound, providing a solid numerical foundation for the subsequent evidence-grounded generation discussed in the following section.

Moreover, the robustness of program-aided reasoning facilitates the distillation of advanced capabilities into smaller, more efficient models, which is crucial for deploying MM-RAG systems in resource-constrained environments. Techniques such as Program-aided Distillation (PaD) introduce reasoning programs to suppress errors in synthetic training data, allowing smaller models to outperform larger counterparts in reasoning tasks [178]. This is complemented by approaches like [179], which train small language models to map natural language questions into formal language descriptions evaluated by symbolic solvers, achieving massive improvements in accuracy without the need for massive parameter counts. The synergy between neural generation and symbolic verification is further highlighted in neurosymbolic frameworks like LINC, where LLMs translate premises into first-order logic for external theorem provers, significantly outperforming pure CoT methods on logical reasoning benchmarks [180].

The evolution of these methods also includes sophisticated prompting strategies that enhance the reliability of code generation itself, creating a self-correcting loop that reinforces the integrity of the reasoning process. For example, [181] proposes generating multiple algebraic expressions or Python functions to solve the same problem in different ways, thereby increasing confidence in the output through cross-verification. Additionally, [182] introduces a markup language to seamlessly integrate CoT with external Python REPLs, enabling LLMs to rectify errors within their reasoning chains dynamically. These advancements collectively underscore a pivotal shift in generative architectures: moving from opaque, probabilistic reasoning to transparent, verifiable, and precise program-aided workflows. By grounding the generation process in executable code, MM-RAG systems can achieve a level of factual faithfulness and numerical precision that is indispensable for high-stakes applications in finance, healthcare, and scientific research. This precision serves as a critical prerequisite for the next stage of evolution: ensuring that these accurate computations are tightly anchored to specific visual regions to prevent hallucinations and enhance interpretability.

4.4 Evidence-Grounded Generation and Visual Anchoring

4.4 Evidence-Grounded Generation and Visual Anchoring

While Section 4.3 established that program-aided and symbolic reasoning can resolve the arithmetic inaccuracies and logical inconsistencies inherent in Large Language Models (LLMs), precise computation alone does not guarantee factual faithfulness in Multimodal Retrieval-Augmented Generation (MM-RAG). A critical failure mode persists even in models equipped with external calculators: the tendency to hallucinate by generating plausible-sounding textual answers that are not grounded in

the retrieved visual evidence. This phenomenon often stems from an excessive reliance on language priors, where the model’s internal parametric knowledge overrides the specific data points or trends present in the input charts and documents as the generation sequence progresses. To bridge the gap between the numerical precision achieved via code execution and the semantic integrity of the final output, advanced generative architectures must incorporate robust mechanisms for evidence-grounded generation and visual anchoring. These mechanisms dynamically link generated tokens to specific regions within charts or documents, utilizing bounding box coordinates and region-specific features to validate reasoning steps, thereby ensuring that the “correct” calculations derived in the previous stage are attributed to the correct visual sources.

The core challenge in this domain lies in bridging the modality gap between continuous visual features and discrete textual tokens. Traditional approaches often treat visual grounding as a separate post-hoc task or rely on coarse global image embeddings, which fail to capture the fine-grained details necessary for precise chart interpretation and document layout analysis. Recent advancements propose integrating visual grounding directly into the generation process to enforce a strict coupling between text and image. For instance, the concept of “Answer Grounding” has evolved to require models not only to produce a textual answer but also to explicitly highlight the relevant visual evidence, such as specific bars in a chart or rows in a table. Frameworks like Dual Visual-Linguistic Interaction (DaVI) exemplify this by introducing interaction mechanisms where a visual-based linguistic encoder incorporates visual features to produce evidence-oriented linguistic representations, while a linguistic-based visual decoder focuses attention on evidence-related regions [183]. This bidirectional flow ensures that the generated text is inextricably linked to the visual features that justified it, preventing the decoupling of logic from data.

To further enforce this alignment during the inference phase, researchers have developed decoding strategies that amplify the influence of visual prompts while suppressing language-only hallucinations. A notable approach is Multi-Modal Mutual-Information Decoding (M3ID), which operates by maximizing the mutual information between the generated tokens and the reference image. By penalizing tokens that have low mutual information with the visual context, M3ID effectively reduces the model’s reliance on language priors, thereby suppressing hallucinations without requiring additional training labels [38]. This method can be paired with Direct Preference Optimization (DPO) to further refine the model’s behavior, ensuring that the generation process remains tethered to the visual evidence throughout the sequence, complementing the deterministic accuracy provided by the program-aided methods discussed previously.

Beyond decoding strategies, the architectural design of the reasoning pipeline plays a crucial role in achieving deep visual anchoring. The “Chain of Manipulations” paradigm, implemented in models like CogCoM, encourages LVLMs to solve problems through a series of explicit operations on the visual input, such as zooming in or grounding specific entities [184]. Each manipulation acts as an interpretable step that grounds the subsequent reasoning in concrete visual evidence, allowing users to trace error causes if the final answer is incorrect. Similarly, the use of fine-grained reward modeling, as seen in ViGoR, provides a signal to optimize the model’s grounding capabilities. By leveraging human evaluations or automated metrics that assess the alignment between text spans and image regions, these frameworks significantly enhance the visual grounding of LVLMs over pre-trained baselines [185].

The integration of spatial reasoning modules is another vital component for handling the complex layouts of financial reports and scientific charts. Models that employ spatial memory and multi-step reasoning, such as the Multimodal Spatial Region Reasoner (MSRR), initialize a 2D spatial memory with object regions and iteratively score them against the query [186]. This allows the model to

perform dynamic localization, refining its focus on relevant regions as the reasoning process unfolds. Furthermore, techniques like Contrastive Region Guidance (CRG) offer a training-free method to guide models toward relevant regions by contrasting outputs produced with and without visual prompts, effectively factoring out biases inherent in the model’s prior knowledge [187].

Addressing the issue of faithfulness also requires mechanisms to detect and correct inconsistencies between the generated text and the visual context, creating a self-correcting loop similar to the verification steps in program-aided reasoning. The “Chain-of-Verification” (CoVe) method, although originally designed for text, has inspired multimodal adaptations where the model plans verification questions to fact-check its draft against the visual evidence before generating the final response [188]. In the multimodal context, this involves verifying whether the claimed data points or trends actually exist in the retrieved chart regions. Additionally, counterfactual reasoning has been employed to enhance robustness; by implanting counterfactual thoughts and verifying them against the visual input, models can learn to distinguish between supported facts and hallucinated content [189].

Finally, the precision of visual anchoring is often limited by the quality of the underlying feature representations. Methods that utilize query-modulated refinement networks, such as QRNet, adjust intermediate features in the visual backbone based on the query semantics, ensuring that the extracted features are optimal for the specific grounding task [32]. This dynamic adjustment helps align the visual features with the linguistic requirements of the query, reducing the inconsistency that often leads to unfaithful generation. Moreover, iterative reasoning frameworks that denoise noisy boxes conditioned on query semantics, such as language-guided diffusion models, provide a generative approach to progressively refine the grounding boxes, enhancing the matching between the query and the visual region [190].

In summary, evidence-grounded generation and visual anchoring represent a paradigm shift from static retrieval to dynamic, interactive reasoning, serving as the essential counterpart to the computational precision offered by program-aided methods. By tightly coupling textual generation with spatial localization, utilizing mutual information decoding, and employing iterative verification loops, MM-RAG systems can significantly reduce hallucinations and provide trustworthy, interpretable answers for complex chart and document understanding tasks. However, the efficacy of these grounding mechanisms and the quality of the reasoning chains they support remain heavily contingent upon the examples provided to the model. This necessity drives the transition to the next critical area of research: automated demonstration selection and prompt optimization, which aims to systematically curate the input context to maximize the performance of these grounded reasoning architectures.

4.5 Automated Demonstration Selection and Prompt Optimization

4.5 Automated Demonstration Selection and Prompt Optimization

While Section 4.4 established mechanisms for grounding generation in visual evidence to suppress hallucinations, the efficacy of these Multimodal Retrieval-Augmented Generation (MM-RAG) systems remains heavily contingent upon the quality of reasoning demonstrations provided during the in-context learning phase. Even with robust visual anchoring, Large Vision-Language Models (LVLMs) require precise guidance to navigate the complex logical structures inherent in charts and documents. Although Chain-of-Thought (CoT) prompting has revolutionized complex reasoning, the manual construction of high-quality, domain-specific demonstrations poses a significant

bottleneck, particularly given the heterogeneity of visual layouts and textual content found in financial reports, scientific papers, and legal documents. Consequently, recent research has pivoted towards automated methodologies for demonstration selection and prompt optimization, aiming to eliminate manual engineering while enhancing generalization across diverse multimodal scenarios. This focus on optimizing the *input* reasoning context serves as a critical precursor to the modular architectural shifts discussed in Section 4.6, where distinct cognitive capabilities are decoupled.

A primary challenge in automated selection is ensuring that chosen demonstrations are not only semantically relevant but also structurally diverse enough to cover the vast solution space of chart and document queries. Traditional approaches often rely on simple similarity metrics, which can lead to redundant examples that fail to illuminate distinct reasoning pathways. To address this, diversity-aware sampling strategies have emerged as a critical component. For instance, the concept of “Diversity of Thought” suggests that leveraging variations in the input prompt itself, rather than just decoding strategies, can significantly improve reasoning abilities [191]. By soliciting feedback from the model to ideate different approaches suitable for a specific problem, systems can ensemble diverse prompts to achieve superior performance on complex planning and graph-based reasoning tasks, which are analogous to navigating the structured dependencies in document layouts. Furthermore, methods like SEER formulate exemplar selection as a Knapsack Integer Linear Program, explicitly incorporating diversity constraints to prioritize exemplars with desirable attributes while respecting prompt capacity budgets [192]. This is particularly vital for HybridQA tasks involving both unstructured text and structured tables, where a narrow set of demonstrations might miss critical inter-modal reasoning patterns.

Beyond diversity, the underlying reasoning pattern of a demonstration plays a more crucial role than mere semantic similarity or accuracy. The Pattern-Aware CoT approach posits that considering factors such as step length and the specific logic flow within intermediate steps can effectively mitigate bias induced by homogeneous demonstrations [193]. In the context of chart understanding, where a query might require trend analysis followed by specific data point extraction, selecting demonstrations that mirror this specific multi-step pattern is essential for guiding the model correctly. Complementing this, Latent Skill Discovery (RSD) introduces an unsupervised learning mechanism to create latent space representations of reasoning skills, allowing the system to select examples that demonstrate the precise skills required for a given query, such as arithmetic calculation or spatial relation inference, without requiring manual annotation [194].

Iterative bootstrapping represents another powerful paradigm for refining demonstration quality autonomously, moving beyond static pools of examples to enable self-correction and the generation of increasingly sophisticated reasoning chains. The Iter-CoT framework utilizes iterative bootstrapping to autonomously rectify errors in generated reasoning chains, simultaneously selecting challenging yet answerable questions as exemplars to enhance generalizability across varying difficulty levels [195]. This is particularly relevant for MM-RAG systems that must handle queries ranging from simple fact retrieval to complex multi-hop reasoning across document pages. Similarly, synthetic prompting techniques leverage a few handcrafted examples to prompt the model to generate new examples through a backward and forward process, effectively expanding the demonstration pool with high-quality, solvable instances [196]. This reduces the dependency on costly human labeling while ensuring the demonstrations are tailored to the specific nuances of the target domain.

The automation of prompt construction extends to meta-learning and heuristic-driven strategies to handle distribution shifts and domain specificity. Diversity-Aware Meta Visual Prompting (DAM-VP) addresses distribution shifts by clustering downstream datasets into homogenous subsets, each

optimized with its own prompt initialized by a meta-prompt learned across multiple datasets [197]. This divide-and-conquer design is highly applicable to document understanding, where different document types (e.g., invoices vs. research papers) exhibit distinct structural characteristics requiring specialized prompting strategies. Furthermore, heuristic-driven approaches like Heuristic-Driven Link-of-Analogy (HD-LoA) transform haphazard example selection into a methodical process that emphasizes task-specific heuristics, enabling models to draw analogies from known situations to handle unseen classes in document-level event extraction [198].

Retrieval mechanisms have also been integrated directly into the CoT selection process to dynamically choose optimal demonstrations based on cross-modal similarities. Research indicates that using retrieval to dynamically select demonstration examples, combined with stratified sampling to promote diversity, can substantially improve the performance of large multimodal models on complex reasoning benchmarks [199]. This aligns with the broader finding that active learning principles, specifically those prioritizing similarity and low uncertainty, outperform random sampling or pure uncertainty-based methods when selecting in-context examples [200]. Moreover, the necessity of human-generated demonstrations is being challenged by self-contemplation strategies, where models generate their own demonstrations based on internal competence, achieving results comparable to hand-crafted sets while significantly reducing external data dependency [201].

Finally, advanced optimization techniques such as metaheuristics and mixture-of-experts architectures are being employed to refine prompts further, pushing the boundaries of what automated systems can achieve without gradient access. The Plum framework utilizes metaheuristic algorithms like genetic algorithms and simulated annealing to discover effective, human-understandable prompts in a black-box setting, offering a general solution for prompt optimization without gradient access [202]. Additionally, ConstitutionalExperts demonstrates that learning unique prompts for different semantic regions of data using a mixture-of-experts architecture can outperform single-prompt optimization, suggesting that adaptive, region-specific prompting is key for handling the multifaceted nature of visual documents [203]. Collectively, these automated selection and optimization techniques form the backbone of next-generation MM-RAG systems, enabling them to adapt dynamically to the intricate reasoning demands of chart and document understanding. By automating the creation of high-fidelity reasoning contexts, these methods lay the groundwork for the transition toward modular architectures, where such optimized prompts can be strategically deployed across specialized sub-agents to solve even more complex deductive tasks.

4.6 Modular Architectures and Decomposition for Complex Deduction

4.6 Modular Architectures and Decomposition for Complex Deduction

Building upon the automated demonstration selection and prompt optimization strategies discussed in Section 4.5, which refine the *input* context for reasoning, the field is now witnessing a fundamental architectural shift from monolithic pipelines to modular frameworks. While optimized prompts provide high-quality guidance, relying on a single Large Language Model (LLM) to simultaneously decompose complex queries, retrieve heterogeneous visual evidence, perform logical synthesis, and generate final answers often leads to cognitive overload and reasoning failures. This is particularly acute in chart and document understanding, where financial reports, scientific papers, and technical manuals demand distinct cognitive capabilities for tasks ranging from optical character recognition (OCR) correction to trend detection in time-series data. Modular architectures ad-

dress these limitations by decoupling problem decomposition from solution generation, enabling specialized sub-agents to handle discrete tasks within a cohesive, scalable reasoning pipeline. This transition from optimizing the prompt to restructuring the agent architecture serves as the critical bridge to the efficiency and consistency challenges explored in Section 4.7.

A foundational principle of these emerging architectures is the explicit separation of the “decomposer” and the “solver.” Research indicates that problem decomposition and solution generation are distinct capabilities that yield superior performance when addressed by separate modules rather than a single monolithic LLM [204]. By isolating the decomposition phase, systems can employ specialized models to break down intricate, multi-step queries regarding document layouts or chart trends into manageable sub-problems. This modularization facilitates precise coordination: a decomposer identifies key concepts and structural dependencies, while a solver executes specific reasoning steps, a strategy shown to significantly outperform baselines where one model attempts to manage the entire cognitive load [205]. In the context of Multimodal Retrieval-Augmented Generation (MM-RAG), this separation is paramount; a dedicated module can first analyze the visual structure of a document to identify relevant regions, while a separate module handles the extraction and verification of textual data within those regions, ensuring that the retrieval process is grounded in accurate structural understanding.

Furthermore, modular frameworks facilitate the seamless integration of diverse reasoning strategies and external tools, which is essential for robust document analysis. The “Planner-Reasoner-Executor-Reflector” (PRER) framework exemplifies how agent-based decomposition can model the mathematical and logical solving process by assigning specific roles to different components [206]. In a chart understanding scenario, the “Planner” might determine that a query requires both OCR extraction and statistical trend analysis, the “Executor” would invoke specific tools for these tasks, and the “Reflector” would validate the consistency of the extracted data against the visual representation. This approach mitigates the risk of hallucinations, as the reflection module can cross-verify outputs before they are synthesized into a final response. Similarly, the “IdealGPT” framework demonstrates the efficacy of iterative decomposition in vision-language tasks, utilizing an LLM to generate sub-questions, a vision-language model (VLM) to provide sub-answers, and another LLM to reason toward a final conclusion, repeating this cycle until confidence thresholds are met [207].

The adaptability of modular systems also enables dynamic strategy selection based on task complexity, directly addressing the variability inherent in document datasets. The “Adaptive-Solver” framework illustrates how a system can modulate its solving strategies by switching between different models, prompting methods, or decomposition granularities depending on the difficulty of the problem [208]. In document understanding, a simple query might be handled by a lightweight module, while a complex query involving multi-hop reasoning across multiple pages triggers a more sophisticated chain of specialized agents. This efficiency is further enhanced by frameworks like “DefInt,” which employs a default-interventionist strategy where smaller models handle routine deductions, and larger, more capable models are invoked only when low-confidence scenarios arise, thereby optimizing computational resources without sacrificing accuracy [209].

Moreover, modular architectures support the integration of symbolic solvers and program-aided reasoning to ensure factual precision, a necessity when interpreting numerical data in charts where generative models often struggle with exact arithmetic. Frameworks such as “Logic-LM” and “SYRELM” integrate LLMs with symbolic solvers, where the LLM acts as a translator to formulate problems into a formal language that a deterministic solver can execute [210] [179]. This hybrid approach is particularly valuable for MM-RAG systems processing financial tables or scientific graphs, where arithmetic errors in the generation phase can lead to severe misinterpretations. By

delegating calculation to a symbolic engine, the modular pipeline ensures that the logical synthesis of retrieved evidence remains mathematically sound, bridging the gap between linguistic flexibility and numerical rigor.

Finally, the modularity of these architectures fosters robustness through self-correction and peer review mechanisms, creating a defense-in-depth against reasoning errors. Systems like “SCREWS” and those employing multi-agent debate allow for iterative refinement of outputs, where different modules or agents critique and revise each other’s contributions [211] [212]. In the context of document understanding, one agent might specialize in verifying OCR accuracy, while another checks for logical consistency in the generated summary. This collaborative dynamic prevents the propagation of errors that often plague end-to-end models. Additionally, the “CRAFT” framework highlights the importance of creating and retrieving from specialized toolsets, enabling LLMs to adapt to unseen domains by dynamically assembling the necessary modular components for a specific task [213]. By decoupling these capabilities, modular architectures provide a scalable and flexible foundation for next-generation MM-RAG systems. However, while these decomposed pipelines offer superior reasoning depth, they introduce new challenges regarding computational overhead and logical consistency across multiple handoffs, topics that are critically examined in the following section on efficiency and consistency in multimodal reasoning pipelines.

4.7 Efficiency and Consistency in Multimodal Reasoning Pipelines

4.7 Efficiency and Consistency in Multimodal Reasoning Pipelines

While the modular architectures discussed in the previous section provide a robust framework for decomposing complex deduction tasks, their practical deployment in real-world document and chart understanding scenarios introduces significant challenges regarding computational overhead and logical fidelity. The transition from theoretical modularity to production-ready Multimodal Retrieval-Augmented Generation (MM-RAG) systems necessitates a delicate balance: maintaining the depth of multi-step reasoning enabled by specialized agents while adhering to the stringent latency and cost constraints of high-resolution visual processing. Addressing the trade-offs between the granular reasoning capabilities of modular systems and computational efficiency requires a multifaceted approach involving adaptive inference strategies, advanced model compression techniques, and rigorous consistency metrics.

A primary strategy for enhancing efficiency within these pipelines involves optimizing prompt structures and decoding processes to reduce the verbosity often inherent in sequential modular interactions. Traditional sequential prompting, where intermediate results are generated step-by-step before reaching a final answer, can lead to disconnected outputs and increased latency, particularly when multiple modules exchange information. To mitigate this, prompt sketching has been proposed as a paradigm where the model predicts values for multiple variables within a template simultaneously, rather than generating text sequentially. This approach allows the decoder to score follow-up instructions during generation, optimizing the overall template likelihood and granting greater control over the reasoning framework. Research indicates that prompt sketching can outperform standard chain-of-thought schemes in zero-shot settings across various reasoning tasks, including arithmetic and general question answering, by reducing the generation of unnecessary intermediate tokens [214]. Furthermore, the compression of textual prompts themselves is critical; leveraging relation-aware graphs to extract key information elements can significantly reduce prompt length while maintaining or even improving task performance, thereby lowering the computational burden on the generator [215].

Beyond prompt engineering, model compression via knowledge distillation (KD) offers a viable pathway to deploy compact models capable of sophisticated reasoning without the heavy resource demands of large-scale modular ensembles. The challenge lies in transferring the complex reasoning capabilities of massive teacher models—often the “solvers” in modular frameworks—to smaller student models without significant performance degradation. Recent advancements suggest that distilling specific components of the reasoning process, such as problem decomposition, is more effective than attempting to distill the entire solution generation capability. For instance, separating the decomposition phase from the solving phase allows for the creation of smaller, cost-efficient models that handle task breakdown, which can then be paired with larger solvers only when necessary [216]. Additionally, integrating rationales into the distillation process has proven essential. Methods that incorporate intermediate rationales—generated by teachers or derived from external tools like OCR—as supervision signals enable small student models to achieve accuracy gains comparable to much larger counterparts in visual document understanding tasks [217]. Specialized distillation frameworks, such as those utilizing multi-chain-of-thought consistency, further enhance the robustness of these smaller models by enforcing agreement among diverse reasoning paths, ensuring that the distilled knowledge retains the logical fidelity required for complex chart and document analysis [218].

Consistency in multimodal reasoning is equally critical, as the increased number of handoffs in modular pipelines can amplify the risk of hallucinations and logical contradictions, severely undermining the trustworthiness of generated answers. Ensuring that a model’s generated rationale faithfully supports its final prediction is a non-trivial challenge, especially when dealing with conflicting information retrieved from diverse document sources. Techniques such as counterfactual reasoning objectives during distillation have been developed to prevent student models from ignoring rationales, thereby fostering self-consistent behavior where the model respects its own reasoning steps [219]. Moreover, the ability to resolve conflicts between parametric memory and retrieved external knowledge is vital. Frameworks that treat knowledge editing as a constrained decoding problem can adaptively substitute new knowledge into reasoning steps, ensuring coherence and pertinence when documents contradict the model’s internal priors [220]. In scenarios involving contradictory information, relying on source preferences and defeasible reasoning strategies becomes necessary to navigate inconsistencies effectively, a capability that standard models often lack without specific fine-tuning or architectural adjustments [221].

To further optimize efficiency, adaptive inference mechanisms allow systems to dynamically allocate computational resources based on problem complexity, complementing the static modularity described previously. Rather than applying a uniform reasoning strategy to all queries, adaptive solvers can modulate their approach by switching models, altering prompting methods, or adjusting the granularity of problem decomposition. This dynamic strategy selection ensures that simple queries are handled with minimal overhead, while complex, multi-hop reasoning tasks receive the necessary depth of analysis [208]. Similarly, algorithms that determine inference termination moments adaptively can skip unnecessary deep layers for easier instances, significantly reducing computational resources while maintaining performance across diverse tasks [222].

Finally, the evaluation of these efficiency and consistency measures requires metrics that go beyond simple accuracy, particularly when validating the output of complex, multi-agent systems. Assessing the reliability of benchmarking results themselves is crucial, as inefficient evaluation protocols can obscure true model capabilities. Methods that intelligently reduce computation costs during evaluation without compromising reliability, such as those measuring the decision impact on reliability, provide a framework for validating that efficiency gains do not come at the expense

of robust reasoning [223]. By integrating prompt sketching, targeted knowledge distillation, adaptive resource allocation, and rigorous consistency checks, MM-RAG systems can achieve the dual goals of high-efficiency inference and trustworthy, deep reasoning required for next-generation document and chart understanding applications, laying the groundwork for the specialized evaluation benchmarks discussed in the subsequent section.

5 Optimization, Robustness, and Trustworthiness

5.1 Iterative Self-Reflection and Multi-Agent Debate for Hallucination Mitigation

5.1 Iterative Self-Reflection and Multi-Agent Debate for Hallucination Mitigation

While retrieval augmentation provides the foundational evidence necessary for Multimodal Retrieval-Augmented Generation (MM-RAG) in chart and document understanding, the mere presence of relevant context does not guarantee its correct integration. Models frequently struggle to reconcile retrieved visual evidence with their parametric knowledge, often leading to hallucinations where fabricated details persist despite contradictory data. To bridge the gap between passive retrieval and active factual grounding, recent research has shifted towards dynamic internal verification mechanisms. Specifically, iterative self-reflection frameworks and multi-agent debate strategies have emerged as critical paradigms, empowering systems to critique their own outputs and engage in adversarial reasoning to converge on factually accurate interpretations before finalizing a response.

A cornerstone of this paradigm shift is the concept of self-reflection, where the model is trained not only to generate answers but also to evaluate the necessity of retrieval and the quality of its own generations. The Self-RAG framework exemplifies this approach by introducing special “reflection tokens” that allow a language model to adaptively retrieve passages on demand and critically reflect on its generated content [224]. In the context of visually rich documents, such reflection tokens can be extended to verify whether specific data points in a chart or sections in a document actually support the generated claim. This method significantly enhances factuality and citation accuracy, outperforming standard retrieval-augmented models by enabling the system to distinguish between situations where external evidence is required and those where internal knowledge suffices [224]. Furthermore, the integration of self-evaluation capabilities allows models to steer themselves toward factuality without relying exclusively on human-annotated data, as demonstrated by approaches that leverage self-evaluation components to validate responses based on internal knowledge [225]. These self-corrective loops serve as a primary defense against hallucination, setting the stage for more complex, multi-perspective verification strategies.

Building upon single-agent reflection, multi-agent debate (MAD) frameworks have emerged as a powerful strategy to overcome the cognitive constraints and stubborn biases inherent in individual models. In complex document reasoning tasks, a single agent may become trapped in a line of incorrect reasoning or fail to abandon an initially formed but erroneous hypothesis. The Multi-Agent Debate with Retrieval Augmented (MADRA) framework addresses this by incorporating the retrieval of prior knowledge directly into the debate process, effectively breaking cognitive constraints and enhancing reasoning capabilities [226]. By enabling agents to autonomously select pertinent evidence through a self-selection module, MADRA minimizes the impact of irrelevant or noisy data, a critical feature when dealing with the high-dimensional and often cluttered layouts of financial reports or scientific papers [226].

The efficacy of multi-agent interactions is further highlighted in the Multi-Agent Debate Refinement (MADR) framework, which specifically targets the generation of faithful explanations for fact-checking. MADR leverages multiple Large Language Models (LLMs) acting as agents with diverse roles in an iterative refining process, ensuring that the final explanation undergoes rigorous validation against provided evidence [227]. This is particularly vital for chart understanding, where explaining the trend of a graph requires strict adherence to the visual data rather than plausible-sounding but fabricated narratives. By enforcing a structured debate where agents challenge each other’s interpretations, the system significantly reduces the likelihood of unfaithful elements, thereby advancing the credibility of automated document analysis [227].

However, the implementation of multi-agent systems is not without its complexities. Research indicates that while multi-agent debating systems hold promise, their performance is highly sensitive to hyperparameter settings and the specific protocols employed, such as the level of agreement required between agents [228]. Simple debate formats may not always outperform other prompting strategies like self-consistency unless carefully tuned to encourage divergent thinking and prevent the “Degeneration-of-Thought” problem, where agents prematurely converge on incorrect solutions [212]. To mitigate this, frameworks like Mirror propose a multiple-perspective self-reflection method that guides agents toward diverse yet plausibly reliable reasoning trajectories by encouraging diversity in the directions generated by a “Navigator” agent and seeking agreement among strategically induced perturbations [229].

Moreover, the intersection of retrieval and debate offers unique advantages for handling knowledge conflicts, a common scenario in MM-RAG where retrieved document content contradicts the model’s parametric memory. Studies on realistic knowledge conflicts suggest that models often exhibit a parametric bias, failing to update their answers even when presented with conflicting evidence [43]. Multi-agent debates can serve as a mechanism to resolve these conflicts by forcing agents to explicitly argue for the retrieved evidence over internal priors, thus reducing the tendency to ignore contextual information. Additionally, the integration of retrieval uncertainty quantification, such as that proposed in CONFLARE, can inform the debate process by signaling when retrieved context is insufficient, prompting agents to express uncertainty rather than hallucinating an answer [230].

In summary, the synergy between iterative self-reflection and multi-agent debate represents a critical advancement in mitigating hallucinations in MM-RAG systems by establishing robust internal verification loops. By transitioning from static generation to dynamic, interactive verification processes, frameworks like Self-RAG, MADRA, and MADR provide the necessary architectural scaffolding to ensure that generated interpretations of charts and documents are grounded in retrieved evidence. While these methods effectively reduce internal reasoning errors and resolve knowledge conflicts, they primarily operate on the logic and consistency of the generation process itself. Consequently, to achieve comprehensive fidelity, these interactive mechanisms must be complemented by explicit, external verification standards. This necessity leads naturally to the development of evidence-based verification and factual error correction mechanisms, which focus on structuring reasoning into executable logic and systematically correcting deviations from factual truth, as discussed in the following section.

5.2 Evidence-Based Verification and Factual Error Correction Mechanisms

5.2 Evidence-Based Verification and Factual Error Correction Mechanisms

While Section 5.1 established dynamic internal verification loops through self-reflection and multi-agent debate to mitigate reasoning biases, these interactive mechanisms primarily operate on the logical consistency of the generation process. To achieve comprehensive fidelity, such internal checks must be complemented by explicit, external verification standards that rigorously anchor generated claims to retrieved visual evidence. This section shifts focus from *how* models reason to *what* grounding standards they must meet, detailing mechanisms for evidence-based verification and systematic factual error correction. These approaches aim to ensure that every generated claim in chart and document understanding is strictly grounded in retrieved data, providing robust pathways to correct deviations from factual truth before the system confronts the physical imperfections of real-world inputs discussed in Section 5.3.

A primary strategy for enhancing verification involves decomposing complex queries into executable logical structures, thereby reducing reliance on the generative model’s internal parametric knowledge for factual accuracy. Program-guided reasoning has emerged as a powerful paradigm for this purpose. By leveraging the in-context learning capabilities of Large Language Models (LLMs), complex claims can be translated into First-Order Logic (FOL) clauses or executable programs consisting of specialized sub-tasks [231]. This approach, exemplified by frameworks like Program-Guided Fact-Checking (ProgramFC), allows the system to decompose a multifaceted claim into simpler verifiable components that are solved using a shared library of functions [232]. In the context of multimodal documents, this translates to generating code or logical sequences that explicitly extract values from charts, compare trends across tables, or verify spatial relationships before synthesizing a final answer. This decomposition not only improves accuracy but also provides explicit, human-readable explanations of the reasoning process, making the verification traceable and debuggable [232]. Furthermore, extending this logic to neuro-symbolic frameworks allows models to ground abstract concepts across diverse domains, including 2D images and 3D scenes, by executing logic-based programs with trainable domain-specific grounding modules [233].

Beyond structural decomposition, addressing factual errors often requires innovative training methodologies, particularly when paired data containing both false claims and their corrected versions are scarce. A notable advancement in this area is the concept of learning to inject factual errors to improve correction capabilities. The “Learning to Inject Factual Errors” (LIFE) framework proposes a distantly supervised method that operates on a “mask-corrupt-correct” paradigm [234]. Instead of relying on expensive manual annotations of errors, this approach trains a corruptor model to deliberately introduce factual inconsistencies into correct text derived from visual evidence. This process generates a substantial volume of synthetic paired data, which is then used to train a corrector model. By circumventing the bottleneck of explicit factual error identification, LIFE enables models to learn robust correction strategies that outperform traditional baselines and even large proprietary models in specific factual consistency metrics [234]. This technique is particularly relevant for MM-RAG systems dealing with noisy OCR outputs or ambiguous chart labels, where the ability to recognize and rectify subtle factual distortions is essential.

Another crucial aspect of evidence-based verification is ensuring that the retrieved evidence itself is sufficient and has not been manipulated. Systems must be capable of verifying claims even

when faced with insufficient or adversarial evidence. Research into probing factually grounded content transfer introduces the notion of “factual ablation” as a metric for consistency [235]. This concept posits that a grounded model should demonstrate a lower likelihood of producing a specific output when presented with a less relevant or ablated grounding document. By measuring the divergence in model behavior when key evidence is removed or altered, researchers can quantify the degree to which a generation is truly anchored in the provided context rather than relying on spurious correlations or pre-trained biases. This is complemented by approaches that optimize evidence retrieval based on utility rather than simple relevance, incorporating feedback from the claim verifier to ensure that retrieved document segments effectively support the verification process [236].

Furthermore, the integration of fine-grained reward modeling has shown promise in refining the visual grounding of Large Vision-Language Models (LVLMs). Frameworks such as ViGoR utilize fine-grained rewards to penalize hallucinations of non-existent scene elements and incorrect attribute inference, significantly enhancing the alignment between generated text and visual input [185]. Similarly, the development of phrase-critic models allows for the refinement of generated explanations by scoring how well candidate phrases are grounded in the image, effectively acting as a filter against ungrounded justifications [237]. These verification layers can be combined with iterative refinement processes, such as those found in diffusion-based grounding models, where predictions are progressively denoised and corrected based on language guides to achieve higher precision in locating visual evidence [190].

In summary, the landscape of evidence-based verification and factual error correction in MM-RAG is characterized by a shift towards explicit reasoning structures, synthetic data augmentation for error correction, and rigorous metrics for grounding consistency. By integrating program-guided decomposition [232], distant supervision techniques like LIFE [234], and utility-driven retrieval feedback [236], modern systems are becoming increasingly resilient to hallucinations. These mechanisms collectively ensure that MM-RAG systems do not merely generate plausible-sounding text but produce outputs that are faithfully anchored in the complex, often noisy, visual realities of charts and documents. However, the efficacy of these verification strategies is fundamentally contingent upon the quality of the input visual data. If the retrieved evidence itself is corrupted by physical degradations or OCR errors, even the most sophisticated program-guided reasoning or factual ablation techniques will fail. Consequently, ensuring robustness against noisy modalities and OCR imperfections has emerged as a critical research frontier, necessitating a multi-layered defense strategy that spans from low-level image restoration to high-level textual post-correction, as discussed in the following section.

5.3 Robustness Against Noisy Modalities and OCR Imperfections

5.3 Robustness Against Noisy Modalities and OCR Imperfections

While Section 5.2 established mechanisms for verifying factual claims and correcting logical hallucinations, the efficacy of these verification strategies is fundamentally contingent upon the quality of the input visual data. In real-world deployments, documents are rarely pristine; they frequently suffer from a multitude of degradations including uneven illumination, shadows, blur, noise, wrinkles, and low resolution. These physical artifacts directly propagate errors through the Optical Character Recognition (OCR) pipeline, leading to garbled text tokens that severely degrade the semantic alignment and reasoning capabilities of downstream generative models. If the retrieved evidence itself is corrupted by noise, even the most sophisticated program-guided reasoning or fac-

tual ablation techniques described previously will fail. Consequently, ensuring robustness against noisy modalities and OCR imperfections has emerged as a critical research frontier, necessitating a multi-layered defense strategy that spans from low-level image restoration to high-level textual post-correction, thereby laying a clean foundation for the efficient long-context processing discussed in Section 5.4.

A primary line of defense involves enhancing the visual quality of document images prior to text extraction. This stage addresses the root cause of many OCR failures by employing advanced deep learning techniques to reverse physical degradations. A significant challenge in this domain is the removal of shadows and uneven lighting, which are common in camera-captured documents. Recent advancements have moved beyond simple thresholding to sophisticated intrinsic image decomposition methods. For instance, physically constrained learning-based approaches have been developed to estimate document reflectance directly, effectively disentangling shading from text content even under complex, multi-illuminant conditions. Such methods have demonstrated practical applicability by achieving substantial improvements in Character Error Rate (CER) when used as a pre-processing step for OCR pipelines [238]. Furthermore, specialized frameworks leveraging frequency-aware networks and large-scale real-world datasets have been proposed to handle high-resolution document shadow removal, preserving fine font details that are often lost in traditional restoration attempts [239]. The integration of diffusion models has further revolutionized this space; frameworks like DocDiff utilize residual diffusion to recover high-frequency text edges, while ShadowDiffusion integrates degradation priors to eliminate boundary artifacts, offering state-of-the-art performance in restoring readability [240; 241].

Beyond shadow removal, addressing general noise, blur, and resolution loss is equally vital to ensuring that the visual tokens fed into memory-compressed architectures remain informative. Unsupervised and blind restoration techniques are particularly valuable given the scarcity of paired clean-noisy data in real-world scenarios. Unified end-to-end unsupervised models have been introduced to remove diverse noise types, including salt-and-pepper noise and watermarks, without requiring ground-truth pairs, thereby significantly improving OCR accuracy on scanned pages [242]. Similarly, generative adversarial networks (GANs) and diffusion probabilistic models have been adapted for document enhancement tasks such as binarization, deblurring, and watermark removal, with some approaches like DECDM enabling domain adaptation without paired training data to preserve data privacy [243; 244]. For low-resolution inputs, cascaded detail-preserving networks and prompt-based super-resolution methods have shown efficacy in upscaling document images while maintaining character integrity, directly translating to higher word-level accuracy in subsequent recognition tasks [245; 246]. Additionally, geometric distortions caused by camera capture, such as page curvature and perspective shifts, are addressed by progressive rectification pipelines that iteratively refine content alignment, ensuring that text lines remain straight and readable for OCR engines [247].

When visual restoration is insufficient or computationally prohibitive, robustness must be enforced at the textual level through OCR post-correction mechanisms. Traditional post-processing relying on dictionaries or static language models often fails to capture context-specific errors or orthographic variations, especially in historical or specialized corpora. To overcome this, neural approaches combining recurrent and convolutional architectures with novel attention mechanisms have been proposed to correct transcription errors at the character level, achieving dramatic reductions in word error rates for historical texts [248]. Furthermore, the scarcity of manually corrected training data can be mitigated by leveraging text repetitions within large document collections. By estimating error distributions from repeating text spans and generating synthetic training examples, denoising

autoencoders can be trained to correct OCR outputs without extensive manual annotation, proving effective for large-scale digitization projects [249].

For massive document collections where individual processing is cost-prohibitive—a scenario increasingly common as context windows expand—batch-processing strategies offer a scalable solution. Recognizing that documents within a collection often share structural similarities and repeated vocabulary, clustering schemes can group erroneous words across multiple documents. This allows for collective correction, either automatically or via human-in-the-loop verification, drastically reducing the manual effort required to clean large archives compared to sequential processing [250]. Additionally, synthetic data generation pipelines have become indispensable for training robust NLP models that can tolerate OCR noise. By simulating realistic degradation levels and generating corresponding noisy-clean text pairs, researchers can train Named Entity Recognition (NER) and other downstream models to maintain high accuracy even when fed imperfect OCR outputs [24]. Finally, detection networks specifically designed to filter non-textual elements and handle corruption in business documents help prevent erroneous detections that often cascade into severe OCR failures, ensuring that only valid text regions are processed [26; 25]. Together, these visual and textual strategies form a comprehensive shield against the pervasive noise and imperfections inherent in real-world document understanding. By securing the fidelity of the input modality, these methods ensure that the subsequent stages of evidence-based verification and efficient context compression operate on reliable data, enabling MM-RAG systems to achieve greater overall fidelity and trustworthiness.

5.4 Context Compression and Efficiency for Long-Document Processing

5.4 Context Compression and Efficiency for Long-Document Processing

Having established in Section 5.3 the critical necessity of securing input fidelity against visual noise and OCR imperfections, we now address the subsequent challenge: processing these cleaned yet voluminous document contexts efficiently. The effective handling of extensive document corpora—such as multi-page financial reports, scientific manuscripts, and legal archives—remains a paramount bottleneck in Multimodal Retrieval-Augmented Generation (MM-RAG). As input sequences expand to accommodate high-resolution visual tokens and lengthy text, the quadratic computational complexity and linear memory expansion inherent to standard Transformer architectures create severe constraints. To overcome these limitations without compromising the fidelity required for complex chart and document reasoning, recent research has pivoted toward architectural innovations that prioritize dynamic context compression, adaptive memory management, and efficient inference strategies. These advancements lay the essential groundwork for the model compression and trustworthiness trade-offs discussed in Section 5.5.

A primary avenue for enhancing efficiency involves the deployment of temporary, inference-time adaptation mechanisms that circumvent the overhead of permanent parameter updates or massive Key-Value (KV) cache storage. Traditional approaches often struggle to maintain coherent context over long generations due to hardware limitations. However, novel methods propose embedding contextual information directly into temporary Low-Rank Adaptation (LoRA) modules during the generation process. For instance, utilizing temporary LoRA modules allows the system to progressively train on previously generated text, effectively preserving contextual knowledge within a lightweight adapter that is discarded post-generation [251]. This approach not only mitigates the need for storing extensive KV caches but also significantly reduces latency and memory usage, making it highly suitable for processing long-form document narratives where maintaining

discourse-level coherence is critical. Furthermore, parameter-efficient fine-tuning strategies have been extended to support long-context adaptation specifically. Techniques such as LongLoRA demonstrate that by combining shifted sparse attention with trainable embeddings, it is possible to extend the effective context window of pre-trained models to handle hundreds of thousands of tokens with limited computational costs [252]. These methods suggest that for MM-RAG systems, adapting the model to “read” longer documents can be achieved more efficiently through sparse attention patterns and low-rank updates rather than full retraining.

Parallel to adaptation strategies, memory-cached recurrence mechanisms offer a robust solution for paragraph-level understanding and long-range dependency modeling. The core concept involves compressing accumulating context into a compact memory space, allowing models to operate within restricted memory environments while retaining access to historical information. Compressed Context Memory systems, for example, integrate lightweight conditional LoRA modules to recursively compress attention key-value pairs, facilitating inference in streaming settings with effectively unlimited context lengths [253]. This is particularly relevant for document understanding tasks where the model must synthesize information across disparate sections of a report. Similarly, architectures inspired by operating system memory management, such as virtual context management, intelligently move data between fast and slow memory tiers to provide the appearance of a large context window [254]. By treating the document as a file system and dynamically retrieving relevant passages, these systems overcome the static limitations of fixed context windows. Moreover, the integration of differentiable memory caches, as seen in Cached Transformers, extends the self-attention mechanism to attend to both past and current tokens via a recurrent gating unit, thereby enhancing the model’s receptive field for long-range dependencies [255].

Another critical dimension of efficiency in long-document processing is the optimization of the retrieval component itself, specifically through generative retrieval approaches that eliminate the need for heavy, external indexing structures. Traditional dense retrieval relies on maintaining massive vector indices, which can be cumbersome and slow to update in dynamic document environments. Generative retrieval models, conversely, encode document identifiers directly within the model’s parameters, predicting them based on the query. However, applying this to dynamic corpora requires continual learning capabilities. The CLEVER model addresses this by introducing incremental product quantization and memory-augmented learning, allowing the system to incrementally index new documents without forgetting previous knowledge [129]. This eliminates the overhead of rebuilding external indices and streamlines the retrieval pipeline for evolving document collections. Additionally, frameworks like Re3val enhance generative retrieval by incorporating contextual reranking and reinforcement learning, ensuring that the retrieved document pointers are optimized for the downstream generation task [256].

Beyond architectural changes, algorithmic optimizations in attention mechanisms play a vital role in reducing the memory footprint of long-context processing. The observation that only a small subset of tokens, often referred to as “Heavy Hitters,” contributes disproportionately to attention scores has led to the development of eviction policies that dynamically retain only the most critical context. The Heavy Hitter Oracle (H2O) implements such a policy, significantly reducing KV cache memory usage while maintaining generation quality [257]. Similarly, adaptive KV cache compression techniques profile attention heads to discard non-essential tokens, achieving substantial memory reductions without the need for resource-intensive fine-tuning [258]. These compression strategies are essential for MM-RAG systems processing high-resolution images and lengthy text, where the sheer volume of visual and textual tokens can quickly exhaust available GPU memory. Furthermore, methods that leverage the structural properties of specific domains, such as com-

pressing dialogue turns into “conversational attention sinks,” demonstrate that domain-specific compression can yield massive speedups and memory savings [259].

In conclusion, the landscape of long-document processing in MM-RAG is being reshaped by a convergence of temporary adaptation modules, recurrent memory compression, and generative retrieval paradigms. By moving away from brute-force context expansion and toward intelligent, dynamic management of information density, these innovations enable systems to reason over massive, structured visual corpora with unprecedented efficiency. The synergy between temporary LoRA adapters [251], compressed memory architectures [253], and continual generative retrieval [129] provides a comprehensive toolkit for building scalable document understanding systems. However, while these architectural strategies optimize inference-time efficiency, they do not inherently reduce the static model size or address the deployment constraints of edge devices. This necessitates a deeper examination of static model compression techniques and their profound implications for model trustworthiness, which will be explored in the following section.

5.5 Model Compression, Distillation, and Trustworthiness Trade-offs

5.5 Model Compression, Distillation, and Trustworthiness Trade-offs

While Section 5.4 detailed architectural innovations for managing long contexts through dynamic memory and efficient attention, the practical deployment of Multimodal Retrieval-Augmented Generation (MM-RAG) systems in resource-constrained environments—such as mobile devices and edge servers—necessitates a different class of optimization: static model compression. Techniques including quantization, pruning, and knowledge distillation have become essential for reducing the computational footprint of these systems. However, the pursuit of such efficiency introduces a critical tension with model trustworthiness, a dimension encompassing factual accuracy, bias mitigation, and robustness against adversarial perturbations. In the specific domain of chart and document understanding, where fine-grained visual reasoning and precise data extraction are paramount, the degradation of trustworthiness due to aggressive compression can manifest as severe hallucinations, misinterpretation of complex visual layouts, or the amplification of societal biases embedded in training corpora.

Recent empirical studies reveal that the relationship between compression intensity and trustworthiness is non-linear and highly dependent on the specific technique employed. Quantization, which reduces the precision of model weights and activations, serves as a particularly nuanced tool in this regard. Research indicates that moderate quantization, such as reducing models to 4-bit precision, can often preserve the trustworthiness of original high-capability models with negligible impact on safety dimensions [260]. In some instances, moderate quantization acts as a regularizer, unexpectedly improving specific trustworthiness metrics related to ethics and fairness by smoothing the loss landscape and reducing overfitting to noisy training data [260]. However, this beneficial effect has strict limits; extreme quantization to very low bit levels, such as 3 bits or lower, tends to significantly degrade trustworthiness, leading to a marked increase in hallucination rates and a collapse in the model’s ability to adhere to safety constraints [260]. This suggests that while efficiency can be gained without sacrificing reliability up to a certain threshold, crossing into extreme compression regimes poses unacceptable risks for high-stakes applications like medical record analysis or legal document review. Furthermore, the cost of compression extends to parametric knowledge, where aggressive techniques can degrade a model’s ability to recall factual information even when general perplexity remains stable [261].

In contrast to quantization, pruning techniques, which remove redundant connections or neurons,

appear to pose a more immediate threat to model trustworthiness. Evidence suggests that model pruning can significantly degrade trustworthiness even at moderate sparsity levels, such as 50% [260]. This degradation is often attributed to the disproportionate impact of pruning on underrepresented features within the data distribution. Pruning has been shown to amplify existing algorithmic biases by disproportionately affecting performance on “Compression Identified Exemplars” (CIE), a subset of data points that are often atypical or belong to minority groups [262]. Similarly, research on “Pruning Identified Exemplars” (PIEs) confirms that compression disproportionately impacts model performance on the underrepresented long-tail of the data distribution [263]. In multimodal settings, this could translate to a compressed model failing to correctly interpret charts with unusual layouts or documents containing non-standard fonts, thereby introducing systematic errors that are difficult to detect through standard accuracy metrics alone. Furthermore, the “Junk DNA Hypothesis” posits that small-magnitude weights, which are primary targets for magnitude-based pruning, actually encode vital knowledge essential for tackling difficult downstream tasks [264]. Removing these weights can result in an irreparable loss of nuanced reasoning capabilities required for complex document understanding, a deficit that cannot be fully recovered even with subsequent fine-tuning.

Knowledge distillation, the process of transferring knowledge from a large teacher model to a smaller student model, presents a distinct set of trade-offs. While generally effective for maintaining performance, distillation is not immune to trustworthiness pitfalls, particularly regarding bias propagation. If a student model is distilled using biased training data, it may inherit or even exacerbate the biases of the teacher, regardless of whether the teacher itself was debiased [265]. However, distillation has also been observed to inhibit the memorization of specific training examples, potentially improving generalization and reducing the risk of data leakage, which is a crucial aspect of privacy-preserving MM-RAG systems [266]. Additionally, some studies suggest that distillation can act as a regularization technique that reduces toxicity and bias in generative models, offering a pathway to fairer compressed models if carefully managed [267]. Moreover, knowledge distillation methods have been demonstrated to be orthogonal to other efficiency-enhancing methods, allowing them to be combined effectively without conflicting performance improvements [268].

The interplay between compression and adversarial robustness further complicates the trustworthiness landscape, creating a bridge to the security challenges discussed in the subsequent section. While some research suggests that quantization can improve robustness against simple gradient-based attacks by introducing noise that masks gradients, other studies warn of a “false sense of security” where quantization merely obscures vulnerabilities without resolving them [269]. The relationship between model compression and adversarial robustness remains complex, with evidence showing that conclusions about robustness are often application-dependent and influenced by the specific compression pipeline used [270]. Recent investigations reveal that inconsistencies in robustness findings often arise from whether robust optimization is performed during the quantization stage [271]. Moreover, compressed models, particularly those with high activation sparsity, have been found to be more vulnerable to hardware faults and bit-flip errors, which can be exploited by adversarial actors to induce catastrophic failures [272]. The combination of multiple compression techniques, such as pruning and quantization, can yield synergistic size reductions but requires careful calibration to avoid compounding their respective negative impacts on reliability [273]. Additionally, universal adversarial perturbations can transfer differently between pruned and full models, suggesting that compression techniques can alter systemic vulnerabilities in ways that may either blunt or expose models to specific attack vectors [274].

Ultimately, deploying compressed MM-RAG systems for chart and document understanding re-

quires a holistic evaluation framework that moves beyond benign task performance. Practitioners must rigorously assess compressed models across multiple trustworthiness dimensions, including fairness, robustness, and factual consistency, rather than relying solely on accuracy metrics which often hide disparate impacts on specific data subsets [260]. New metrics such as label loyalty and probability loyalty have been proposed to better measure how closely a compressed model mimics the original, providing a more nuanced view of compression effects beyond standard accuracy [275]. The findings underscore that while moderate quantization offers a viable path to efficient and trustworthy deployment, extreme compression and indiscriminate pruning can exacerbate hallucinations and biases, necessitating a cautious, evidence-based approach to model optimization. This careful calibration of internal model integrity is a prerequisite for defending against external threats, setting the stage for the discussion on defending against adversarial evidence and synthetic disinformation in the following section.

5.6 Defending Against Adversarial Evidence and Synthetic Disinformation

5.6 Defending Against Adversarial Evidence and Synthetic Disinformation

Building upon the internal integrity challenges posed by model compression discussed in Section 5.5, the deployment of Multimodal Retrieval-Augmented Generation (MM-RAG) systems in high-stakes domains such as finance, healthcare, and legal analysis introduces a distinct layer of external security vulnerabilities: adversarial evidence and synthetic disinformation. While the previous section addressed how efficiency optimizations can inadvertently erode trustworthiness, this section examines how malicious actors actively exploit the retrieval-generation pipeline. Unlike traditional text-only systems, MM-RAG pipelines are uniquely susceptible to attacks that manipulate both the retrieval corpus and the visual-textual alignment, leading to the generation of hallucinated or maliciously altered outputs. A primary threat vector involves the injection of poisoned documents or manipulated visual evidence into the knowledge base, which can mislead the retrieval mechanism and subsequently corrupt the generative process. Research has demonstrated that attackers can inject a minimal number of poisoned texts into a massive database to force a Large Language Model (LLM) to generate specific, attacker-chosen answers for targeted questions, achieving success rates as high as 90% with only five poisoned samples [276]. This form of knowledge poisoning is particularly dangerous in multimodal contexts where synthetic disinformation can be embedded within charts, diagrams, or document layouts that appear visually authentic to human observers but contain subtle semantic perturbations designed to deceive automated retrievers.

The vulnerability of MM-RAG systems extends beyond simple text injection to include sophisticated evidence manipulation attacks that target the fact-verification capabilities of the system. Adversaries can tamper with online evidence by camouflaging relevant information or planting misleading content that aligns superficially with a claim while fundamentally altering its veracity [277]. In multimodal settings, this manifests as “retrieval poisoning,” where attackers craft documents that are visually indistinguishable from benign sources but contain hidden triggers or semantic inconsistencies that guide the application toward generating incorrect responses [278]. Furthermore, the robustness of these systems is challenged by low-level perturbations, such as typographical errors or minor visual distortions, which can disproportionately disrupt the synergy between retrieval and generation components. Studies utilizing genetic attacks have shown that simulating real-world noisy documents through low-level perturbations can devastate the performance of RAG pipelines, highlighting the fragility of current systems against even minor textual inaccuracies [279].

To mitigate these threats, recent research has proposed defense mechanisms centered on credibility-

aware generation and robust verification protocols, shifting the paradigm from passive consumption of retrieved data to active validation. A promising approach is Credibility-aware Generation (CAG), a framework designed to equip models with the ability to discern and process information based on its inferred credibility rather than treating all retrieved contexts as equally reliable [280]. By transforming training data to reflect varying levels of credibility, CAG enables models to maintain robust performance even when faced with noisy or manipulated documents, effectively reducing the impact of flawed information introduced during the retrieval phase. This is crucial for MM-RAG systems operating in environments where synthetic disinformation may be prevalent, as it allows the model to weigh evidence dynamically and resist the propagation of adversarial signals into the final generated output.

In addition to credibility scoring, robust verification protocols are essential for detecting and neutralizing poisoned corpora before they influence the generation process. Defense strategies must account for the possibility of adaptive adversaries who can bypass single-metric detection systems. For instance, multi-metric investigation approaches have been developed to identify poisoned model updates in federated learning settings by creating complex optimization problems for attackers, a principle that can be adapted for verifying the integrity of retrieved multimodal documents [281]. Moreover, the concept of “initiative defense” suggests proactively degrading the performance of malicious manipulation models by injecting imperceptible perturbations into target data, thereby preventing the successful engendering of adversarial content before it reaches the retrieval index [282]. While originally proposed for facial manipulation, this proactive stance offers a conceptual blueprint for hardening document corpora against synthetic disinformation campaigns.

The challenge of defending against adversarial evidence is further compounded by the transferability of attacks across different model architectures and modalities. Mutual-modality adversarial attacks leverage the alignment between visual and textual embedding spaces to generate perturbations that are highly transferable, posing a significant risk to black-box MM-RAG systems [283]. Consequently, defense mechanisms must be model-agnostic and capable of operating without explicit knowledge of the underlying retrieval or generation architectures. Techniques such as dimensionality reduction have shown promise in thwarting patch-based attacks by projecting samples onto lower-dimensional spaces that retain essential information while filtering out adversarial noise, a strategy that could be extended to protect against localized manipulations in chart and document images [284].

Furthermore, the security of MM-RAG systems requires rigorous evaluation against diverse attack vectors, including those that exploit implementation bugs rather than just algorithmic weaknesses. Research indicates that exploiting bugs in machine learning implementations can enable strictly more powerful attacks than classic adversarial techniques, necessitating a broader security posture that includes fuzzing and formal verification of the entire retrieval-generation pipeline [285]. As the landscape of synthetic disinformation evolves, future defenses must integrate continuous monitoring, adaptive credibility assessment, and multi-modal verification to ensure the trustworthiness of MM-RAG systems. The development of standardized benchmarks for adversarial robustness in multimodal retrieval, similar to those emerging for text-only systems, will be critical for advancing the state of the art in defending against these sophisticated threats. Ultimately, building trustworthy MM-RAG systems requires a holistic approach that addresses vulnerabilities at every stage, from data ingestion and indexing to retrieval scoring and final generation, ensuring resilience against an ever-expanding array of adversarial tactics.

6 Domain-Specific Applications and Case Studies

6.1 Financial Intelligence and Regulatory Compliance

6.1 Financial Intelligence and Regulatory Compliance

Transitioning from the life-critical demands of healthcare, the domain of financial intelligence and regulatory compliance presents a similarly complex yet distinct frontier for Multimodal Retrieval-Augmented Generation (MM-RAG) systems. Financial ecosystems are characterized by an overwhelming volume of heterogeneous data, ranging from structured quantitative tables in earnings reports to unstructured textual narratives in Management’s Discussion and Analysis (MDA) sections, and complex regulatory filings such as SEC Form 10-K and 8-K documents. The sheer complexity of these hybrid documents, which often span dozens of pages with intricate layouts, poses significant hurdles for traditional natural language processing pipelines. Recent advancements demonstrate that integrating MM-RAG with domain-specific fine-tuning and sophisticated retrieval strategies is essential for unlocking accurate quantitative extraction and interpreting nuanced regulatory constraints.

A primary application of MM-RAG in this sector is the automated analysis of financial reports to support investment decisions and risk assessment. Traditional machine learning and deep learning algorithms have historically struggled with stock trend prediction due to a lack of interpretability and an inability to integrate real-time textual information from financial news or reports. To address these limitations, frameworks like AlphaFin have been developed, which combine traditional research datasets with real-time financial data and handwritten chain-of-thought reasoning to benchmark retrieval-augmented stock-chain frameworks [286]. These systems leverage the textual understanding capabilities of Large Language Models (LLMs) while mitigating hallucinations through robust retrieval mechanisms, ensuring that predictions are grounded in verifiable financial evidence. Furthermore, the extraction of numerical data from lengthy financial reports has been significantly enhanced by frameworks such as Automated Financial Information Extraction (AFIE), which specializes in comprehending hybrid long-documents containing both text and tables. Empirical validations of AFIE on models like GPT-3.5 and GPT-4 have shown substantial accuracy increases compared to naive methods, proving that targeted architectural interventions can effectively parse complex financial structures [287].

The efficacy of these systems is heavily dependent on the quality of the retrieval component, particularly when dealing with the dense and specialized vocabulary of finance. Research indicates that generic embedding models often fail to capture the semantic nuances of domain-specific queries. Consequently, the integration of domain-specific fine-tuning for both embedding models and the generator LLM has emerged as a best practice. Studies utilizing the FinanceBench dataset reveal that combining a fine-tuned embedding model with a fine-tuned LLM yields superior accuracy compared to using generic models, with the gains from fine-tuned embeddings being particularly pronounced. Moreover, employing iterative reasoning mechanisms on top of the RAG pipeline further elevates performance, enabling systems to approach human-expert quality in answering complex financial inquiries [288]. This is complemented by advanced chunking strategies that move beyond simple paragraph-level segmentation. By dissecting documents based on structural elements—such as headers, tables, and specific financial metrics—retrieval precision is markedly improved, as this approach respects the inherent hierarchy of financial reporting [289].

Regulatory compliance introduces an additional layer of complexity, requiring systems to navigate voluminous guidelines and ensure strict adherence to legal standards. While often associated with

pharmaceutical protocols, the principles of the Question and Answer Retrieval Augmented Generation (QA-RAG) model are directly transferable to financial regulation where precision is paramount, demonstrating that specialized RAG architectures can significantly outperform conventional methods in accuracy [290]. The need for multi-perspective retrieval is also critical in legal and compliance contexts, where a single query may require viewing information through various regulatory lenses. The MVRAG framework addresses this by utilizing intention-aware query rewriting from multiple domain viewpoints, thereby enhancing retrieval precision and the reliability of the final inference in knowledge-dense domains like law and medicine, which share structural similarities with financial regulation [291].

Beyond extraction and compliance, MM-RAG facilitates deeper analytical insights, such as sentiment analysis and topic modeling within financial disclosures. Fine-tuned models, such as Llama 2 adapted for financial news, can perform multitask analysis including sentiment extraction and entity recognition, providing structured outputs that serve as predictive features for quantitative models [292]. Additionally, topic modeling techniques like Latent Dirichlet Allocation have been applied to thousands of regulatory filings to categorize content and evaluate stock market reactions to specific disclosure topics, revealing that different types of news stories elicit statistically significant but varied market responses [293].

However, the deployment of these systems is not without risks, mirroring the safety concerns found in clinical settings. The integrity of the retrieval database is paramount, as vulnerabilities to retrieval poisoning attacks could lead to malicious responses in critical financial applications. Attackers can craft visually indistinguishable documents to mislead RAG systems, highlighting the need for robust verification protocols in financial MM-RAG deployments [278]. Furthermore, privacy concerns regarding the leakage of proprietary financial data from the retrieval datastore necessitate careful system design to prevent unauthorized data extraction via prompt injection [294]. Despite these challenges, the convergence of fine-tuning, multi-view retrieval, and hybrid query mechanisms establishes a robust foundation for the next generation of financial intelligence tools. Just as healthcare moves toward autonomous agents for clinical dialogue, the financial sector is poised to leverage these advancements for more accurate, transparent, and compliant decision-making processes in global markets.

6.2 Clinical Decision Support and Multimodal EHR Analysis

6.2 Clinical Decision Support and Multimodal EHR Analysis

Bridging the gap between the quantitative rigor of financial intelligence and the imperative for human safety, the domain of clinical decision support presents a uniquely high-stakes frontier for Multimodal Retrieval-Augmented Generation (MM-RAG) systems. Just as financial ecosystems grapple with heterogeneous data, healthcare is characterized by the complex integration of Electronic Health Records (EHRs), which encompass longitudinal time-series vital signs, tabular laboratory results, unstructured narrative notes, and medical imaging. Unlike traditional unimodal approaches that analyze these modalities in isolation, MM-RAG frameworks are designed to unify them, recognizing that patient health trajectories are inherently multimodal. Relying on a single data source often yields an incomplete clinical picture, leading to suboptimal predictive performance. Recent advancements demonstrate that effectively fusing these diverse data streams is critical for enhancing the accuracy of mortality prediction, disease progression modeling, and medication safety protocols, setting a precedent for the precision required in subsequent domains like legal reasoning.

A primary challenge in deploying MM-RAG for EHR analysis lies in navigating the structural and semantic heterogeneity of medical data. Structured elements, such as ICD codes and lab values, exist in rigid tabular formats, whereas clinical notes contain rich, unstructured semantic information that is often noisy and verbose. To address this disconnect, researchers have developed sophisticated fusion architectures that transcend simple concatenation. For instance, the MINGLE framework employs a two-level infusion strategy to integrate medical concept semantics from clinical notes with structured data using hypergraph neural networks, effectively learning complex interactions between different data types to generate robust visit representations [295]. Similarly, the MedM-PLM approach utilizes a cross-modal module within a transformer architecture to explicitly model the intrinsic interactions between structured clinical codes and unstructured narratives, acknowledging that these modalities are documented during the same encounter and inform one another [296]. Furthermore, handling the temporal irregularity inherent in ICU data—where measurements are taken at varying frequencies—is crucial. Frameworks like those proposed in [297] introduce gating mechanisms and time-attention layers to dynamically incorporate irregular time-series data alongside clinical note sequences, ensuring that the temporal dynamics of patient deterioration are accurately captured.

Beyond mere data fusion, the incorporation of external medical knowledge via Knowledge Graphs (KGs) serves as a cornerstone for reliable MM-RAG systems in healthcare, mirroring the need for verified evidence seen in regulatory compliance. Hallucinations in generative models pose severe risks in clinical settings, potentially leading to diagnostic errors or unsafe medication recommendations. To mitigate this, MM-RAG systems increasingly leverage KGs to ground generation in verified medical facts. The REALM framework exemplifies this by utilizing a retrieval-augmented approach where an LLM extracts task-relevant entities from long-context notes and aligns them with a professionally labeled external knowledge graph, PrimeKG. This alignment ensures consistency with clinical standards and significantly reduces hallucinations in mortality and readmission predictions [298]. Similarly, the KindMed framework induces knowledge from myriad external sources to construct medical knowledge graphs, which are then fused with patient historical admissions via hierarchical sequence learning to recommend precise and safe medication combinations [299]. By bridging the gap between patient-specific data and generalized medical ontologies, these systems provide an explainable diagnostic pathway that enhances trust among healthcare providers [300].

The mitigation of hallucinations is further reinforced through advanced reasoning frameworks and self-reflection mechanisms, establishing a safety paradigm that parallels the strict adherence required in legal precedents. While standard LLMs may generate plausible but factually incorrect summaries, RAG-driven approaches combined with self-critique loops offer a more robust solution. Studies have shown that RAG-LLM based Clinical Decision Support Systems (CDSS) significantly boost the accuracy of medication error identification, particularly when operated in a “co-pilot” mode alongside human experts [301]. Moreover, the JMLR framework introduces a novel joint training mechanism where the retrieval model and the LLM are trained synchronously. This approach enhances the system’s ability to retrieve relevant clinical guidelines and reason over them, outperforming conventional pre-training methods and reducing the computational resources required for high-fidelity medical question answering [302]. The benchmark MIRAGE has further validated that combining diverse medical corpora with optimized retrievers can elevate the performance of smaller models to match state-of-the-art large models, while simultaneously mitigating the “lost-in-the-middle” effect often seen in long-context processing [303].

Despite these advancements, significant challenges remain regarding data privacy and the handling

of missing modalities, issues that resonate with the data security concerns in finance and the evidentiary completeness required in law. In real-world scenarios, not all patients have complete records across all modalities. Frameworks like DrFuse tackle this by disentangling shared and unique features across modalities, allowing for effective prediction even when certain data types, such as medical images, are missing [304]. Additionally, privacy concerns necessitate the development of decentralized learning approaches and synthetic data generation techniques. Methods such as PromptEHR utilize prompt learning to generate high-fidelity synthetic EHRs that preserve longitudinal patterns and cross-modality connections while minimizing privacy risks [305]. As the field evolves, the focus is shifting towards autonomous agents capable of active document exploration and multi-turn reasoning. This progression ensures that MM-RAG systems not only retrieve information but also engage in dynamic, evidence-based clinical dialogue, laying a foundational framework for the rigorous, evidence-backed reasoning systems discussed in the legal domain.

6.3 Legal Precedent Retrieval and Case-Based Reasoning

6.3 Legal Precedent Retrieval and Case-Based Reasoning

Following the critical need for factual grounding in clinical decision support, the legal domain presents a parallel imperative for precision, where the integration of Multimodal Retrieval-Augmented Generation (MM-RAG) marks a paradigm shift from simple keyword matching to sophisticated, evidence-backed reasoning systems. At the heart of this transformation lies the adaptation of Case-Based Reasoning (CBR), a methodology that solves new problems by retrieving and adapting solutions from similar past cases. In the context of MM-RAG, CBR provides a structured framework for indexing vast legal corpora, evaluating similarity across multi-dimensional feature spaces, and generating arguments that are strictly grounded in judicial precedent. The synergy between CBR’s explicit reliance on historical cases and RAG’s generative capabilities addresses the critical need for interpretability and factual accuracy in legal question answering, mirroring the safety constraints observed in healthcare applications [306].

A fundamental challenge in legal retrieval is defining “relevance,” which extends beyond textual semantic similarity to include alignment in legal facts, statutory applications, and judicial reasoning. Traditional retrieval methods often struggle with the length and structural complexity of legal documents, where key determinants of relevance may be buried within hundreds of tokens. To overcome this, advanced MM-RAG pipelines incorporate multi-view retrieval strategies that analyze cases from diverse perspectives, such as legal facts, dispute focuses, and statutory laws. This approach is exemplified by datasets and frameworks that utilize multi-view similarity measurements to capture the full complexity of legal cases, ensuring that retrieved precedents share not just surface-level text but deep legal elements [307]. Furthermore, the introduction of multi-view RAG frameworks tailored for knowledge-dense domains leverages intention-aware query rewriting to enhance retrieval precision, allowing the system to simulate the multi-perspective analysis performed by human legal experts [291].

To address the intricate connectivity within legal corpora, structured case indexing has evolved to explicitly model relationships such as case-to-case references and shared legal charges. Graph-based approaches, such as inductive graph learning models, construct global case graphs to represent these semantic and legal relationships, enabling the retrieval system to exploit the network structure of case law rather than treating documents as isolated entities [308]. Complementing this, causal learning frameworks have been proposed to account for the mediation effect of law articles on case matching decisions. By treating law article embeddings as instrumental variables, these models

decompose case representations into law-related and law-unrelated components, thereby improving the fidelity of similarity judgments in accordance with legal logic [309]. Additionally, structure-aware pre-trained language models have been developed to capture long-distance dependencies and key legal elements within document structures, significantly outperforming general-purpose models in distinguishing relevant precedents [310].

The generation phase in legal MM-RAG benefits immensely from these rigorous retrieval mechanisms. By retrieving a set of highly similar source cases, the system can enforce similarity between the query’s components and the evidence base, leading to higher quality generated answers [306]. To ensure the generated arguments are not only fluent but also legally sound, recent frameworks incorporate logic rules and neuro-symbolic reasoning. For instance, neural-symbolic enhanced legal case retrieval systems learn case-level and law-level logic rules to provide faithful, interpretable explanations for why specific cases were retrieved and how they support the generated conclusion [311]. This explainability is further enhanced by methods that extract rationales using inverse optimal transport, aligning sentences based on both semantics and legal characteristics to produce transparent matching results [312].

Query reformulation plays a pivotal role in bridging the gap between user intent and the dense language of legal texts. Large Language Models (LLMs) are increasingly employed to identify and extract salient content from long query cases, effectively filtering out noise and focusing retrieval models on key legal features [313]. Techniques such as prompt-based input reformulation define legal facts and issues as key features, encoding them in a way that preserves global context while respecting model input limits [314]. Moreover, unsupervised approaches leveraging event extraction have demonstrated state-of-the-art performance in prior case retrieval, offering a scalable solution for identifying relevant precedents without extensive labeled data [315].

Despite these advancements, challenges remain in handling the temporal evolution of legal principles and the adaptability of retrieved solutions. Frameworks like PILOT address the temporal shift in case law by incorporating modules specifically designed to handle the evolution of legal contexts over time, ensuring that retrieved precedents are current and applicable [316]. Furthermore, adaptation-guided retrieval mechanisms utilize metric Markov Random Fields to estimate the adaptability of solutions, enlarging the set of retrievable cases that can be effectively revised to solve the target problem [317]. The integration of these sophisticated retrieval and reasoning components into MM-RAG pipelines marks a significant step toward autonomous legal agents capable of constructing robust, evidence-backed arguments. As the field progresses, these systems lay the groundwork for handling even more complex, multi-modal scientific data, transitioning from structured legal precedents to the synthesis of diverse experimental results and visual data found in scientific literature.

6.4 Scientific Literature Synthesis and Materials Discovery

6.4 Scientific Literature Synthesis and Materials Discovery

Building on the structured, evidence-backed reasoning frameworks established in legal Case-Based Reasoning, the application of Multimodal Retrieval-Augmented Generation (MM-RAG) in scientific research represents a paradigm shift in how researchers navigate, synthesize, and extract knowledge from the exponentially growing corpus of scientific literature. While legal domains rely on precedent and statutory logic, scientific domains such as biomedical research and materials science face unique challenges driven by the sheer volume of data and the critical complexity of multimodal information embedded in charts, diagrams, and experimental results. Traditional search engines and standard

Large Language Models (LLMs) often struggle with the “long-tail” nature of scientific knowledge, where rare but critical discoveries are obscured by an abundance of common concepts, and the inability to reason over non-textual data leads to incomplete or hallucinated insights. MM-RAG systems address these challenges by integrating advanced retrieval mechanisms with generative capabilities, enabling precise navigation through dense literature and high-fidelity data distillation, thereby transitioning from the structured retrieval of legal precedents to the synthesis of diverse experimental results and visual data.

A primary challenge in scientific literature synthesis is the information overload problem, particularly in biomedical research where new associations between entities like drugs, genes, and diseases are constantly emerging. Standard embedding-based retrieval methods often fail to capture these long-tail connaissances because they tend to cluster around over-represented concepts, neglecting rare but pivotal findings. To mitigate this, researchers have developed graph-enhanced retrieval strategies that leverage knowledge graphs to downsample dominant clusters and surface obscure connections. For instance, the integration of biomedical knowledge graphs with LLMs has demonstrated a substantial improvement in retrieving relevant context for complex queries, effectively empowering open-source models to tackle domain-specific questions with accuracy comparable to proprietary systems [318; 319]. Furthermore, frameworks like HyKGE utilize hypothesis generation to extend exploration directions within knowledge graphs, optimizing the interaction between LLMs and retrieved knowledge to ensure diverse and reliable medical responses [320]. These approaches highlight the necessity of moving beyond simple semantic similarity to structured, graph-based reasoning that can handle the intricate dependencies inherent in scientific data, mirroring the multi-view and graph-based strategies seen in legal contexts but adapted for biological networks.

In the realm of materials science, the stakes for accuracy are even higher, as hallucinations in property prediction or synthesis procedures can lead to irreproducible experimental results and wasted resources. LLMs inherently lack long-term memory and often possess outdated or mixed knowledge regarding material properties, leading to significant errors in tasks such as predicting bandgaps or formation energies. The LLaMP framework exemplifies a robust MM-RAG solution designed specifically for this domain. By employing multiple data-aware reasoning-and-acting agents that dynamically interact with computational databases like the Materials Project, LLaMP can fetch real-time data, process higher-order structures such as crystal lattices, and summarize multi-step synthesis procedures without requiring extensive fine-tuning. This approach has been shown to drastically reduce hallucination rates, correcting massive percentage errors in formation energy predictions and eliminating fabricated structural strains, thereby establishing a nearly hallucination-free pathway for exploring materials informatics [321]. The success of such systems underscores the critical role of grounding generative models in authoritative, up-to-date external data sources to ensure scientific reproducibility, a principle that becomes even more stringent in the highly specialized technical standards discussed in subsequent sections.

Beyond text and structured databases, scientific literature is replete with visual data, including complex charts, phase diagrams, and microscopy images, which contain high-fidelity experimental results that are often lost in text-only processing pipelines. Effective MM-RAG systems must therefore possess the capability to parse and reason over these visual modalities, distinguishing them from the text-heavy documents of law or the rigid hierarchies of technical standards. The development of foundation models like BiomedCLIP, pretrained on millions of scientific image-text pairs, provides a crucial backbone for understanding the semantic alignment between biomedical images and their textual descriptions. Such models enable the retrieval of relevant visual evidence to support generated answers, bridging the gap between visual experimental data and natural lan-

guage queries [322]. Additionally, specialized agents like PaperQA have been designed to perform information retrieval across full-text scientific articles, assessing the relevance of sources and providing answers with proper provenance. These agents excel in benchmarks requiring the synthesis of information from multiple papers, matching the performance of expert human researchers in navigating complex scientific landscapes [323].

The extraction of structured data from unstructured scientific text remains a bottleneck for building comprehensive knowledge bases. While general-purpose LLMs show promise, they often struggle with the nuanced syntax of chemical formulas and physical properties compared to specialized models. However, recent evaluations suggest that with appropriate prompting strategies or fine-tuning, LLMs can achieve high precision in Named Entity Recognition and Relation Extraction tasks, facilitating the automated construction of materials databases from literature [324]. Complementary efforts like the Functional Materials Knowledge Graph (FMKG) demonstrate how LLMs can be utilized to extract millions of entities from decades of research papers, organizing unstructured information into structured triples that cover synthesis, properties, and applications, thus laying the groundwork for more sophisticated text-mining systems [325].

Furthermore, the ability to handle multi-hop queries is essential for scientific discovery, as answers often require synthesizing information dispersed across multiple documents or data modalities. Existing RAG systems frequently falter when faced with such complexity, necessitating the development of benchmarks and frameworks specifically designed to evaluate and enhance multi-hop reasoning capabilities. The introduction of datasets like MultiHop-RAG has highlighted the inadequacy of current systems in retrieving and reasoning over multiple pieces of supporting evidence, driving the community toward more advanced iterative retrieval and reasoning architectures [326]. Similarly, in materials discovery, the vision of “self-driving laboratories” relies on MM-RAG agents that can not only retrieve knowledge but also formulate hypotheses, design experiments, and interpret results in a continuous loop, effectively acting as autonomous collaborators in the scientific process [327].

In conclusion, the integration of MM-RAG in scientific literature synthesis and materials discovery is transforming the research workflow by enabling precise retrieval of long-tail knowledge, robust reasoning over multi-hop queries, and accurate distillation of data from both text and visual modalities. By combining graph-based retrieval, dynamic agent interactions, and multimodal foundation models, these systems are overcoming the limitations of static LLMs, reducing hallucinations, and accelerating the pace of innovation in critical scientific domains. As these technologies mature, they promise to unlock new frontiers in biomedical understanding and materials design. However, while broad scientific domains benefit from these general advancements, highly specialized verticals—such as telecommunications standards and traditional medicine—present distinct challenges regarding jargon density and structural rigidity, necessitating the niche adaptation strategies explored in the following section.

6.5 Technical Standards and Niche Domain Adaptation

6.5 Technical Standards and Niche Domain Adaptation

While the previous section highlighted how MM-RAG accelerates discovery in broad scientific domains like biomedicine and materials science, its application in highly specialized verticals—such as telecommunications standards and traditional medicine—presents a distinct set of challenges. These niche domains are characterized by dense technical jargon, rigid hierarchical structures, rapidly evolving knowledge bases, and a scarcity of high-quality training data suitable for stan-

dard vector embedding models. Consequently, the conventional reliance on dense vector similarity, often effective in general contexts, proves insufficient here, necessitating innovative adaptation strategies that prioritize semantic precision and structural fidelity over broad contextual matching.

A primary obstacle in these domains is the unsuitability of generic vector embeddings for capturing the nuanced relationships inherent in specialized corpora. In fields like traditional medicine, where concepts are deeply rooted in historical texts and specific cultural contexts, standard embedding models trained on general web data fail to correlate document relatedness with human expert assessment. Research has demonstrated that in such niche domains, document embeddings often correlate more strongly with simple token overlaps rather than true semantic similarity, leading to poor retrieval performance. To address this, novel approaches such as Prompt-RAG have been proposed, which operate without the need for vector embeddings entirely. Instead, this paradigm leverages natural language prompts to guide the retrieval process, effectively bypassing the limitations of generic embedding spaces. Empirical evaluations in Korean Medicine have shown that such prompt-based methods can outperform both conventional vector-based RAG systems and standalone large language models in terms of relevance and informativeness, highlighting the critical need for embedding-free or hybrid retrieval mechanisms in specialized verticals [328].

The telecommunications sector offers another compelling case study for niche domain adaptation, particularly due to the intricate nature of technical standard documents, such as those produced by the 3rd Generation Partnership Project (3GPP). These documents are not only highly technical but also evolve rapidly, with frequent updates and inter-dependent specifications that create a dynamic knowledge landscape. Standard RAG pipelines often struggle to keep pace with these changes or to correctly interpret the rigid logic and precise terminology required for compliance and engineering tasks. The Telco-RAG framework was specifically developed to navigate these challenges, offering an open-source solution tailored to the idiosyncrasies of telecommunications standards. By addressing the specific structural and semantic requirements of telecom documentation, Telco-RAG provides a blueprint for implementing robust RAG systems in other technical domains where precision and up-to-date information are paramount [329].

Furthermore, the issue of “less popular” or low-frequency knowledge is pervasive in technical standards and niche sciences. Large language models typically exhibit a performance drop when dealing with entities or concepts that appear infrequently in their pre-training corpora. While fine-tuning on synthetic data can offer some improvements, retrieval-augmented generation remains a superior strategy for handling these long-tail concepts, provided the retrieval mechanism is adequately optimized. Studies comparing fine-tuning and RAG approaches indicate that while fine-tuning boosts performance across various popularity levels, RAG surpasses other methods when equipped with advanced retrieval and data augmentation techniques. This suggests that for niche domains, the focus should remain on enhancing the retrieval component to effectively surface rare but critical technical details [330].

Handling rapidly evolving technical corpora also demands strategies that go beyond static indexing. In environments where documents are continuously updated, such as regulatory standards or emerging scientific fields, the ability to incrementally update knowledge without catastrophic forgetting is essential. The challenge is compounded by the fact that standard dense retrieval methods may suffer from the “lost-in-the-middle” effect or fail to distinguish between outdated and current specifications. Advanced frameworks that incorporate self-reflection loops and domain-specific instruction sets have shown promise in mitigating these issues. For instance, systems designed for biomedical domains, which share similar characteristics of complexity and rapid evolution with technical standards, utilize customized reflective tokens to assess generated explanations and retrieve

domain-specific documents dynamically. This self-reflective capability ensures that the system acts more like a domain expert, verifying the consistency of retrieved information against the query’s specific constraints [331].

Additionally, the structural complexity of technical documents often requires retrieval mechanisms that can understand hierarchical relationships and entity dependencies. Just as organizational charts in corporate settings require nested understanding, technical specifications often follow strict tree-like hierarchies. Representing entities as a tree structure can significantly augment the context provided to the generator. The Tree-RAG (T-RAG) approach exemplifies this by using a tree structure to represent entity hierarchies, generating textual descriptions that enrich the context for queries pertaining to specific entities within the organization. This method has been shown to perform better than simple RAG or fine-tuning implementations in real-world enterprise deployments, suggesting that incorporating structural metadata is a vital strategy for niche domain adaptation [332]. This structural awareness serves as a crucial bridge to the broader enterprise applications discussed next, where similar hierarchical complexities exist within multicultural organizational frameworks.

Finally, the integration of multi-view retrieval strategies is crucial for domains where a single perspective is insufficient to capture the full scope of a technical problem. In knowledge-dense fields like law and medicine, which share traits with technical standards, multi-view RAG frameworks utilize intention-aware query rewriting from multiple domain viewpoints to enhance retrieval precision. By considering different semantic forms and specific domain knowledge perspectives, these systems can uncover relevant information that single-view retrievers might miss, thereby improving the interpretability and reliability of the generated responses [291]. As MM-RAG systems continue to penetrate these specialized verticals, the convergence of prompt-based retrieval, structural awareness, and dynamic adaptation strategies will be essential for overcoming the inherent limitations of generic models. These advancements lay the necessary groundwork for the next frontier: scaling MM-RAG across the diverse linguistic and structural landscapes of global enterprise knowledge management.

6.6 Enterprise Knowledge Management and Multicultural Contexts

6.6 Enterprise Knowledge Management and Multicultural Contexts

Building upon the structural awareness and dynamic adaptation strategies identified in niche technical domains, the deployment of Multimodal Retrieval-Augmented Generation (MM-RAG) systems within corporate environments represents a critical frontier for scaling these capabilities across diverse linguistic and organizational landscapes. While Section 6.5 highlighted the necessity of tree-structured representations for technical hierarchies, modern enterprises present a parallel yet distinct challenge: managing vast, heterogeneous knowledge bases that span multiple languages, complex document hierarchies, and varying levels of workforce literacy. In this context, knowledge is not merely textual; it is embedded in organizational charts, multilingual human resources manuals, financial reports with embedded visualizations, and technical documentation that often mixes structured data with free-form narrative. Effective MM-RAG implementations must therefore bridge the gap between global information availability and localized comprehension, ensuring that insights are delivered with both speed and cultural relevance. This transition from specialized verticals to broad enterprise ecosystems sets the stage for understanding the universal robustness and security challenges discussed in the subsequent section.

A primary concern in multinational corporations is the integration of diverse linguistic environ-

ments, where employees interact with systems in their native languages while the underlying knowledge base may be dominated by English or other high-resource languages. Standard RAG models frequently struggle in these scenarios due to disparities in pre-training data quality and quantity across languages. To address this, recent approaches have focused on enhancing multilingual information retrieval through sophisticated data feeding strategies and cross-lingual alignment techniques. Specific implementations have demonstrated success by integrating a diverse array of tools to facilitate the seamless adoption of RAG models across languages, ensuring that information delivery is tailored to unique multicultural requirements [333]. These systems often leverage high-resource languages to augment the capabilities of dialogue generation in low-resource settings, utilizing adversarial training retrievers and fusion-in-decoder generators to align semantics across different linguistic domains [334]. Furthermore, the use of prompts augmented by retrieval cross-lingually has shown significant improvements in zero-shot performance for low-resource languages, proving that retrieving semantically similar sentences from high-resource languages can serve as effective context for generating accurate responses [335].

Beyond language, the structural complexity of enterprise documents necessitates advanced retrieval mechanisms that understand entity hierarchies and relationships, echoing the hierarchical challenges seen in technical standards but applied to corporate governance. In large organizations, entities such as departments, job roles, and product lines exist within nested tree structures that flat vector embeddings often fail to capture adequately. To overcome this limitation, innovative frameworks like Tree-RAG have been developed to represent entity hierarchies explicitly. By utilizing a tree structure to generate textual descriptions that augment the context during query processing, such systems can better navigate the intricate relationships within an organization’s hierarchy, outperforming simple RAG or fine-tuning implementations in domain-specific question answering [332]. This structural awareness is crucial for tasks involving internal document hierarchies, where understanding the relationship between a specific policy and the broader organizational unit is essential for accurate reasoning. Additionally, context tuning mechanisms have been proposed to fetch relevant information that improves both tool retrieval and plan generation, employing signals such as numerical and categorical data to rank context items effectively [336].

The operational mode of MM-RAG systems in enterprises also requires careful calibration between full autonomy and co-pilot assistance, a balance critical for maintaining trust in high-stakes corporate decisions. While autonomous agents offer the promise of rapid information synthesis, the risk of hallucination and the need for accountability often necessitate a human-in-the-loop approach. Systems designed for enterprise use must balance the efficiency of automated retrieval with the reliability of human oversight. Research indicates that validation of RAG systems is often only feasible during operation, suggesting that robustness evolves through continuous interaction rather than static design [337]. In this regard, human-in-the-loop pipelines that guide large language models with revision instructions have proven effective, allowing human feedback to serve as a supervision signal that enhances the model’s output through in-context learning [338]. This hybrid approach ensures that while the system leverages the speed of AI for initial draft generation or data retrieval, critical decisions and final outputs remain subject to human verification, thereby mitigating risks associated with erroneous responses.

Moreover, the security and privacy of proprietary enterprise data remain paramount, foreshadowing the detailed security analysis in the following section. Unlike public domain applications, corporate RAG systems often handle sensitive information that cannot be exposed to external APIs, driving a strong trend toward on-premise deployments and the use of open-source models fine-tuned on private organizational documents. Experiences from building such systems highlight that while con-

structuring a basic RAG pipeline is straightforward, achieving robustness and reliability requires deep domain customization and a thorough understanding of the specific application context [332]. The effectiveness of these systems is further influenced by the choice between fine-tuning and retrieval augmentation. Studies comparing these approaches in industrial domains, such as agriculture, suggest that while fine-tuning can embed specific geographic or domain knowledge into the model weights, RAG provides superior flexibility for updating information and handling low-frequency entities, with the combination of both yielding the highest accuracy [339].

Finally, the evaluation of enterprise MM-RAG systems must extend beyond standard accuracy metrics to include measures of faithfulness, latency, and user satisfaction across different cultural contexts. The “lost-in-the-middle” effect, where models struggle to utilize information placed in the middle of long contexts, poses specific challenges for retrieving details from lengthy corporate manuals or legal documents [303]. Therefore, optimizing chunking strategies, query expansion, and re-ranking algorithms is essential for maintaining high retrieval quality in these dense information environments [340]. As enterprises continue to digitize their knowledge assets, the development of MM-RAG systems that are culturally aware, structurally intelligent, and operationally flexible will be indispensable for fostering efficient and inclusive organizational learning. However, as these systems become more integral to critical business operations, the vulnerabilities inherent in their retrieval and generation components—ranging from data leakage to knowledge corruption—demand rigorous scrutiny, a topic explored in depth in the next section on cross-domain robustness and security.

6.7 Cross-Domain Lessons on Robustness and Security

6.7 Cross-Domain Lessons on Robustness and Security

Building upon the operational complexities and cultural nuances of enterprise deployments discussed previously, the expansion of Multimodal Retrieval-Augmented Generation (MM-RAG) systems into high-stakes verticals such as finance, healthcare, and law reveals a convergent set of security vulnerabilities and robustness challenges that transcend specific domain boundaries. While applications in these sectors differ in their data modalities and regulatory constraints, the underlying attack surfaces—ranging from data leakage to knowledge corruption—exhibit striking similarities. A primary lesson learned from cross-domain analysis is the critical vulnerability of RAG pipelines to prompt injection attacks, which blur the distinction between retrieved data and system instructions. In real-world scenarios, adversaries can exploit the instruction-following capabilities of large language models to extract sensitive information or manipulate outputs by injecting malicious content into the retrieval context. Research has demonstrated that “Follow My Instruction and Spill the Beans” attacks can achieve near-perfect success rates in extracting verbatim text from datastores, highlighting that the integration of external knowledge significantly expands the attack surface beyond the model’s parametric knowledge [294]. Furthermore, the threat of “Indirect Prompt Injection” poses a severe risk where adversaries remotely compromise applications by strategically injecting prompts into data sources likely to be retrieved, effectively allowing data to act as arbitrary code execution [341].

Beyond the direct manipulation of the generation phase, the integrity of the knowledge base itself is paramount, extending the structural concerns of enterprise hierarchies to issues of active sabotage. Cross-domain case studies underscore the prevalence of knowledge poisoning attacks, where adversaries inject a small number of corrupted documents into the retrieval corpus to force the model into generating attacker-chosen responses. The “PoisonedRAG” framework illustrates that

injecting merely five poisoned texts into a database containing millions of entries can yield attack success rates as high as 90%, rendering standard retrieval mechanisms insufficient for ensuring factual consistency [276]. This vulnerability is exacerbated in multi-agent or collaborative settings, where the “spiral of silence” or indirect propagation of malice can occur. In such environments, a single compromised agent can generate prompts that subtly influence other agents within the society to produce harmful outputs, creating a cascade of misinformation that is difficult to trace back to its origin [342]. Additionally, even minor textual inaccuracies or low-level perturbations in retrieved documents, such as typographical errors simulated by genetic attacks, can devastate the synergy between retrieval and generation components, leading to systemic failures in reasoning [279].

Data privacy remains another critical cross-domain concern, particularly in sectors utilizing federated learning or distributed retrieval architectures, which are often necessary to address the on-premise requirements noted in enterprise contexts. In vertical federated learning setups common in healthcare and finance, the risk of label leakage and feature reconstruction from shared intermediate embeddings or gradients is a persistent threat. Attacks such as “CAFE” have shown that catastrophic data leakage can occur even when large batch sizes are used as a defense, allowing adversaries to recover private training data from aggregated gradients [343]. Similarly, the vulnerability of systems to “Human-Imperceptible Retrieval Poisoning” demonstrates that attackers can craft documents that appear benign to human reviewers but are designed to mislead the retrieval mechanism, thereby compromising the application’s output without triggering obvious alarms [278]. The risk is further compounded in multi-turn interactions, where prompt leakage can expose both task instructions and contextual knowledge, with black-box defenses often struggling to reduce attack success rates to acceptable levels [344].

To build robust, domain-aware MM-RAG systems capable of withstanding these sophisticated threats, several best practices emerge from these cross-domain lessons. First, the implementation of an “Instruction Hierarchy” is essential, where models are trained to prioritize privileged system instructions over untrusted user or retrieved data, thereby mitigating the impact of prompt injections [345]. Second, robust verification protocols must be integrated to detect and neutralize poisoned retrieval contexts before they influence generation. This includes employing diversity-aware sampling and consistency checks to identify anomalies in retrieved documents. Third, privacy-preserving techniques such as local differential privacy and secure multi-party computation should be standard in distributed settings to prevent gradient leakage and ensure data sovereignty [346]. Finally, the development of comprehensive evaluation frameworks that simulate diverse attack vectors, including indirect injections and subpopulation poisoning, is crucial for assessing system resilience prior to deployment [347]. By addressing these common failure points through layered defenses and rigorous evaluation, practitioners can enhance the trustworthiness of MM-RAG systems across all domains, laying the groundwork for secure adoption in even the most sensitive applications.

7 Benchmarks, Evaluation Metrics, and Ecosystems

7.1 Comprehensive Benchmarks for Chart and Document Understanding

7.1 Comprehensive Benchmarks for Chart and Document Understanding

The rapid proliferation of Multimodal Large Language Models (MLLMs) has necessitated the development of rigorous evaluation frameworks capable of assessing their proficiency in visually rich

domains. While early benchmarks primarily focused on general object recognition or simple captioning, the current landscape demands specialized datasets that probe complex visual reasoning, fine-grained text comprehension, and the ability to synthesize information across heterogeneous modalities. This subsection surveys pivotal benchmarks that have redefined the standards for evaluating MLLMs in the specific contexts of chart analysis and document understanding. These resources establish the foundational ground truth necessary for the subsequent discussion on retrieval precision and adaptive decision-making, highlighting unique methodological contributions and addressing specific gaps in the existing literature.

A primary challenge in chart understanding lies in the discrepancy between synthetic, template-based questions and the nuanced, multi-step reasoning required in real-world data analysis. Traditional datasets often rely on metadata or direct data point annotations, which allow models to bypass true visual comprehension. To overcome these limitations, [348] introduces a comprehensive benchmark specifically designed to evaluate the data reliability and visual reasoning capabilities of MLLMs without relying on auxiliary metadata. Covering 42 categories and over 2,000 charts, [348] compels models to derive values by leveraging inherent visual elements such as color legends, coordinate systems, and spatial relationships, much like a human analyst. Furthermore, it proposes an enhanced evaluation metric, Acc+, which facilitates objective assessment without the need for costly human or LLM-based intervention, thereby revealing significant limitations in current models’ ability to interpret complex chart structures accurately.

Complementing this focus on reasoning, [349] expands the evaluation scope by constructing a multi-modal evaluation set that spans 18 chart types and 22 disciplinary topics. This benchmark is instrumental in rigorously testing the capacity of off-the-shelf MLLMs to engage in interpretable pattern recognition and domain-specific reasoning. The associated work demonstrates that specialized pre-training and reasoning strategies are essential, as general-purpose models often fail to match the performance of dedicated architectures like ChartVLM on these complicated tasks. Similarly, [177] addresses the scarcity of reasoning-heavy training data by leveraging LLMs to generate step-by-step rationales and sub-questions. This approach underscores the necessity of benchmarks that not only test final answers but also evaluate the intermediate logical steps required to solve complex numerical problems in charts, a capability that extractive models frequently lack.

Beyond standalone charts, the integration of visual data within broader document contexts presents a distinct set of challenges, particularly regarding layout complexity and cross-referencing. [4] introduces a novel task that shifts the focus from isolated chart analysis to document-level question answering. By integrating charts into 50,000 synthetic documents with diverse styles, [4] evaluates a model’s ability to first perform document layout analysis to locate relevant charts and then execute complex reasoning and common-sense inference. This benchmark is critical for assessing how well models can navigate the non-linear structure of visually rich documents and synthesize information from both textual narratives and graphical depictions, laying the groundwork for evaluating retrieval mechanisms that must identify and extract such cross-modal evidence.

In the realm of engineering and technical documentation, [350] offers a unique perspective by combining textual design requirements with CAD images and engineering drawings. Unlike benchmarks where the image and text are tightly coupled, [350] features document-grounded questions where inputs originate from different sources, simulating real-world engineering workflows. This setup rigorously tests a model’s ability to comprehend rules, ensure compliance, and extract specific requirements, revealing substantial gaps in current MLLMs’ abilities to apply detailed technical constraints to visual designs.

Furthermore, the prevalence of text-rich visual content in real-world applications, such as maps, web pages, and dense reports, requires benchmarks that specifically target optical character recognition (OCR) and text-centric visual comprehension. [351] addresses this need by introducing a benchmark comprising 2.3K multiple-choice questions across charts, maps, and webs. [351] emphasizes scenarios characterized by extensive embedded text, providing a robust evaluation of how well MLLMs can handle the interplay between dense textual information and visual layout. The evaluation of 34 prominent models on this benchmark highlights that even state-of-the-art systems struggle with the intricacies of text-rich environments, indicating a critical area for future improvement where retrieval errors could severely impact downstream generation.

Additional benchmarks further diversify the evaluation landscape by targeting specific structural and procedural nuances. [352] provides a massive dataset derived from academic papers, enabling the evaluation of multi-turn dialogues about scientific graphs, which is essential for research assistance applications. Meanwhile, [353] focuses on multipanel images, testing a model’s ability to comprehend distinct subfigures within a single layout, a common feature in posters and web screenshots. Finally, [2] targets the hierarchical semantic relations in multi-page research articles, pushing the boundaries of document understanding beyond single-page analysis. Collectively, these specialized benchmarks provide a granular view of MLLM capabilities, moving the field beyond simple accuracy metrics toward a deeper understanding of reasoning fidelity, structural awareness, and domain adaptability. However, while these benchmarks define *what* needs to be understood, they primarily evaluate end-to-end performance. To build robust Multimodal Retrieval-Augmented Generation (MM-RAG) systems, it is equally crucial to evaluate *how* information is retrieved and when external knowledge is actually necessary, a transition that leads directly to the metrics for retrieval precision and adaptive decision-making discussed in the following section.

7.2 Metrics for Retrieval Precision and Adaptive Decision-Making

7.2 Metrics for Retrieval Precision and Adaptive Decision-Making

While the preceding section established the foundational benchmarks that define *what* models must understand in chart and document contexts, it primarily focused on end-to-end performance. To build robust Multimodal Retrieval-Augmented Generation (MM-RAG) systems, it is equally critical to evaluate *how* information is retrieved and *when* external knowledge is necessary. The evaluation of retrieval components within MM-RAG systems for chart and document understanding therefore requires a paradigm shift from traditional static metrics to dynamic, context-aware protocols. In visually rich domains, the cost of retrieving irrelevant information extends beyond reduced efficiency; it acts as a significant catalyst for hallucination, where the generative model attempts to synthesize answers from non-existent or contradictory visual evidence. Consequently, modern evaluation frameworks must rigorously assess three critical dimensions: the adaptive necessity of retrieval, the quantification of uncertainty in retrieved contexts, and the robustness of the system against noisy or irrelevant documents. These dimensions serve as the bridge between the static capabilities measured in Section 7.1 and the generative fidelity analyzed in Section 7.3.

A primary challenge in MM-RAG is determining when external retrieval is actually required. Indiscriminate retrieval can introduce noise that degrades performance, particularly when the model’s parametric knowledge is sufficient. To address this, the concept of Adaptive Retrieval-Augmented Generation (ARAG) has emerged, necessitating benchmarks that evaluate a system’s ability to dynamically decide whether to query an external database. The [354] benchmark provides a crucial framework for this assessment. It comprises a dataset of questions covering new world and

long-tail knowledge where the necessary information is absent from the language model’s internal parameters, making external retrieval mandatory for correct answers. This benchmark reveals that calibration-based methods often rely heavily on threshold tuning, while standard prompting strategies frequently fail to guide Large Language Models (LLMs) in making reliable retrieval decisions. The findings from [354] underscore the need for metrics that measure not just the accuracy of the final answer, but the precision of the decision-making process regarding retrieval necessity. In the context of chart understanding, this translates to evaluating whether a system correctly identifies when a query requires analyzing a specific visual element versus when it can be answered based on general chart literacy.

Beyond the decision to retrieve, quantifying the uncertainty associated with the retrieved content is paramount for ensuring trustworthiness. In complex document layouts and dense charts, retrieval models may return passages with high similarity scores that are semantically irrelevant or contradictory. The [230] framework introduces a rigorous method for applying conformal prediction to quantify this retrieval uncertainty. By establishing a similarity score cutoff threshold based on a user-specified error rate, [230] ensures that the true answer is captured within the retrieved context with a statistically guaranteed confidence level. This approach is particularly vital for MM-RAG systems processing financial reports or scientific papers, where missing a critical data point due to low retrieval confidence can lead to severe factual errors. Furthermore, the integration of Bayesian methodologies, as discussed in [355], allows for the calibration of uncertainty within the retrieval pipeline itself. By leveraging techniques such as LIME and SHAP to analyze black-box reranker models, this approach generates importance scores that serve as supplementary relevance metrics, enhancing the interpretability and reliability of the retrieval process.

Robustness against irrelevant or noisy documents represents the third pillar of evaluation. In real-world scenarios, retrieval systems often encounter documents that are topically related but factually incorrect or entirely unrelated to the specific query intent. The [356] dataset addresses this by evaluating LLM robustness across diverse languages using two distinct subsets: one containing non-relevant passages and another with relevant ones. The metrics derived from [356], specifically the hallucination rate on non-relevant subsets and the error rate on relevant subsets, provide a comprehensive view of a model’s failure modes. Studies using this benchmark have shown that many models struggle to balance these capacities, often hallucinating answers when no relevant evidence is present or failing to recognize relevant passages when they exist. This duality is critical for chart analysis, where a model might fabricate data trends from a noisy background or miss a subtle annotation in a complex figure.

Additionally, the reliability of retrieval rankings must be assessed through the lens of uncertainty and calibration. The work presented in [357] argues that capturing a model’s belief in its own relevance scores is critical for adapting to new document distributions. By employing an efficient Bayesian framework, this research demonstrates that incorporating uncertainty information significantly improves ranking effectiveness and confidence calibration. Similarly, [358] introduces a risk control framework that utilizes a qualify model for uncertainty estimation and a decision model for selective output, moving beyond traditional exact match metrics to risk-related evaluations. These approaches suggest that future MM-RAG evaluation protocols must include metrics that penalize overconfidence in low-quality retrievals, ensuring that systems can abstain from answering or flag uncertain results when the retrieved visual or textual evidence is insufficient.

Finally, the evaluation of retrieval precision must account for the dynamic nature of document corpora and the potential for adversarial perturbations. Research such as [279] highlights the vulnerability of RAG systems to minor textual errors and noisy documents, which can devastate

system performance. This necessitates robustness metrics that test systems against synthetic disinformation and low-level perturbations common in scanned documents or poorly OCR-ed charts. Moreover, the [359] framework offers a method to predict retrieval quality without human judgments by decomposing the task into relevance subtasks, thereby enabling more scalable and interpretable evaluation of retrieval systems. Collectively, these metrics and frameworks form a robust ecosystem for evaluating the retrieval components of MM-RAG, ensuring that systems are not only accurate but also adaptive, uncertainty-aware, and resilient in the face of complex, visually rich data environments. By establishing rigorous standards for retrieval precision and decision-making, this section lays the necessary groundwork for the subsequent analysis of generation faithfulness, where the integrity of the retrieved context directly dictates the reliability of the synthesized output.

7.3 Evaluating Generation Faithfulness and Visual Grounding

7.3 Evaluating Generation Faithfulness and Visual Grounding

While the preceding section established rigorous metrics for assessing retrieval precision and adaptive decision-making, the efficacy of a Multimodal Retrieval-Augmented Generation (MM-RAG) system ultimately hinges on the fidelity of its generative output. In the domain of chart and document understanding, the consequences of retrieval errors are compounded by generation failures; even perfectly retrieved contexts can yield incorrect answers if the model fails to ground its reasoning in the visual evidence. Unlike standard text-based question answering where hallucinations may involve factual inaccuracies, errors here often manifest as a complete disconnect between the visual data and the textual narrative. Consequently, evaluating generation faithfulness and visual grounding requires a paradigm shift from simple accuracy scores to metrics that rigorously assess whether a model’s reasoning is truly anchored in the retrieved visual context. This subsection analyzes emerging frameworks designed to quantify these properties, focusing on the dual requirements of faithfulness (adherence to facts) and plausibility (logical reliance on relevant visual regions), thereby bridging the gap between reliable retrieval and trustworthy synthesis.

A foundational challenge in this area is defining what constitutes “grounded” generation, particularly when distinguishing between models that guess correctly based on language priors and those that genuinely interpret visual data. Traditional Visual Question Answering (VQA) metrics often obscure this distinction. To address this, the concept of Faithful and Plausible Visual Grounding (FPVG) has been introduced as a rigorous criterion for evaluation. FPVG assesses whether a model not only identifies question-relevant objects within a scene but also demonstrably relies on the information contained in those specific objects to formulate its answer [360]. In the context of complex financial charts or scientific diagrams, this distinction is critical; a model might correctly state a trend value but fail to ground this claim in the specific data series depicted, leading to brittle systems that collapse under slight layout perturbations. The FPVG metric provides a straightforward yet powerful mechanism to quantify this reliance, ensuring that the visual modality is not merely decorative but integral to the reasoning process, thus validating the utility of the retrieval step discussed previously.

Beyond structural grounding, the factual consistency of generated text relative to the input image remains a primary concern. Large Vision-Language Models (LVLMs) are prone to generating plausible-sounding but factually incorrect details, a phenomenon known as visual hallucination. To measure this without relying on expensive human annotations for every query, reference-free metrics have gained traction. For instance, FAITHSCORE offers a fine-grained approach by decomposing generated answers into atomic facts and verifying their consistency against the input image [361].

This method is particularly valuable for chart understanding, where a single sentence might contain multiple claims about data points, trends, and comparisons. By evaluating faithfulness at the atomic level, FAITHSCORE can pinpoint exactly where a model deviates from the visual evidence, providing a more nuanced diagnostic than binary accuracy metrics. Similarly, the VALOR-EVAL framework extends this by incorporating both faithfulness and coverage, addressing the trade-off where models might produce highly faithful but overly sparse responses [362]. This holistic view is essential for document understanding tasks where omitting key details from a dense table or figure can be as detrimental as fabricating information.

The evaluation landscape also increasingly leverages Large Language Models (LLMs) themselves as judges to assess factuality, especially when gold-standard answers are unavailable or difficult to define for open-ended document queries. The UFO framework proposes a unified approach where LLMs verify facts against various sources, including reference documents and search results, making it adaptable for Retrieval-Augmented Generation workflows [363]. In MM-RAG systems, this capability allows for the automated verification of claims made about retrieved document chunks, ensuring that the synthesized response does not contradict the source material. Furthermore, metrics like QUIP-Score have been developed to specifically measure the extent to which generated answers are directly found in underlying text corpora, encouraging models to ground their responses in explicit evidence rather than parametric memory [364]. This is particularly relevant for legal and medical document analysis, where citing the exact source of a claim is a requirement for trustworthiness and directly complements the uncertainty quantification strategies outlined in the previous section.

However, relying solely on output consistency is insufficient if the underlying reasoning process is flawed. Recent research highlights the need to evaluate the internal consistency and reasoning capabilities of models. Studies have shown that models often lack the ability to generate coherent explanations, leading to inaccurate responses even when the final answer appears correct [365]. To mitigate this, frameworks like Chain-of-Verification (CoVe) introduce a multi-step evaluation process where the model plans verification questions to fact-check its own drafts before finalizing the response [188]. While primarily a mitigation strategy, the principles of CoVe inform evaluation protocols that reward models for demonstrating self-correction and explicit verification steps. Additionally, the detection of hallucinations in domain-specific contexts, such as medical Q&A, has led to specialized benchmarks like K-QA, which utilize NLI-based metrics to approximate precision and recall in clinical information extraction [366]. These domain-aware metrics are crucial for assessing MM-RAG systems in high-stakes environments where generic faithfulness scores may overlook critical nuances, setting the stage for the deeper cognitive assessments discussed in the following section.

The complexity of visual hallucinations in multi-modal settings further necessitates specialized diagnostic suites. HallusionBench, for example, is designed to disentangle language hallucination from visual illusion by using control groups of images and questions to test logical consistency [367]. This is vital for chart understanding, where visual illusions (e.g., misleading axis scales) can trick models into generating factually wrong interpretations even if the text generation itself is fluent. Moreover, the identification of specific hallucination types, such as contextual guessing or numeric discrepancies, allows for more targeted evaluations [368]. By categorizing errors, researchers can better understand whether a failure stems from poor OCR, misalignment of visual tokens, or a lack of domain knowledge.

Finally, the robustness of these evaluation metrics against adversarial inputs and spurious correlations is an active area of investigation. Benchmarks like CorrelationQA assess how models handle

images that are highly relevant yet inconsistent with the expected answer, revealing an “instinctive bias” that leads to hallucinations [40]. Evaluating generation faithfulness in MM-RAG thus requires a multi-faceted approach that combines atomic fact verification, structural grounding analysis, self-consistency checks, and domain-specific validity tests. As the field evolves, the integration of these diverse metrics into unified evaluation ecosystems will be essential for developing chart and document understanding systems that are not only accurate but also trustworthy and transparent in their reasoning. This focus on the integrity of the generated output provides the necessary foundation for the subsequent exploration of hierarchical reasoning and higher-order cognitive capabilities, where the complexity of tasks demands not just factual correctness, but deep logical synthesis.

7.4 Assessing Hierarchical Reasoning and Cognitive Capabilities

7.4 Assessing Hierarchical Reasoning and Cognitive Capabilities

Building upon the rigorous assessment of generation faithfulness and visual grounding established in the preceding section, it becomes evident that factual consistency alone is insufficient for high-stakes chart and document understanding. While Section 7.3 ensures that a model’s output is anchored in retrieved visual evidence, it does not fully evaluate the depth, structure, or logical sophistication of the reasoning process itself. A system may produce a faithful summary of a single data point yet fail to synthesize complex trends across multiple charts or deduce causal relationships within a technical document. Consequently, the evaluation paradigm must expand from verifying “what” is generated to analyzing “how” the model arrives at its conclusions. This subsection explores advanced frameworks designed to probe hierarchical comprehension, sequential algorithmic logic, and deep domain expertise, moving beyond binary correctness to assess the structural integrity of multimodal reasoning.

A pivotal development in this direction is the adaptation of educational taxonomies to benchmark visual-language models, most notably through the lens of Bloom’s Taxonomy. Traditional Visual Question Answering (VQA) datasets often prioritize lower-level cognitive skills such as remembering and understanding, neglecting the critical abilities to analyze, evaluate, and create. The [369] dataset explicitly addresses this gap by constructing a hierarchical graph representation of picture stories that map to different levels of comprehension. This framework allows for a graded evaluation of model performance, revealing that while state-of-the-art models like GPT-4V demonstrate improved accuracy across all levels, they exhibit a significant performance drop—up to 38.0%—when tasked with advanced comprehension requiring synthesis and evaluation. Furthermore, [369] highlights a concerning tendency in current models to bypass visual inputs entirely for higher-level tasks, relying instead on parametric knowledge or linguistic priors, which undermines the core premise of multimodal grounding. This finding underscores the necessity of benchmarks that enforce strict adherence to visual evidence while demanding complex cognitive operations, a requirement that is paramount for MM-RAG systems interpreting dense financial reports or scientific diagrams where superficial pattern matching is inadequate.

Complementing hierarchical comprehension is the imperative to assess sequential and algorithmic reasoning, particularly when documents contain procedural information or charts depicting dynamic processes. Standard static evaluation protocols often fail to test a model’s ability to maintain context over a sequence of dependent decisions, a capability essential for navigating non-linear table structures or following flowchart logic branches. The [370] introduces an interactive evaluation protocol designed to test sequential reasoning in algorithmic contexts, such as depth-first search or binary search. In this setup, the availability of information is contingent upon the model’s previous

actions, necessitating robust memory retention and strategic planning. The findings from [370] reveal that closed-source models generally outperform open-source counterparts in these sequential tasks, yet even top-tier models struggle with consistency when the reasoning chain extends beyond a few steps. For MM-RAG applications, where extracting data might require iteratively refining queries based on intermediate findings, such sequential reasoning capabilities are indispensable. The benchmark also notes that naively providing interactive examples can inadvertently hurt few-shot performance, suggesting that the integration of sequential reasoning into RAG pipelines requires careful prompt engineering and architectural support rather than simple context concatenation.

Beyond structural and sequential logic, the assessment of graduate-level domain expertise represents the frontier of cognitive evaluation for AI systems. In fields like biology, physics, and chemistry, the ability to reason over complex charts and technical documentation requires a depth of knowledge that surpasses general web-scraped corpora. The [371] offers a rigorous testbed for this capability, comprising questions written by domain experts that are deliberately designed to be “Google-proof.” The difficulty of [371] is evidenced by the fact that highly skilled non-expert validators achieve only 34% accuracy despite unrestricted web access, while state-of-the-art AI systems struggle to reach 39%. This benchmark is crucial for MM-RAG because it simulates real-world scenarios where the retrieved context may contain highly specialized information that requires deep synthesis rather than surface-level extraction. The gap between expert human performance (65-74%) and current AI capabilities highlights a significant challenge: ensuring that retrieval-augmented systems do not hallucinate plausible-sounding but factually incorrect interpretations of complex scientific data. The integration of such high-difficulty benchmarks into the evaluation ecosystem forces the development of more robust verification mechanisms and better alignment between retrieved evidence and generative outputs, bridging the gap between factual grounding and true expert-level reasoning.

Contrasting these advanced frameworks with standard accuracy metrics reveals profound discrepancies in how model intelligence is measured. Traditional metrics often yield saturated scores on easy tasks, masking fundamental deficiencies in logical consistency and causal reasoning. For instance, models may achieve high accuracy on simple fact-retrieval questions while failing completely on tasks requiring the composition of multiple atomic skills or the understanding of feasibility constraints. The limitations of standard evaluation are further illuminated by studies showing that models often lack the ability to spontaneously generalize atomic skills to complex reasoning tasks without specific training strategies like hierarchical curriculum learning [372]. Moreover, the reliance on static datasets can lead to overfitting on spurious correlations, whereas interactive and hierarchical benchmarks like those discussed above provide a more diagnostic view of model failures. By adopting frameworks that explicitly target hierarchical comprehension, sequential logic, and domain expertise, the community can better identify the specific cognitive bottlenecks hindering the deployment of MM-RAG systems. However, even a model capable of deep reasoning and hierarchical synthesis remains vulnerable if it lacks foundational reliability against adversarial inputs or distributional shifts. Thus, establishing cognitive depth is a necessary precursor to, but not a substitute for, the rigorous stress-testing of robustness and bias addressed in the following section.

7.5 Robustness, Bias, and Adversarial Evaluation

7.5 Robustness, Bias, and Adversarial Evaluation

While the preceding section highlighted the necessity of evaluating hierarchical reasoning and cognitive depth, high-level logical capabilities are insufficient if a system lacks foundational reliability. The deployment of Multimodal Retrieval-Augmented Generation (MM-RAG) systems in high-stakes domains such as healthcare, finance, and legal analysis necessitates a rigorous evaluation framework that extends beyond standard accuracy and cognitive metrics to address vulnerability. Traditional benchmarks often fail to capture a system’s susceptibility to adversarial inputs, inherent dataset biases, and the distributional shifts encountered in real-world scenarios. Consequently, robustness, bias, and adversarial evaluation have emerged as critical pillars for ensuring the trustworthiness of multimodal understanding systems. This subsection synthesizes current methodologies and benchmarks designed to expose these weaknesses, drawing heavily from advancements in general machine learning and visual question answering that are directly applicable to the complex reasoning required in chart and document understanding.

A primary concern in the adversarial evaluation of multimodal systems is their susceptibility to deceptive prompts and hallucinated reasoning, which can undermine even models with strong hierarchical capabilities. Models trained to align visual and textual modalities can be easily misled by prompts containing false premises or non-existent visual elements, leading to confident but factually incorrect generations. To quantify this vulnerability, researchers have introduced specialized benchmarks such as MAD-Bench, which curated 850 test samples across categories like non-existent objects, spatial relationships, and visual confusion to empirically assess model resilience [373]. The findings from such studies reveal a stark performance gap; while state-of-the-art models like GPT-4V demonstrate moderate robustness, many open-source and instruction-tuned models suffer catastrophic failures, with accuracy plunging to between 5% and 35% on deceptive inputs [373]. This highlights a critical deficiency in current training paradigms, where models prioritize pattern matching over genuine visual grounding, a flaw that is particularly dangerous in MM-RAG systems where retrieved evidence might be subtle or easily misinterpreted under adversarial pressure.

Beyond adversarial prompting, the issue of bias in dataset construction poses a fundamental threat to the fairness and generalizability of multimodal models, complicating the assessment of domain expertise discussed previously. Biases often arise from spurious correlations in training data, causing models to rely on shortcut features rather than learning robust semantic representations. The introduction of datasets like Biased MNIST has been instrumental in illustrating how models exploit hidden biases, failing to generalize when tested against multiple forms of bias simultaneously [374]. Research indicates that many bias mitigation algorithms are ineffective because they are overly sensitive to the choice of tuning set and often fail to scale across diverse bias sources [374]. Furthermore, conventional approaches that down-weight biased instances based on the assumption that the main model behaves similarly to a biased proxy have been shown to be flawed, potentially wasting valuable training data without addressing the root cause of the bias [375]. In the context of document and chart understanding, such biases could manifest as systematic errors in interpreting specific demographic groups in medical records or financial trends in specific market sectors, necessitating more nuanced metrics that go beyond simple demographic breakdowns to include outlier-based identification of harmed populations [376].

The reliability of MM-RAG systems is further compromised by the discrepancy between static benchmark distributions and the dynamic, often unpredictable nature of production data, a phenomenon known as context shift. This challenge parallels the need for dynamic adaptation mech-

anisms explored in advanced psychometric approaches. Standard evaluation protocols frequently underestimate model brittleness because they do not account for semantically meaningful changes in the underlying data generation process [377]. To address this, the community has moved towards Out-of-Distribution (OOD) testing, which evaluates model performance on data that diverges significantly from the training distribution. Benchmarks like WILDS provide a curated collection of datasets reflecting real-world distribution shifts, such as changes in hospital protocols for tumor identification or varying camera traps for wildlife monitoring, demonstrating that standard training methods yield substantially lower OOD performance [378]. Similarly, in the domain of tabular and structured data often found in documents, the Wild-Tab benchmark reveals that many sophisticated OOD generalization methods struggle to maintain performance on unseen industrial data, with simple Empirical Risk Minimization sometimes proving more robust [379].

Moreover, the evaluation of robustness must consider the specific failure modes of models under various types of distribution shifts, not just unknown classes. The BROAD benchmark categorizes five distinct types of distribution shifts and exposes the inconsistency of existing OOD detection methods, which often only detect anomalies they were explicitly designed to expect [380]. This limitation is particularly pertinent for MM-RAG systems that must navigate evolving document formats and novel chart types. Additionally, the integrity of evaluation itself is under scrutiny, with studies highlighting how data contamination in benchmarks can inflate performance scores, leading to a false sense of security regarding model capabilities [381]. The practice of using OOD test sets for model selection or relying on explicit knowledge of OOD split construction, as seen in some VQA benchmarks, further undermines the validity of generalization claims [382].

Finally, a holistic assessment of reliability requires integrating multiple dimensions of robustness, including adversarial robustness, calibration, and detection capabilities, into a unified framework. Recent proposals suggest evaluating systems across five key properties: in-distribution accuracy, distribution-shift robustness, adversarial robustness, calibration, and OOD detection, arguing that optimizing for one metric does not necessarily constrain others [383]. This comprehensive approach serves as a vital bridge to the advanced psychometric and qualitative paradigms discussed in the following section, ensuring that the metrics used are not only rigorous but also reflective of real-world constraints. By adopting rigorous benchmarks like MAD-Bench, Biased MNIST, and WILDS, and by adhering to strict evaluation protocols that prevent data leakage and account for context shifts, the research community can better ensure that multimodal understanding systems are not only accurate and cognitively capable but also robust, fair, and trustworthy in the face of operational uncertainties.

7.6 Advanced Evaluation Paradigms and Psychometric Approaches

7.6 Advanced Evaluation Paradigms and Psychometric Approaches

While Section 7.5 established the critical necessity of robustness against adversarial inputs, biases, and distribution shifts, ensuring foundational reliability is only the first step toward truly trustworthy Multimodal Retrieval-Augmented Generation (MM-RAG) systems. High-level logical capabilities and resistance to failure modes must be measured with equal rigor. Consequently, the field is undergoing a paradigm shift from static, single-metric benchmarks toward advanced evaluation frameworks that incorporate psychometric principles, qualitative diagnostics, and dynamic adaptation mechanisms. These emerging approaches aim to transcend simple performance ranking, offering actionable insights into the latent cognitive traits of models and ensuring that MM-RAG systems for chart and document understanding are not merely accurate, but scientifically validated,

interpretable, and adaptable to the non-stationary nature of real-world data.

A cornerstone of this evolution is the integration of psychometrics—the science of measuring latent psychological traits—into AI benchmarking to address the theoretical vacuums of traditional tests. Conventional benchmarks often lack a rigorous foundation for comparing model performance against human populations or for assessing the validity and difficulty of individual test items. To bridge this gap, the **PATCH** framework introduces a novel methodology for Psychometrics-Assisted benchmarking, leveraging established theories to measure latent constructs such as mathematical proficiency or visual reasoning ability [384]. By anchoring evaluations in item difficulty and population references, **PATCH** reveals that standard practices can yield divergent conclusions compared to psychometrics-based assessments, thereby necessitating a more scientifically grounded comparison between large multimodal models and diverse human cohorts. Building on this dynamic philosophy, the **DyVal 2** framework employs meta-probing agents inspired by psychometric theory to automatically transform static evaluation problems into configurable, dynamic probes of cognitive abilities [385]. This approach facilitates a multifaceted analysis of model capabilities, uncovering implicit correlations between cognitive skills and demonstrating a “Matthew effect” where strengths in one domain strongly correlate with others, thus providing a deeper understanding of model architecture than static scores alone.

Complementing these quantitative advances is the growing recognition that scalar metrics are insufficient for diagnosing specific failure modes or guiding targeted improvements in complex MM-RAG pipelines. Quantitative scores often obscure fine-grained behavioral nuances, forcing developers to rely on heuristic adjustments. In response, **QualEval** proposes a paradigm that augments quantitative metrics with automated qualitative evaluation, utilizing powerful large language model reasoners and linear programming solvers to generate human-readable insights and fine-grained visualizations [386]. Acting as a “data-scientist-in-a-box,” **QualEval** accelerates model refinement by pinpointing specific weaknesses that aggregate scores miss. Similarly, the **CheckEval** framework addresses evaluation ambiguity by decomposing criteria into detailed sub-aspects and constructing checklists of Boolean questions, a structured approach that significantly enhances interpretability and correlation with human judgments [387]. These tools represent a pivotal move toward evaluation systems that actively participate in the iterative refinement of MM-RAG architectures rather than serving as passive rankers.

Furthermore, advanced evaluation must account for the non-stationary nature of real-world document and chart distributions, a challenge that static datasets fail to capture. The concept of “context shift” describes semantically meaningful changes in underlying data processes that induce model brittleness in production environments, paralleling the distribution shift concerns raised in Section 7.5 but focusing here on adaptive measurement strategies [377]. To mitigate this, dynamic benchmarking strategies are essential. While frameworks like **Dynamic Quality-Diversity Search** maintain an archive of solutions updated in response to environmental changes, the principles underlying them suggest a need for MM-RAG evaluation protocols that continuously evolve to target emerging model weaknesses [388]. The **Zeno** framework supports this by providing an interactive environment for behavioral evaluation, enabling practitioners to visualize and test AI systems across diverse use cases to discover systematic failures, such as biases or safety issues, that aggregate metrics overlook [389]. This shift toward “living” benchmarks ensures that evaluation remains relevant as models and data distributions evolve.

Finally, the reliability and fairness of the evaluation process itself require rigorous scrutiny, particularly regarding the stability of model rankings and the integrity of automated judges. Research indicates that leaderboard rankings can be adversarially manipulated or may not reflect equitable

performance across different data subsets. The study **How Robust are Model Rankings** demonstrates that top-performing models on standard leaderboards may not be optimal for specific applications, advocating for leaderboard customization based on sample difficulty and user focus areas to ensure equitable evaluation [390]. Additionally, the widespread use of LLMs as evaluators introduces systematic biases, such as position bias, where response ordering significantly skews results [391]. This necessitates calibration frameworks, such as those proposed in **Calibrating LLM-Based Evaluator**, which align LLM-based scoring with human preferences through multi-stage refinement [392]. Moreover, the **Metrics Reloaded** framework emphasizes the importance of problem-aware metric selection, guiding researchers to choose validation metrics that truly reflect domain interests rather than defaulting to potentially misleading standard measures [393].

In summary, the future of evaluating MM-RAG for chart and document understanding lies in adopting these advanced, multidimensional paradigms that bridge the gap between foundational robustness (Section 7.5) and the systemic ecosystem challenges discussed next (Section 7.7). By integrating psychometric rigor via **PATCH** and **DyVal 2**, embracing qualitative diagnostics through **QualEval** and **CheckEval**, and ensuring dynamic adaptability and fairness as highlighted by **Zeno** and **How Robust are Model Rankings**, the community can develop evaluation ecosystems that truly reflect the complexity and stakes of real-world multimodal reasoning. However, while these sophisticated frameworks offer a path forward, their practical realization is currently hindered by fundamental vulnerabilities within the existing benchmarking infrastructure, a critical tension that sets the stage for the discussion on systemic gaps and future directions in the following section.

7.7 Critical Gaps and Future Directions in Benchmarking Ecosystems

7.7 Critical Gaps and Future Directions in Benchmarking Ecosystems

While Section 7.6 outlined the emergence of sophisticated psychometric and dynamic evaluation paradigms, the practical realization of these advanced frameworks is currently hindered by fundamental systemic vulnerabilities within the existing benchmarking ecosystem. Despite the rapid proliferation of benchmarks for Multimodal Retrieval-Augmented Generation (MM-RAG) in chart and document understanding, the field remains fragmented, with critical gaps that threaten the validity of reported progress. Addressing these deficiencies is a prerequisite for the effective deployment of the rigorous methods described previously.

A primary and pervasive challenge undermining these advanced paradigms is the risk of data contamination, which fundamentally compromises the integrity of performance metrics. As models are increasingly trained on web-scale corpora, there is a high probability that test sets from popular benchmarks have been inadvertently included in pre-training data, allowing models to “cheat” via memorization rather than demonstrating genuine reasoning capabilities [394]. This issue is exacerbated by the opacity of training datasets and the difficulty in measuring contamination levels, leading to inflated assessments of model effectiveness [395]. Recent longitudinal analyses suggest that contamination trends are increasing over time, necessitating rigorous contamination detection mechanisms and the development of dynamic, contamination-resilient evaluation frameworks [396]. Without addressing this, the psychometric rigor and dynamic adaptability advocated in Section 7.6 may yield illusory progress driven by data leakage rather than architectural advancements.

Furthermore, the field lacks unified and standardized metrics specifically tailored for the unique complexities of multimodal RAG systems, a gap that limits the utility of current diagnostic tools. Existing evaluation practices often rely on disparate metrics that assess retrieval precision, generation faithfulness, or grounding accuracy in isolation, failing to capture the holistic interplay between

these components. While frameworks like ARES attempt to automate RAG evaluation across dimensions such as context relevance and answer faithfulness, they often depend on lightweight judges that may not fully grasp the nuances of visual evidence in complex documents [397]. Similarly, existing benchmarks frequently overlook the multi-granular nature of document understanding, where answers may depend on synthesizing information from layout structures, OCR text, and visual charts simultaneously. The absence of a comprehensive, unified metric suite hinders fair comparison across different system architectures and obscures specific failure modes, such as the tendency of models to hallucinate when retrieved visual context is ambiguous or noisy [398]. This fragmentation stands in contrast to the integrated, multi-faceted analysis promised by frameworks like QualEval and DyVal 2.

A critical gap also exists in the linguistic and cultural diversity of current benchmarks, which restricts the generalizability of even the most robust evaluation models. The majority of existing datasets are predominantly English-centric, creating a significant blind spot for MM-RAG applications in global, multicultural environments. This limitation is particularly acute in enterprise and regulatory domains where documents often contain mixed languages and culturally specific formatting. The introduction of benchmarks like IGLUE highlights the necessity of evaluating transfer learning across modalities, tasks, and languages, revealing that current models struggle significantly in zero-shot multilingual settings [399]. Moreover, the lack of culturally situated evaluation data means that models may fail to interpret charts or documents correctly when they involve region-specific conventions or non-Latin scripts [400]. Future ecosystems must prioritize the creation of multilingual and cross-cultural benchmarks that reflect the diversity of real-world document corpora, ensuring that MM-RAG systems are robust and equitable across different linguistic communities [401].

Another pivotal direction for future research is the shift from static, one-off question-answering evaluations to interactive and agent-based protocols, extending the dynamic principles introduced in Section 7.6. Traditional benchmarks typically present a fixed query and expect a single generated response, failing to capture the iterative nature of real-world document exploration where users may refine queries, request clarifications, or navigate through multi-page reports. Emerging frameworks like KIEval propose knowledge-grounded interactive evaluation methods that utilize multi-round dialogues to distinguish between mere recall and deep comprehension, offering a more rigorous test of a model’s reasoning capabilities [402]. Additionally, the rise of autonomous agents capable of tool use and web navigation necessitates evaluation suites that assess behavioral safety and risk awareness in dynamic environments, as seen in benchmarks like R-Judge [403]. Static benchmarks are insufficient for evaluating the planning, self-correction, and long-horizon reasoning required for advanced MM-RAG agents, highlighting the need for the “living” and adaptive testing environments discussed earlier.

Finally, the reliability of the evaluation process itself remains under scrutiny, particularly regarding the widespread use of LLMs as judges, which directly impacts the trustworthiness of the quantitative and qualitative insights generated by tools like CheckEval and QualEval. While LLM-based evaluators offer scalability, they are prone to position bias, where the ranking of candidate responses can be manipulated simply by altering their order in the prompt [391]. This introduces a layer of uncertainty in benchmark results, especially when human annotation is scarce or expensive. Future ecosystems must incorporate robust calibration strategies, such as balanced position sampling and multi-evidence verification, to ensure that automated evaluations align closely with human judgments [404]. Moreover, there is a growing need for “living” benchmarks that can evolve alongside model capabilities, incorporating adversarial examples and updated data to prevent saturation and

ensure continuous progress [405]. Addressing these gaps through unified metrics, diverse data, interactive protocols, and reliable evaluation methodologies is essential for building trustworthy and capable MM-RAG systems that can truly leverage the advanced paradigms of the future.

8 Open Challenges, Ethical Considerations, and Future Directions

8.1 Privacy-Preserving Retrieval and Data Sovereignty

8.1 Privacy-Preserving Retrieval and Data Sovereignty

While the following section addresses the ethical imperatives of bias and fairness, the foundational prerequisite for deploying Multimodal Retrieval-Augmented Generation (MM-RAG) systems in high-stakes domains such as healthcare, finance, and law is the rigorous protection of data sovereignty and sensitive information. Unlike traditional centralized machine learning paradigms where data is aggregated into a single repository, MM-RAG systems often operate across distributed environments where data owners—such as hospitals, legal firms, or financial institutions—cannot share raw documents due to strict regulatory frameworks like the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA). The fundamental tension lies in enabling powerful retrieval and reasoning capabilities over private, visually rich corpora without compromising the confidentiality of the underlying data or the queries issued by users. As noted in recent surveys, while Federated Learning (FL) has emerged as a promising solution to facilitate distributed collaborative learning without disclosing original training data, retaining data and computation on-device is often insufficient for guaranteeing privacy, as model parameters exchanged between parties can still conceal sensitive information [406].

To address these vulnerabilities, advanced cryptographic techniques and secure computation protocols must be integrated into the retrieval and generation pipelines of MM-RAG systems. One critical approach involves the adoption of Secure Multi-Party Computation (MPC), which allows multiple parties to jointly compute functions over their inputs while keeping those inputs private. In the context of distributed retrieval, MPC can enable the aggregation of relevance scores or the execution of complex query logic across disparate document silos without any single party gaining access to another’s raw data or specific query intents. For instance, frameworks leveraging MPC primitives, such as secret sharing, have been proposed to create decentralized aggregation mechanisms that provide strict privacy guarantees against curious aggregators or colluding data owners [407]. Furthermore, the integration of MPC with FL has been shown to enhance robustness against model poisoning attacks while maintaining differential privacy, a dual requirement essential for trustworthy MM-RAG systems in adversarial environments [408].

Homomorphic Encryption (HE) represents another cornerstone for preserving data sovereignty in MM-RAG applications. HE allows computations to be performed on encrypted data, producing an encrypted result that, when decrypted, matches the result of operations performed on the plaintext. This capability is particularly vital for protecting the privacy of medical images and electronic health records (EHRs) during the retrieval and analysis phases. Research has demonstrated the feasibility of using HE-based protocols to protect deep learning models from adversarial collaborators in medical contexts, ensuring that neither the model weights nor the intermediate activations leak sensitive patient information [409]. Moreover, recent advancements have enabled efficient Area Under the Curve (AUC) calculations in horizontal federated learning systems using fully homomorphic encryption, achieving high accuracy and computational robustness even with large numbers of participating parties, thereby addressing privacy leaks that often occur during the evaluation

phase of distributed models [410].

The challenge of data sovereignty is further complicated by the heterogeneous nature of multimodal data, which includes text, tables, charts, and images. In sectors like finance and healthcare, data is often tabular or visual, and recent studies have highlighted that federated learning systems processing such data are vulnerable to sophisticated reconstruction attacks. For example, generative adversarial networks (GANs) have been successfully used to reconstruct private tabular data from shared model updates, indicating that standard FL protocols are inadequate for protecting structured visual and textual data [411]. Similarly, in medical imaging, adversaries have been able to reconstruct private images from shared gradients, underscoring the urgent need for enhanced defense mechanisms beyond simple parameter sharing [412]. Consequently, future MM-RAG architectures must incorporate defense-in-depth strategies, combining differential privacy with cryptographic secure aggregation to mitigate these reconstruction risks effectively.

Compliance with data sovereignty regulations also necessitates mechanisms for “data localization,” where data processing occurs strictly within geographic or institutional boundaries. Federated learning naturally aligns with this requirement by keeping data local; however, ensuring that the global model itself does not memorize or leak sensitive attributes requires rigorous privacy-preserving techniques. Approaches such as Local Differential Privacy (LDP) have been adapted for industrial cross-silo settings to enforce stronger privacy guarantees even when entities are untrusted, demonstrating high model performance under tight privacy budgets [346]. Additionally, the concept of Confidential and Private Collaborative (CaPC) learning has been introduced to address scenarios where both confidentiality (preventing explicit data sharing) and privacy (preventing implicit inference) are required, allowing parties to improve their local models without training a central model or joining datasets [413].

Looking forward, the evolution of privacy-preserving retrieval for MM-RAG will likely depend on hybrid frameworks that balance computational efficiency with rigorous security guarantees. Systems must be designed to handle the “right to be forgotten,” enabling the removal of specific client contributions from a trained model without full retraining, a capability that is becoming increasingly mandated by privacy regulations [414]. Furthermore, as MM-RAG systems scale to include autonomous agents capable of interactive document exploration, the attack surface expands, necessitating robust verification protocols and accountability mechanisms, potentially leveraging blockchain technologies to audit model updates and penalize malicious behavior [415]. Ultimately, achieving true data sovereignty in multimodal domains requires a paradigm shift from merely protecting data at rest to securing the entire lifecycle of retrieval, reasoning, and generation through a synergy of cryptographic innovation and regulatory-aware system design. Establishing these secure foundations is critical, as even the most privacy-compliant system fails its ethical mandate if it propagates societal biases, a challenge explored in the subsequent discussion on bias mitigation and accountability.

8.2 Bias Mitigation and Accountability in Multimodal Reasoning

8.2 Bias Mitigation and Accountability in Multimodal Reasoning

While the preceding section established the cryptographic and regulatory foundations necessary for data sovereignty, establishing these secure foundations is critical, as even the most privacy-compliant system fails its ethical mandate if it propagates societal biases. The deployment of Multimodal Retrieval-Augmented Generation (MM-RAG) systems in high-stakes domains such as healthcare, legal analysis, and financial reporting introduces profound ethical challenges, par-

ticularly concerning the propagation and amplification of societal biases. Unlike unimodal text systems, MM-RAG architectures must navigate a complex landscape where biases are embedded not only in linguistic corpora but also in visual representations, chart annotations, and the retrieval mechanisms that bridge these modalities. The sources of bias in these systems are multifaceted, originating from skewed training data, flawed retrieval rankings, and the inherent stereotypical associations learned by multimodal encoders. Addressing these issues requires a fundamental shift from purely performance-driven optimization to comprehensive accountability frameworks that prioritize fairness, transparency, and human oversight.

A primary source of bias in MM-RAG stems from the training corpora used to pre-train foundational models. Large-scale datasets scraped from the internet often reflect historical inequalities and stereotypical associations regarding gender, race, and socioeconomic status. Research has demonstrated that Large Language Models (LLMs) can systematically disadvantage names associated with racial minorities and women when generating advice or making predictions, indicating that social biases are deeply ingrained in the model parameters themselves [416]. In a multimodal context, this extends to visual data where models may amplify gender biases present in image datasets, leading to skewed interpretations of charts or documents depicting specific demographic groups [417]. Furthermore, automatic evaluation metrics, which are often used to guide model training and selection, have been found to encode similar societal biases. For instance, model-based evaluation metrics for image captioning may favor captions that align with gender stereotypes, thereby perpetuating these biases during the reinforcement learning process [418]. This creates a dangerous feedback loop where biased metrics validate biased generations, making it difficult to detect and mitigate unfair outcomes in chart and document understanding tasks.

The retrieval mechanism in MM-RAG systems presents another critical vector for bias propagation, acting as a gatekeeper that determines which evidence influences the generator’s reasoning. If retrieval models are biased, they can systematically exclude or down-rank evidence relevant to marginalized groups, effectively silencing alternative perspectives before generation even begins. Studies on neural ranking models have shown that they can significantly intensify gender bias compared to traditional retrieval methods, often due to the use of biased pre-trained embeddings [419]. Additionally, societal biases resonating in retrieved contents can reinforce existing stereotypes, necessitating robust measurement frameworks to audit the fairness of ranking outputs [420]. The uncertainty inherent in retrieval scores further complicates this issue; without accounting for predictive uncertainty, ranking systems may confidently promote biased documents while suppressing fairer alternatives. Approaches that leverage uncertainty estimates have shown promise in mitigating bias in ranked lists without sacrificing utility, suggesting that probabilistic perspectives are essential for fair retrieval in multimodal systems [421].

To address these challenges, the field must move towards comprehensive accountability metrics and human-centered transparency frameworks. Traditional comparative fairness notions, which rely on comparing outcomes between groups, are often difficult to apply in practice due to the subjective nature of fairness and the complexity of multimodal interactions. A paradigm shift towards non-comparative justice, where model outcomes are evaluated against a desired standard for each specific input, offers a more practical approach for human auditing [422]. This aligns with the need for human-in-the-loop systems that empower domain experts to identify and mitigate biases through interactive tools. For example, visual analytics tools that utilize causal models allow users to detect unfair causal relationships in data and simulate debiased scenarios, thereby enhancing trust and accountability in algorithmic decisions [423]. Such tools are crucial for MM-RAG systems, where the interplay between visual layout, textual content, and retrieved context creates opaque decision

pathways that require explicit explanation.

Moreover, accountability in MM-RAG demands rigorous auditing processes that go beyond static benchmarks. Given the rapid evolution of models, benchmarks can quickly become obsolete, and biases may manifest in subtle, context-dependent ways that standard tests miss. Frameworks that enlist models to construct challenging, adversarial datasets help keep pace with model evolution and distinguish between genuine bias and model brittleness [424]. Additionally, the transparency of the entire machine learning pipeline, from data sourcing to model deployment, must be quantifiable to ensure that stakeholders can validate the systems they rely on. Methods for quantifying the transparency of process pipelines enable auditors to assess the visibility of contributions and the reliability of data sources, which is vital for maintaining trust in long-term deployments [425].

Finally, mitigating bias in MM-RAG requires acknowledging the limitations of current debiasing techniques. While methods such as oversampling minority groups or applying post-processing constraints can reduce adverse impact, they often come at the cost of model accuracy or fail to address the root causes of bias embedded in the pre-training corpus [426]. Consequently, future research must focus on developing causality-guided debiasing frameworks that address the data-generating processes and internal reasoning mechanisms of models [427]. By integrating these advanced mitigation strategies with robust accountability metrics and human-centered design principles, the community can build MM-RAG systems that are not only powerful but also fair, transparent, and trustworthy. However, the practical realization of these ethical frameworks is contingent upon the system’s ability to operate efficiently at scale; without the capacity to access comprehensive and diverse corpora through scalable indexing, retrieval mechanisms risk becoming narrow or skewed, inadvertently amplifying the very biases they aim to mitigate, a challenge explored in the subsequent discussion on scalability and efficient indexing.

8.3 Scalability and Efficient Indexing for Massive Corpora

8.3 Scalability and Efficient Indexing for Massive Corpora

While the preceding section highlighted the critical need for fairness and accountability in MM-RAG reasoning, the practical realization of these ethical frameworks is contingent upon the system’s ability to operate efficiently at scale. The deployment of Multimodal Retrieval-Augmented Generation (MM-RAG) systems in real-world scenarios involving financial reports, scientific literature, and legal archives faces a formidable barrier: the computational bottleneck associated with indexing and retrieving from massive, multi-page document collections and high-resolution visual archives. As the volume of data expands to billions of documents and features, traditional retrieval paradigms struggle to maintain the dual requirements of low-latency response times and high retrieval accuracy. This scalability challenge is not merely an engineering hurdle but a prerequisite for the robust, transparent, and unbiased systems discussed previously; without efficient access to comprehensive corpora, retrieval mechanisms risk becoming narrow or skewed, inadvertently amplifying the biases they aim to mitigate. Consequently, addressing the trade-off between the semantic power of dense retrieval methods and their prohibitive memory usage and latency when scaling to web-sized corpora creates an urgent need for innovative indexing strategies that can handle dynamic growth without sacrificing performance or fairness.

One of the primary obstacles in scaling MM-RAG is the sheer cost of maintaining up-to-date indices for rapidly evolving corpora, a factor that directly impacts the freshness and relevance of retrieved evidence. In many practical applications, document collections are not static; new charts, updated regulatory filings, and fresh research papers are continuously added. Traditional approaches that

require full re-indexing or re-training of models are computationally unsustainable and can lead to stale knowledge bases that fail to reflect current realities. To address this, the research community has increasingly turned toward incremental workflows and continual learning mechanisms. For instance, the concept of continual learning for generative retrieval has been explored to incrementally index new documents while preserving the ability to answer queries related to both previously and newly indexed content. Approaches such as “Continual Learning for Generative Retrieval over Dynamic Corpora” [129] propose mechanisms like Incremental Product Quantization and memory-augmented learning to encode new documents with low computational cost, thereby mitigating the catastrophic forgetting that often plagues dynamic systems. Similarly, “DSI++: Updating Transformer Memory with New Documents” [135] highlights the severe forgetting events that occur during continual indexing and introduces generative memory strategies to sample pseudo-queries, significantly stabilizing the learning process for large-scale, evolving datasets. These advancements suggest that future MM-RAG systems must move beyond static indices to adopt adaptive, life-long learning architectures capable of seamless integration of new multimodal data, ensuring that accountability frameworks have access to the most current information.

Beyond incremental updates, the architectural design of the index itself requires a paradigm shift to handle the density and complexity of visual data inherent in chart and document understanding. High-dimensional indexing for visual instance search presents unique scalability challenges, where a collection of merely a few million images can generate billions of features. The necessity for systems that provide both dynamicity (handling online insertions) and durability (recovering from crashes) is critical for maintaining trust in high-stakes domains. Research such as “Dynamicity and Durability in Scalable Visual Instance Search” [428] demonstrates that extending scalable disk-based high-dimensional indices to support ACID properties is feasible, enabling transactional integrity even in massive, rapidly growing collections. Furthermore, for extremely large multimedia collections, distributed processing frameworks have proven essential. “Scalable high-dimensional indexing and searching with Hadoop” [429] illustrates how leveraging grid infrastructures can facilitate indexing and searching over tens of billions of descriptors, achieving stable throughput that single-server solutions cannot match. This indicates that scalable MM-RAG for document understanding will likely rely on hybrid architectures that combine sophisticated learned indices with distributed computing backbones, laying the infrastructure necessary for the complex, multi-step interactions envisioned in autonomous agents.

A particularly promising direction for overcoming scalability limits is the transition from explicit external indices to generative retrieval paradigms that utilize semantic identifiers. By encoding the corpus directly into model parameters or generating document identifiers autoregressively, systems can potentially bypass the latency bottlenecks of approximate nearest neighbor search, offering a more fluid interface for future interactive agents. However, scaling generative retrieval to millions of passages remains a significant unsolved challenge, as naive scaling of model parameters often yields diminishing returns. “How Does Generative Retrieval Scale to Millions of Passages” [127] provides a critical empirical study showing that while generative retrieval is competitive on small corpora, scaling to millions requires careful construction of synthetic queries as document representations and acknowledges the current limits of parameter scaling. To mitigate these issues, hybrid approaches like “Generative Dense Retrieval: Memory Can Be a Burden” [128] propose a coarse-to-fine process that combines the deep interaction of generative models with the scalability of dense retrieval, using cluster-level matching to reduce the search space before performing fine-grained retrieval. Additionally, “Model-enhanced Vector Index” [130] introduces a differentiable model-enhanced index that leverages residual quantization to bridge sequence-to-sequence deep retrieval with embedding-based models, achieving better performance with serving latency comparable to

standard dense retrieval.

Efficiency in indexing is also tightly coupled with compression techniques and parameter-efficient adaptation, which are vital for deploying these systems in resource-constrained environments or enabling the rapid decision-making required by autonomous agents. As multi-vector models elevate retrieval quality, they exponentially increase memory requirements, rendering them challenging to scale. Solutions like “ESPN: Memory-Efficient Multi-Vector Information Retrieval” [430] address this by offloading embedding tables to SSDs and utilizing software prefetchers to maintain low latency despite reduced memory footprint. Similarly, binary embedding strategies, as discussed in “Binary Embedding-based Retrieval at Tencent” [431] and “Injecting Domain Adaptation with Learning-to-hash for Effective and Efficient Zero-shot Dense Retrieval” [432], offer substantial reductions in index size and computational cost with minimal accuracy loss. These compression techniques are vital for handling the massive feature spaces generated by high-resolution chart and document images. Moreover, the integration of learned index structures, such as those proposed in “SALI: A Scalable Adaptive Learned Index Framework based on Probability Models” [433], allows indices to adapt to workload skews and improve concurrency, further enhancing the robustness of retrieval systems under heavy load.

In conclusion, achieving scalability for MM-RAG in massive corpora requires a multi-faceted approach that synthesizes incremental learning, generative indexing, distributed computing, and aggressive compression. The future of this field lies in developing unified frameworks that can dynamically adapt to growing multimodal datasets while maintaining the precision necessary for complex document and chart understanding. As evidenced by the progression from static indexing to adaptive, generative, and compressed architectures, the path forward involves treating the index not as a fixed data structure but as a learnable, evolving component of the retrieval system itself. This evolution is the foundational bedrock upon which the next generation of MM-RAG systems will be built: only by solving the challenges of scale and efficiency can we enable the transition from passive retrieval to the autonomous, interactive agents capable of dynamic exploration discussed in the following section.

8.4 Towards Autonomous Interactive Agents and Dynamic Exploration

8.4 Towards Autonomous Interactive Agents and Dynamic Exploration

Building upon the scalable, adaptive indexing architectures discussed in the preceding section, the paradigm of Multimodal Retrieval-Augmented Generation (MM-RAG) for chart and document understanding is undergoing a profound transformation: shifting from static, single-turn question-answering systems toward dynamic, autonomous interactive agents capable of active exploration. While the previous section addressed the foundational challenges of indexing massive, evolving corpora to ensure low-latency access, this section explores how such efficient retrieval backbones empower agents to move beyond passive consumption of fixed document sets. Traditional MM-RAG frameworks typically operate on a rigid retrieve-then-generate basis, formulating answers based on a pre-selected context window. However, complex reasoning tasks in visually rich domains often demand multi-step interactions where the agent must actively navigate document hierarchies, zoom into specific chart regions, or traverse web interfaces to gather sufficient evidence. This evolution necessitates the development of agents endowed with robust planning capabilities, self-correction mechanisms, and the ability to generalize to unseen interface layouts and document

structures, effectively leveraging the dynamic indices described previously to support real-time decision-making.

The transition to autonomous interactive agents is driven by the recognition that real-world document and web environments are partially observable and dynamically structured. Agents can no longer rely solely on pre-processed, flat text representations derived from static indexes; instead, they must engage in active reasoning within open-world environments where information is often hidden behind interaction barriers. Recent research highlights the necessity for agents to actively explore, accumulate, and reason using both newfound and existing information to tackle incomplete-information questions, a capability that distinguishes human-like intelligence from passive pattern matching [434]. In the context of web navigation and document retrieval, this implies that agents must be able to formulate iterative queries, interpret visual cues, and execute actions such as clicking, scrolling, or switching tabs to uncover data points essential for accurate chart interpretation or legal precedent retrieval, thereby transforming the retrieval process from a one-off event into a continuous, exploratory dialogue.

A critical component of this evolution is the development of robust planning frameworks that enable agents to decompose complex goals into executable sub-tasks, utilizing the scalable retrieval infrastructure to fetch relevant context at each step. Hierarchical planning systems have emerged as a promising solution, allowing for centralized strategy formulation alongside decentralized execution. For instance, frameworks like the Hierarchical Auto-Organizing System (HAS) demonstrate how multi-agent organizations can be structured to handle navigation tasks through dynamic group adjustments and multi-modal perception, ensuring that agents can coordinate their actions effectively in complex, unpredictable environments [435]. Similarly, the Investigate-Consolidate-Exploit (ICE) strategy illustrates how agents can evolve by investigating planning trajectories, consolidating them into simplified workflows, and exploiting these pipelines for improved task execution, thereby reducing the computational overhead and enhancing adaptability across different tasks [436]. These planning mechanisms are essential for MM-RAG systems that must navigate deep document trees or multi-page financial reports, where the relevant information is not immediately visible and requires iterative retrieval guided by strategic planning.

Furthermore, the ability to generalize to unseen interface layouts and document structures remains a significant hurdle, even with advanced indexing. Many existing agents struggle when deployed on websites or document formats they have not encountered during training, often failing to map visual elements to actionable commands. To address this, researchers are exploring generalist models that can adapt to diverse environments through unified architectures. The OS-Copilot framework, for example, introduces a self-improving embodied agent capable of interfacing with comprehensive elements in an operating system, including files, web browsers, and third-party applications, demonstrating strong generalization to unseen applications via accumulated skills [437]. In the realm of web navigation, the SEEACT agent leverages Large Multimodal Models (LMMs) like GPT-4V to act as a generalist web agent, though it also highlights that grounding textual plans into specific visual actions remains a major challenge that requires sophisticated strategies combining HTML structure and visual content [438]. The Dual-View Visual Contextualization approach further supports this by contextualizing HTML elements through their visual counterparts in screenshots, enabling agents to better understand task-related contexts in real-world websites [439].

Self-correction and reflection are equally vital for ensuring the reliability of autonomous agents in high-stakes domains, particularly when operating over the vast, noisy corpora enabled by scalable indexing. Without the ability to critique their own actions, agents are prone to error propagation, especially in long-horizon tasks involving multiple retrieval and interaction steps. Frameworks in-

corporating self-reflective memory-augmented planning, such as Self-MAP, employ memory utilization and self-reflection techniques to handle context-dependency issues and improve performance in multi-turn conversational navigation [440]. Additionally, the concept of “learning by asking” empowers agents to resolve ambiguities by actively querying humans or external experts when confidence is low, significantly boosting success rates in navigation and task completion [441]. This interactive capability allows agents to bridge the gap between their internal knowledge and the specific requirements of a complex document or chart, reducing hallucinations and improving factual accuracy in scenarios where automated retrieval alone may be insufficient.

The future of MM-RAG lies in the seamless integration of these capabilities into cohesive systems that can operate autonomously yet collaboratively, supported by the efficient, dynamic indices detailed earlier. Agents must be equipped with the capacity for continuous learning and adaptation, evolving their strategies based on interaction history and feedback. The ReAct meets ActRe framework exemplifies this by enabling the autonomous annotation of agent trajectories, facilitating a closed loop for self-improvement through contrastive self-training [442]. Moreover, the development of toolkits like AgentStudio provides the necessary infrastructure for building and evaluating these general virtual agents in realistic, multimodal settings, covering the entire lifecycle from environment setup to benchmark creation [443]. As we move forward, the focus must shift towards creating agents that not only retrieve and generate but also actively explore, plan, reflect, and adapt. However, realizing the full potential of these autonomous agents requires rigorous validation. The transition from passive retrieval to active exploration introduces new complexities in assessment, necessitating the standardized evaluation frameworks and real-world benchmarks discussed in the following section to ensure these systems are not only capable but also trustworthy and reliable.

8.5 Standardized Evaluation Frameworks and Real-World Benchmarks

8.5 Standardized Evaluation Frameworks and Real-World Benchmarks

The transition toward autonomous, interactive agents described in the preceding section necessitates a parallel evolution in how we assess their capabilities. The rapid advancement of Multimodal Retrieval-Augmented Generation (MM-RAG) systems for chart and document understanding has significantly outpaced the development of robust, standardized evaluation frameworks. While numerous benchmarks exist, they often fail to capture the complexity of real-world deployment scenarios where agents must dynamically navigate, plan, and self-correct. Consequently, current evaluations frequently lead to an overestimation of model capabilities and a lack of actionable insights for improving agent reliability. A critical gap lies in the reliance on static, IID (Independent and Identically Distributed) datasets that do not reflect the dynamic, noisy, and adversarial nature of practical environments. As noted in broader NLU contexts, the trend toward adversarially constructed test sets is necessary but insufficient if the underlying benchmark design fails to measure the specific cognitive abilities required for complex reasoning [444]. For MM-RAG, this necessitates a paradigm shift toward multi-granularity, human-referenced benchmarks that rigorously assess not only answer accuracy but also grounding faithfulness, logical consistency, and resilience against sophisticated adversarial attacks during active exploration.

Current evaluation metrics frequently overlook the nuance of “faithfulness,” a deficiency that becomes particularly acute when autonomous agents generate long-form responses or interpret complex visual contexts. In such scenarios, models may hallucinate data points or cite non-existent document sections while appearing confident. Traditional metrics often rely on exact string matching or simple overlap scores, which fail to detect subtle factual inconsistencies or logical leaps

inherent in multi-step reasoning. To address this, future frameworks must incorporate fine-grained evaluation strategies. For instance, the concept of evaluating faithfulness through the “Longest Supported Subsequence” offers a promising direction by computing the longest non-continuous substring of a claim supported by the context, thereby providing a more granular measure of factual grounding [445]. Furthermore, while human evaluation remains the gold standard for assessing faithfulness, it is often plagued by low inter-annotator agreement and high workload, especially for long documents. Guidelines such as those proposed in LongEval suggest that switching to finer granularities of judgment, such as clause-level analysis, can significantly reduce variance and improve the reliability of human assessments [446]. Extending these principles to multimodal domains is essential, requiring benchmarks that explicitly task annotators with verifying visual grounding and logical deduction steps rather than just final answers, ensuring that the agent’s reasoning chain is as sound as its conclusion.

A significant deficiency in existing benchmarks is the lack of comprehensive robustness testing against adversarial threats, a vulnerability that is magnified in autonomous agents operating in open environments. Most current evaluations assume benign inputs, ignoring the reality that MM-RAG systems in high-stakes domains like healthcare or finance are vulnerable to both digital and physical adversarial attacks. The literature on adversarial robustness highlights that standard accuracy metrics are often misleading and that models may appear robust under specific perturbation intensities while failing catastrophically under others. The proposal of “adversarial hypervolume” as a metric underscores the need to assess robustness comprehensively over a range of perturbation intensities rather than relying on a single accuracy score [447]. In the context of document understanding, this implies creating benchmarks that include multi-granular adversarial attacks, targeting not just individual words but also layout structures, visual elements, and cross-modal alignments. Research has shown that multi-granular perturbations can reveal vulnerabilities that single-level attacks miss, necessitating evaluation frameworks that simulate these complex threat vectors [448].

Moreover, the evaluation of MM-RAG systems must account for the “severity” of errors, not just their frequency, especially when agents are deployed in safety-critical applications. A minor hallucination in a financial chart interpretation by an autonomous agent could have vastly different consequences than a similar error in a casual query. The concept of “Adversarial Severity,” which quantifies the downstream impact of corruptions by measuring semantic error within a hierarchical label space, provides a framework for prioritizing robustness against high-impact failures [449]. Future benchmarks should therefore adopt a multi-dimensional scoring system that weighs errors based on their potential real-world impact, moving beyond binary correct/incorrect classifications. This approach aligns with calls for holistic metrics that combine various perspectives, such as model reliance on specific features and proximity to decision boundaries, to better quantify sample-level vulnerability [450].

The issue of benchmarking validity itself is another pressing concern that threatens the reproducibility of results in this rapidly evolving field. The community faces a proliferation of “benchmarking crimes,” where methodological flaws in evaluation protocols compromise the comparability of results across different agent architectures [451]. In the realm of multimodal understanding, this manifests as inconsistent data preprocessing, unrealistic retrieval constraints, or evaluation metrics that do not align with human judgment. For example, using Large Language Models as evaluators without proper calibration can introduce systematic biases, where the order of response presentation skews the scoring [391]. To mitigate this, standardized frameworks must enforce strict protocols for data curation, metric definition, and evaluator calibration. Initiatives like RobustBench have

demonstrated the value of standardized, unified evaluation pipelines in the image classification domain, providing a model for how the MM-RAG community could establish similar leaderboards that enforce rigorous testing against diverse attack suites [452].

Finally, to support the development of the lifelong learning and adaptive systems envisioned in the next section, real-world benchmarks must be dynamic and evolving to prevent overfitting and to reflect the continuous nature of document corpora. Static datasets quickly become saturated, failing to challenge state-of-the-art models. The integration of generative models to create infinite variations of test cases, as seen in the GREAT Score framework for global robustness evaluation, offers a pathway to scalable and adaptive benchmarking [453]. Additionally, benchmarks should incorporate “unforeseen adversaries” to test generalization capabilities beyond the training distribution, ensuring that systems remain robust when faced with novel attack vectors or domain shifts [454]. By adopting these rigorous, multi-faceted, and human-centric evaluation standards, the field can establish the necessary foundation for developing MM-RAG systems that are not only accurate but also trustworthy, reliable, and safe for deployment in complex, real-world environments, bridging the gap between autonomous capability and the explainable, trustworthy systems of the future.

8.6 Roadmap for Next-Generation Explainable and Trustworthy Systems

8.6 Roadmap for Next-Generation Explainable and Trustworthy Systems

Building upon the rigorous, multi-dimensional evaluation standards established in the preceding section, the trajectory of Multimodal Retrieval-Augmented Generation (MM-RAG) for chart and document understanding must now pivot from assessment to architectural evolution. The transition toward autonomous agents necessitates a strategic roadmap that moves decisively away from static, black-box architectures toward dynamic, transparent, and self-evolving ecosystems. To realize the full potential of these systems in high-stakes domains such as healthcare, law, and scientific research, future development must prioritize three converging pillars: the deep integration of intrinsic explainable AI (XAI) principles, the adoption of lifelong learning mechanisms to handle evolving data landscapes, and the synthesis of unified models that erase the boundaries between perception, retrieval, and reasoning. This roadmap envisions a future where AI agents do not merely retrieve and regurgitate information but engage in cognitively grounded, accountable, and adaptive dialogue with complex visual-textual corpora, directly addressing the faithfulness and robustness gaps identified in current benchmarking practices.

The first critical vector of this roadmap is the transition from post-hoc interpretability to intrinsic explainability, a prerequisite for meeting the “grounding faithfulness” metrics demanded by modern evaluation frameworks. Current MM-RAG systems often suffer from a “black-box” nature where the reasoning path between a retrieved chart segment and a generated conclusion remains opaque, making it difficult to verify claims against the “Longest Supported Subsequence” or detect logical leaps. Future systems must adopt frameworks that explicitly model the cognitive processes of perception, memory, and inference. Inspired by cognitive architectures, next-generation models should incorporate distinct memory modules—working memory for immediate context and long-term memory for accumulated domain knowledge—to facilitate transparent reasoning chains [455]. By mimicking human memory consolidation and retrieval, these systems can provide auditable traces of how specific visual evidence influenced a decision, thereby satisfying the need for clause-level verification

proposed in human evaluation guidelines. Furthermore, the integration of structured knowledge sources, such as knowledge graphs, alongside unstructured document retrieval is essential. This hybrid approach allows models to ground their generation in factual, relational contexts, significantly reducing the risk of hallucinations and enhancing the precision of explanations provided to end-users [456]. The goal is to move beyond simple attention visualization to generating natural language explanations that articulate the “why” and “how” of a system’s deduction, thereby fostering trust among non-expert analysts and aligning with human-centered documentation standards [457].

The second pillar of the roadmap addresses the static nature of current models and datasets through the implementation of lifelong learning capabilities, directly countering the issue of benchmark saturation and distributional shift. As noted in the discussion on dynamic benchmarks, documents and charts are not static; they evolve with new regulations, emerging scientific findings, and shifting corporate strategies. Traditional fine-tuning approaches are ill-suited for this continuous stream of information due to the risk of catastrophic forgetting, where learning new data erases previously acquired knowledge. Future MM-RAG systems must employ dynamic architectures capable of incremental adaptation to ensure they remain robust against “unforeseen adversaries” and domain shifts. Techniques such as dynamic architecture-based learning, which utilizes hierarchically organized prompts to capture knowledge at varying granularities, offer a promising direction for handling unseen tasks and evolving domains without retraining from scratch [458]. Moreover, the incorporation of mixture-of-experts (MoE) architectures can allow models to expand their capacity dynamically, adding specialized experts for new document types or chart styles while regularizing existing parameters to preserve prior knowledge [459]. This capability is crucial for maintaining system relevance in fast-moving fields like materials science or legal precedent, where the ability to continuously accumulate and transfer knowledge from streams of document collections is paramount [460]. By treating the retrieval corpus as a living entity, these systems can adapt to changing “ground truths” and user labeling behaviors over time, ensuring sustained accuracy in historical and contemporary document analysis [461].

The third and perhaps most transformative element of this roadmap is the development of unified models that seamlessly bridge perception, retrieval, and reasoning, thereby mitigating the error propagation inherent in disjointed pipelines. The current paradigm often treats these as separate stages, leading to vulnerabilities that single-level adversarial attacks can exploit. The future lies in embodied and cognitively integrated agents that perceive visual layouts, retrieve relevant contexts, and reason over them within a single, cohesive framework. This requires moving beyond the “big data for small tasks” paradigm toward systems capable of developing “common sense” to solve wide-ranging tasks with limited data, leveraging core domains like causality and intent to understand the deeper semantics of charts and diagrams [462]. Such unified models should be capable of compositional reasoning, decomposing complex document queries into reusable sub-problems and synthesizing solutions in a manner that mirrors human cognitive flexibility [463]. Furthermore, the convergence of Large Language Models with world models or simulators can enhance the system’s ability to reason about physical and spatial relationships inherent in complex engineering drawings or scientific plots, effectively grounding abstract textual queries in concrete visual realities and improving resilience against multi-granular perturbations [464].

Ultimately, achieving trustworthy MM-RAG systems requires a holistic approach that intertwines these three pillars, creating a foundation for the “Hybrid Intelligence” necessary for real-world deployment. It demands a shift where the complementary strengths of human intuition and machine scalability are leveraged to solve complex organizational problems that neither could address

alone [465]. This involves creating interactive environments where humans can steer the model’s reasoning, validate retrieved evidence, and correct logical flaws in real-time, effectively establishing a collaborative semantic inference loop that serves as a continuous feedback mechanism for model improvement [466]. Additionally, robust accountability metrics must be developed to evaluate not just the accuracy of the output, but the integrity of the process, ensuring that systems adhere to ethical standards and data sovereignty requirements while quantifying the severity of potential failures [467]. By following this roadmap, the field can evolve from fragile, task-specific tools to resilient, general-purpose agents capable of navigating the intricacies of visually rich information ecosystems with transparency, adaptability, and unwavering reliability, fully bridging the gap between the evaluation challenges of today and the autonomous capabilities of tomorrow.

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